MIL-M-38510/9D
4 June 1980
SUPERSEDING
MIL-M-38510/9C(USAF)
5 December 1975

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL TTL, SHIFT REGISTERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, shift register microcircuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The complete part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4 bit right-shift, left-shift register
02	5 bit shift register
03	8 bit parallel-out serial shift register
04	8 bit parallel-load shift register
05	4 bit bidirectional shift register
06	4 bit parallel-access shift register

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline, MIL-M-38510, appendix C
A	F-1 (14-pin, 1/4" x 1/4", flat pack)
B	F-3 (14-pin, 3/16" x 1/4", flat pack)
C	D-1 (14-pin, 1/4" x 3/4", dual-in-line pack)
D	F-2 (14-pin, 1/4" x 3/8", flat pack)
E	D-2 (16-pin, 1/4" x 7/8", dual-in-line pack)
F	F-5 (16-pin, 1/4" x 3/8", flat pack)

1.3 Absolute maximum ratings.

Supply voltage range (V_{CC})	-0.5 Vdc to 7.0 Vdc
Input voltage range	-1.5 Vdc at -12 mA to 5.5 Vdc
Storage temperature range	-65° to 150°C
Maximum power dissipation per register, P_D 1/	
Device type 01	422 mWdc
Device type 02	400 mWdc
Device type 03	322 mWdc
Device type 04	372 mWdc
Device type 05	360 mWdc
Device type 06	372 mWdc

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Lead temperature
(soldering, 10 seconds) - 300°C

Thermal resistance, junction to case - - - - - - - - - - - - θ_{JC} = {0.09°C/mW for flat pack; 0.08°C/mW for dual-in-line pack

Junction temperature - T_J = 175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC}) - - - - - - - - - - - - - - - - -	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage - - - - - - - - - - - -	2.0 Vdc
Maximum low level input voltage - - - - - - - - - - - -	0.8 Vdc
Ambient operating temperature range - - - - - - - - - -	-55° to 125°C
Fan out	
Device types 01, 02, 04, 05, and 06	
High logic level - - - - - - - - - - - - - - - - - - -	20
Low logic level -	10
Device type 03	
High logic level - - - - - - - - - - - - - - - - - - -	10
Low logic level -	5
Device type 01	
Low level setup time at mode control with respect to clock 1 input - - - - - - - - -	35 ns minimum
High level setup time at mode control with respect to clock 2 input - - - - - - - - -	35 ns minimum
Low level setup time at mode control with respect to clock 2 input - - - - - - - - -	10 ns minimum
High level setup time at mode control with respect to clock 1 input - - - - - - - - -	10 ns minimum
Width of clock pulse - - - - - - - - - - - - - - - - -	20 ns minimum
Setup time required at serial A, B, C, D inputs - - - - - - - - - - - - - - - - - -	20 ns minimum
Hold time required at serial A, B, C, D inputs - - - - - - - - - - - - - - - - - - -	5 ns minimum
Device type 02	
Minimum clock pulse width - - - - - - - - - - - - - - -	35 ns maximum
Minimum clear pulse width - - - - - - - - - - - - - - -	30 ns maximum
Minimum preset pulse width - - - - - - - - - - - - - - -	30 ns maximum
Serial input setup time - - - - - - - - - - - - - - - -	30 ns minimum
Serial input hold time - - - - - - - - - - - - - - - - -	0 ns minimum
Device type 03	
Minimum clock pulse width - - - - - - - - - - - - - - -	30 ns maximum
Minimum clear pulse width - - - - - - - - - - - - - - -	50 ns maximum
Serial setup time - - - - - - - - - - - - - - - - - - -	15 ns minimum
Serial hold time -	10 ns maximum
Device type 04	
Width of clock input pulse - - - - - - - - - - - - - -	20 ns minimum
Width of load input pulse - - - - - - - - - - - - - - -	25 ns minimum
Clock enable setup time - - - - - - - - - - - - - - - -	30 ns minimum
Parallel input setup time - - - - - - - - - - - - - - -	10 ns minimum
Serial input setup time - - - - - - - - - - - - - - - -	35 ns minimum
Shift setup time -	45 ns minimum
Hold time at serial input - - - - - - - - - - - - - - -	0 ns maximum
Hold time at parallel input - - - - - - - - - - - - - -	25 ns maximum
Device type 05	
Width of clock input pulse - - - - - - - - - - - - - -	20 ns minimum
Width of clear input pulse - - - - - - - - - - - - - -	20 ns minimum
Data input setup time - - - - - - - - - - - - - - - - -	20 ns minimum
Clear input setup time - - - - - - - - - - - - - - - -	25 ns minimum
Hold time at any input - - - - - - - - - - - - - - - -	7 ns minimum
Mode control setup time - - - - - - - - - - - - - - - -	30 ns minimum

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Device type 06	
Width of clock input pulse - - - - - - - - - - - -	16 ns minimum
Width of clear input pulse - - - - - - - - - - - -	12 ns minimum
Shift load input setup time- - - - - - - - - - - -	27 ns minimum
Data input setup time- - - - - - - - - - - - - - -	20 ns minimum
Clear input setup time - - - - - - - - - - - - - -	25 ns minimum
Shift load release time - - - - - - - - - - - - -	10 ns maximum
Data hold time - - - - - - - - - - - - - - - - - -	0 ns minimum

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microcircuits.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth tables and timing diagrams. The truth tables and timing diagrams shall be as specified on figure 2.

3.2.3 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.2.4 Schematic circuits. The schematic circuits shall be as specified on figure 4.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. Lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I and apply over the full recommended ambient operating temperature range, unless otherwise specified.

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TABLE I. Electrical characteristics.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IN} = 2.0 V I_{OH} = -400 µA	02,03	2.4		Volts
		V_{CC} = 4.5 V, V_{IN} = 2.0 V I_{OH} = -800 µA	01,04, 05,06	2.4		Volts
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IN} = 0.8 V I_{OL} = 16 mA	01,02,04, 05,06		0.4	Volts
		V_{CC} = 4.5 V, V_{IN} = 0.8 V I_{OL} = 8 mA	03		0.4	Volts
High level input voltage	V_{IH}	V_{CC} = 4.5 V	01,02,03, 04,05,06	2.0		Volts
Low level input voltage	V_{IL}	V_{CC} = 4.5 V	01,02,03, 04,05,06		0.8	Volts
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -12 mA T_A = 25°C	01,02,03 04,05,06		-1.5	Volts
High level input current at any input except mode control	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01		100	µA
High level input current at mode control	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	01		80	µA
	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01		200	µA
High level input current at any input except preset	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	02		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	02		100	µA
High level input current at preset	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	02		200	µA
	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	02		500	µA
High level input current at any input except clear	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	03		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	03		100	µA
High level input current at clear	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	03		80	µA
	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	03		200	µA
High level input current other than load input	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	04		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	04		100	µA
High level input current, load input	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	04		120	µA
	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	04		300	µA
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V	05,06		40	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	05,06		100	µA
Low level input current at any input except mode control	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	-0.4	-1.6	mA
Low level input current at mode control	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	-0.8	-3.2	mA
Low level input current at any input except preset	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	02	-0.7	-1.6	mA
Low level input current at preset	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	02	-3.0	-8.0	mA
Low level input current at any input except clear	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	03	-0.4	-1.6	mA
Low level input current at clear	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	03	-0.7	-2.6	mA

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits Min	Limits Max	Unit
Low level input current, load input	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	04	-1.2	-3.9	mA
Low level input current other than clock and load input	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	04	-0.4	-1.3	mA
Low level input current, clock input	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	04	-0.4	-1.6	mA
Low level input current other than S_0, S_1, and clock input	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	05	-0.4	-1.3	mA
Low level input current at S_0 and S_1 input	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	05	-0.4	-1.6	mA
Low level input current at clock input	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	05	-0.7	-1.6	mA
Low level input current at clear input	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	06	-0.4	-1.3	mA
Low level input current other than clear and clock inputs	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	06	-0.4	-1.6	mA
Low level input current at clock input	I_{IL3}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	06	-0.7	-1.6	mA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	01	-18	-57	mA
			02,05,06	-20	-57	mA
			03	-10	-27.5	mA
			04	-20	-55	mA
Supply current	I_{CC}	V_{CC} = 5.5 V 2/	01		72	mA
			02		68	mA
			04,05,06		63	mA
Supply current	I_{CC1}	V_{CC} = 5.5 V, $V_{IN(Clock)}$ = 0.4 V 2/	03		44	mA
	I_{CC2}	V_{CC} = 5.5 V, $V_{IN(Clock)}$ = 2.4 V 2/	03		54	mA
Maximum shift frequency	f_{MAX}	V_{CC} = 5 V, C_L = 50 pF ±10%, R_L = 400Ω ±5% (See figure 6)	01	16		MHz
Propagation delay time, low-to-high level from clock 1 or clock 2 to outputs	t_{PLH}			10	42	ns
Propagation delay time, high-to-low level from clock 1 or clock 2 to outputs	t_{PHL}			10	49	ns
Maximum clock frequency	f_{MAX}	V_{CC} = 5 V, C_L = 50 pF ±10%, R_L = 400Ω ±5% (See figure 7)	02	2		MHz
Propagation delay time, low-to-high level, from clock to output	t_{PLH1}			12	56	ns
Propagation delay time, high-to-low level, from clock to output	t_{PHL1}			12	56	ns
Propagation delay time, low-to-high level, from preset to output	t_{PLH2}			12	59	ns
Propagation delay time, high-to-low level, from clear to output	t_{PHL3}			12	77	ns

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
Maximum clock frequency	f_{MAX}	V_{CC} = 5 V C_L = 50 pF ±10% R_L = 800Ω ±5% (See figure 8)	03	18		MHz
Propagation delay time, high-to-low level, clear input to Q outputs	t_{PHL1}			12	63	ns
Propagation delay time, high-to-low level, clock input to Q outputs	t_{PHL2}			10	52	ns
Propagation delay time, low-to-high level, clock input to Q outputs	t_{PLH2}			10	42	ns
Maximum clock frequency	f_{MAX}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 400Ω ±5% (See figure 9)	04	14		MHz
Propagation delay time, low-to-high level, load input to any output	t_{PLH1}			10	40	ns
Propagation delay time, high-to-low level, load input to any output	t_{PHL1}			11	60	ns
Propagation delay time, low-to-high level, clock input to any output	t_{PLH2}			6	37	ns
Propagation delay time, high-to-low level, clock input to any output	t_{PHL2}			10	47	ns
Propagation delay time, low-to-high level, H input to Q_H output	t_{PLH3}			5	27	ns
Propagation delay time, high-to-low level, H input to Q_H output	t_{PHL3}			11	54	ns
Propagation delay time, low-to-high level, H input to $\overline{Q}_H$ output	t_{PLH4}			10	41	ns
Propagation delay time, high-to-low level, H input to $\overline{Q}_H$ output	t_{PHL4}			10	41	ns
Maximum clock frequency	f_{MAX}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 400Ω ±5% (See figure 10)	05	18		MHz
Propagation delay time, high-to-low level output from clear	t_{PHL1}			7	48	ns
Propagation delay time, low-to-high level output from clock	t_{PLH2}			7	36	ns
Propagation delay time, high-to-low level output from clock	t_{PHL2}			7	44	ns

See footnotes at end of table.

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TABLE I. Electrical characteristics - Continued.

Test	Symbol	Conditions	Device type	Limits		Unit
				Min	Max	
Maximum clock frequency	f_{MAX}	V_{CC} = 5.0 V C_L = 50 pF ±10% R_L = 400Ω ±5 (See figure 11)	06	24		MHz
Propagation delay time, high-to-low level output from clear	t_{PHL1}			7	34	ns
Propagation delay time, low-to-high level output from clock	t_{PLH2}			7	28	ns
Propagation delay time, high-to-low level output from clock	t_{PHL2}			7	34	ns

1/ Not more than one output should be shorted at a time.

2/ Device type:

01 - With the outputs open, mode control at 4.5 V, clock pulse applied to both clock inputs, I_{CC} is measured immediately after the application of the clock pulse.

02 - With the outputs open, presets at 4.5 V, I_{CC} is measured with the clock at ground and again with the clock at 4.5 V.

03 - I_{CC} is measured with outputs open, serial inputs grounded, and a momentary ground, then 4.5 V applied to clear.

04 - With the outputs open, serial at ground, clock, clock inhibit, and parallel inputs at 4.5 V, I_{CC} is measured by applying momentary ground, then 4.5 V to shift load prior to measurement.

05 - With all outputs open, inputs A thru D grounded, 5.5 V applied to S_0, S_1, clear, and the serial inputs, I_{CC} is tested by applying clock pulse.

06 - With the outputs open, clear at 5.5 V, shift load, J, $\overline{K}$, and data inputs grounded, I_{CC} is measured by applying clock pulse.

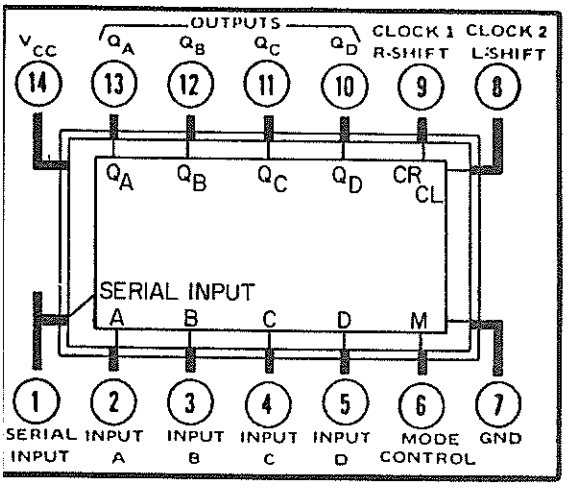
3.5 Electrical test requirements. Electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification and quality conformance by device class are specified in table II. (Subgroups 7 and 8 testing require only a summary of attributes data).

TABLE II. Electrical test requirements.

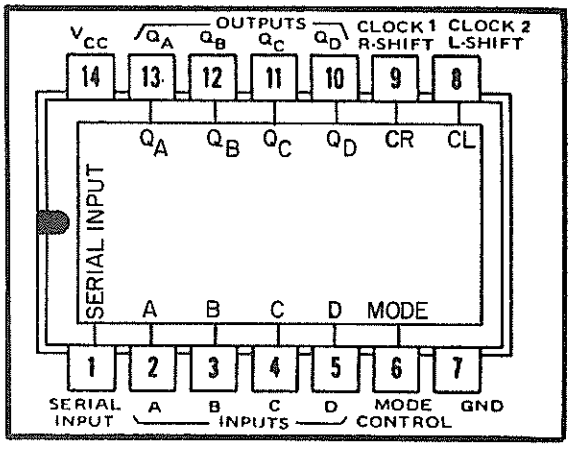
Applicable tests and MIL-STD-883 test methods	Subgroups (see table III) 1/		
	Class S device 2/	Class B device 2/	Class C device 2/
Interim electrical parameters (pre burn-in) (method 5004)	1	1	
Final electrical test parameters method 5004	1*,2,3,7, 9,10,11	1*,2,3, 7,9	1,7
Group A test requirements method 5005	1,2,3,7,8, 9,10,11	1,2,3,7, 9,10,11	1,2,3, 7,9
Group C end-point electrical parameters (method 5005)		1,2,3	1
Additional electrical subgroups for group C periodic inspections			10,11
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3	1

1/ (*) indicates PDA applies to subgroup 1 (see 4.2 c).
2/ Blank spaces indicate tests are not applicable.

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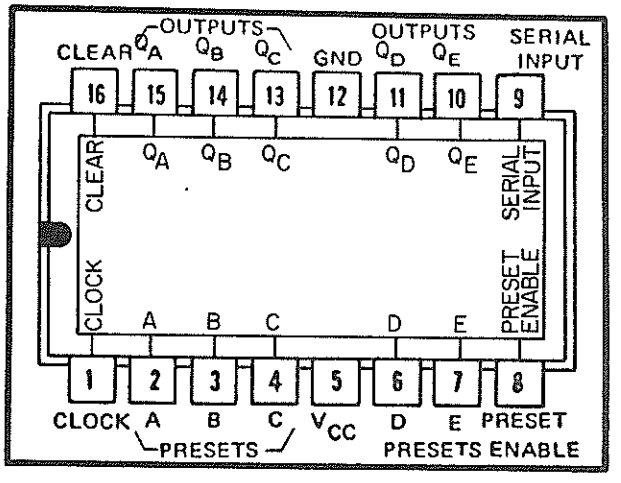


Case A, B and D



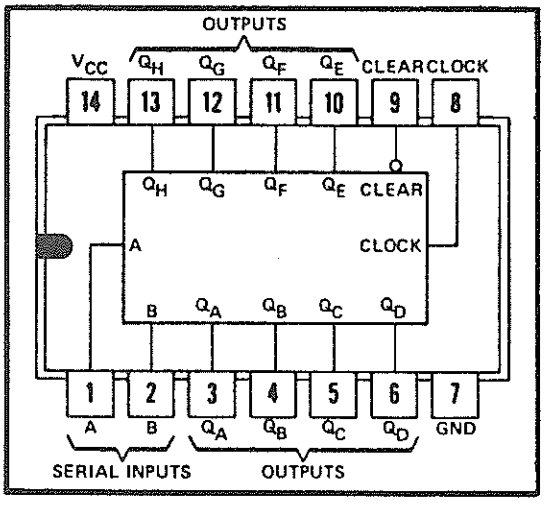
Case C

Device Type 01



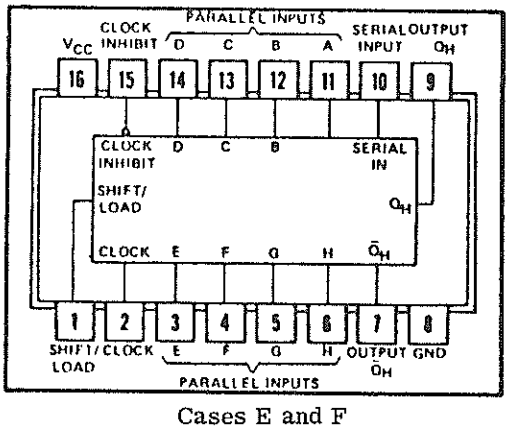
Cases E and F

Device Type 02



Case A, B, C and D

Device Type 03

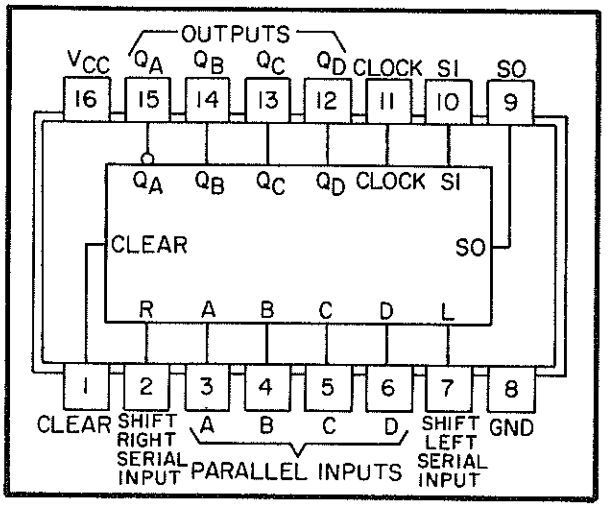


Cases E and F

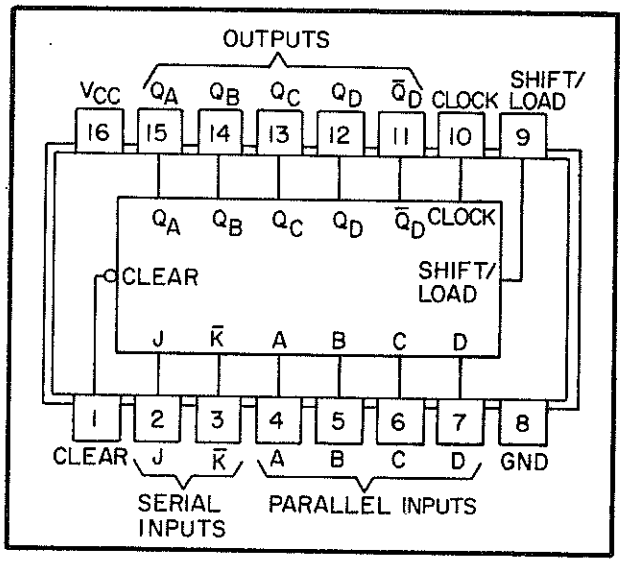
Device Type 04

FIGURE 1. Terminal connections.

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Cases E and F
Device Type 05



Cases E and F
Device Type 06

FIGURE 1. Terminal connections - continued

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Device type 01

INPUTS								OUTPUTS			
MODE CONTROL	CLOCKS 2 (L)	CLOCKS 1 (R)	SERIAL	PARALLEL A	PARALLEL B	PARALLEL C	PARALLEL D	Q_A	Q_B	Q_C	Q_D
H	H	X	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
H	↓	X	X	a	b	c	d	a	b	c	d
H	↓	X	X	Q_B†	Q_C†	Q_D†	d	Q_{Bn}	Q_{Cn}	Q_{Dn}	d
L	L	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
L	X	↓	H	X	X	X	X	H	Q_{An}	Q_{Bn}	Q_{Cn}
L	X	↓	L	X	X	X	X	L	Q_{An}	Q_{Bn}	Q_{Cn}
↑	L	L	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
↓	L	L	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
↓	L	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
↑	H	L	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
↑	H	H	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}

†Shifting left requires external connection of Q_B to A, Q_C to B, and Q_D to C. Serial data is entered at input D.

H = high level (steady state), L = low level (steady state), X = irrelevant (any input, including transitions)
↓ = transition from high to low level, ↑ = transition from low to high level
a, b, c, d = the level of steady-state input at inputs A, B, C, or D, respectively.
Q_{A0}, Q_{B0}, Q_{C0}, Q_{D0} = the level of Q_A, Q_B, Q_C, or Q_D, respectively, before the indicated steady-state input conditions were established.
Q_{An}, Q_{Bn}, Q_{Cn}, Q_{Dn} = the level of Q_A, Q_B, Q_C, or Q_D, respectively, before the most-recent ↓ transition of the clock.

Device type 02

INPUTS									OUTPUTS				
CLEAR	PRESET ENABLE	PRESET A	PRESET B	PRESET C	PRESET D	PRESET E	CLOCK	SERIAL	Q_A	Q_B	Q_C	Q_D	Q_E
L	L	X	X	X	X	X	X	X	L	L	L	L	L
L	X	L	L	L	L	L	X	X	L	L	L	L	L
H	H	H	H	H	H	H	X	X	H	H	H	H	H
H	H	L	L	L	L	L	L	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}	Q_{E0}
H	H	H	L	H	L	H	L	X	H	Q_{B0}	H	Q_{D0}	H
H	L	X	X	X	X	X	L	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}	Q_{E0}
H	L	X	X	X	X	X	↑	H	H	Q_{An}	Q_{Bn}	Q_{Cn}	Q_{Dn}
H	L	X	X	X	X	X	↑	L	L	Q_{An}	Q_{Bn}	Q_{Cn}	Q_{Dn}

H = high level (steady state), L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level
Q_{A0}, Q_{B0}, etc = the level of Q_A, Q_B, etc, respectively before the indicated steady-state input conditions were established.
Q_{An}, Q_{Bn}, etc = the level of Q_A, Q_B, etc, respectively before the most-recent ↑ transition of the clock.

Device type 03

INPUTS				OUTPUTS			
CLEAR	CLOCK	A	B	Q_A	Q_B ...		Q_H
L	X	X	X	L	L		L
H	L	X	X	Q_{A0}	Q_{B0}		Q_{H0}
H	↑	H	H	H	Q_{An}		Q_{Gn}
H	↑	L	X	L	Q_{An}		Q_{Gn}
H	↑	X	L	L	Q_{An}		Q_{Gn}

H = high level (steady state), L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level.
Q_{A0}, Q_{B0}, Q_{H0} = the level of Q_A, Q_B, or Q_H, respectively, before the indicated steady-state input conditions were established.
Q_{An}, Q_{Gn} = the level of Q_A or Q_G before the most-recent ↑ transition of the clock; indicates a one-bit shift.

FIGURE 2. Truth tables and timing diagrams.

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Device type 04

INPUTS					INTERNAL OUTPUTS		OUTPUT
SHIFT/ LOAD	CLOCK INHIBIT	CLOCK	SERIAL	PARALLEL A . . . H	Q_A	Q_B	Q_H
L	X	X	X	a . . . h	a	b	h
H	L	L	X	X	Q_{A0}	Q_{B0}	Q_{H0}
H	L	↑	H	X	H	Q_{An}	Q_{Gn}
H	L	↑	L	X	L	Q_{An}	Q_{Gn}
H	H	↑	X	X	Q_{A0}	Q_{B0}	Q_{H0}

H = high level (steady state), L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level
a . . . h = the level of steady-state input at inputs A thru H, respectively.
Q_{A0}, Q_{B0}, Q_{H0} = the level of Q_A, Q_B, or Q_H, respectively, before the indicated steady state input conditions were established.
Q_{An}, Q_{Gn}, = the level of Q_A or Q_G, respectively, before the most recent ↑ transition of the clock.

Device type 05

INPUTS									OUTPUTS			
CLEAR	MODE		CLOCK	SERIAL		PARALLEL						
	S_1	S_0		LEFT	RIGHT	A	B	C	D	Q_A	Q_B	Q_C	Q_D
L	X	X	X	X	X	X	X	X	X	L	L	L	L
H	X	X	L	X	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}
H	H	H	↑	X	X	a	b	c	d	a	b	c	d
H	L	H	↑	X	H	X	X	X	X	H	Q_{An}	Q_{Bn}	Q_{Cn}
H	L	H	↑	X	L	X	X	X	X	L	Q_{An}	Q_{Bn}	Q_{Cn}
H	H	L	↑	H	X	X	X	X	X	Q_{Bn}	Q_{Cn}	Q_{Dn}	H
H	H	L	↑	L	X	X	X	X	X	Q_{Bn}	Q_{Cn}	Q_{Dn}	L
H	L	L	X	X	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}

H = high level (steady state)
L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level
a, b, c, d = the level of steady-state input at inputs A, B, C, or D, respectively
Q_{A0}, Q_{B0}, Q_{C0}, Q_{D0} = the level of Q_A, Q_B, Q_C, or Q_D, respectively, before the indicated steady-state input conditions were established
Q_{An}, Q_{Bn}, Q_{Cn}, Q_{Dn} = the level of Q_A, Q_B, Q_C, Q_D, respectively, before the most-recent ↑ transition of the clock.

Device type 06

INPUTS									OUTPUTS				
CLEAR	SHIFT/ LOAD	CLOCK	SERIAL		PARALLEL								
			J	$\overline{K}$	A	B	C	D	Q_A	Q_B	Q_C	Q_D	$\overline{Q}_D$
L	X	X	X	X	X	X	X	X	L	L	L	L	H
H	L	↑	X	X	a	b	c	d	a	b	c	d	$\overline{d}$
H	H	L	X	X	X	X	X	X	Q_{A0}	Q_{B0}	Q_{C0}	Q_{D0}	$\overline{Q}_{D0}$
H	H	↑	L	H	X	X	X	X	Q_{A0}	Q_{A0}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
H	H	↑	L	L	X	X	X	X	L	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
H	H	↑	H	H	X	X	X	X	H	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
H	H	↑	H	L	X	X	X	X	$\overline{Q}_{An}$	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$

H = high level (steady state)
L = low level (steady state)
X = irrelevant (any input, including transitions)
↑ = transition from low to high level
a, b, c, d = the level of steady-state input at A, B, C, or D, respectively
Q_{A0}, Q_{B0}, Q_{C0}, Q_{D0} = the level of Q_A, Q_B, Q_C, or Q_D, respectively, before the indicated steady-state input conditions were established
Q_{An}, Q_{Bn}, Q_{Cn} = the level of Q_A, Q_B, or Q_C, respectively, before the most-recent transition of the clock

FIGURE 2. Truth tables and timing diagrams - Continued.

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Device type 01

Positive logic: Mode control = L for right shift.
Mode control = H for left shift or parallel load.

Transfer of information to the output pins occurs when the clock input goes from a logical H to a logical L.

Device type 02

Positive logic: Low input of clear sets all outputs to logical L.

Clear input is independent of clock.
Preset is independent of the clock or clear inputs.

The flip-flops may be independently set to the logical H state by applying a logical H to both the preset input of the specific flip-flop and the common preset input.

Transfer of information to the output pins occurs when the clock input goes from a logical L to a logical H.

The clear input shall be a logical H and the preset input shall be at a logical L when clocking occurs.

The proper information shall appear at the R-S inputs of each flip-flop prior to the rising edge of the clock input voltage waveform.

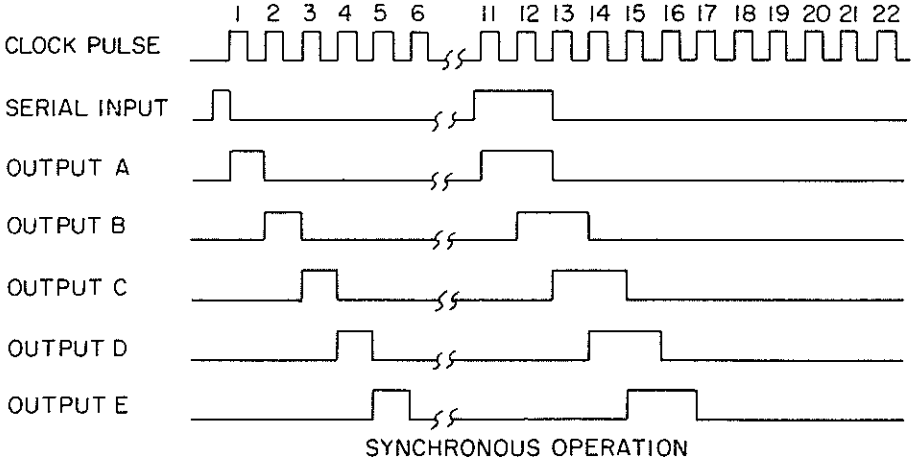
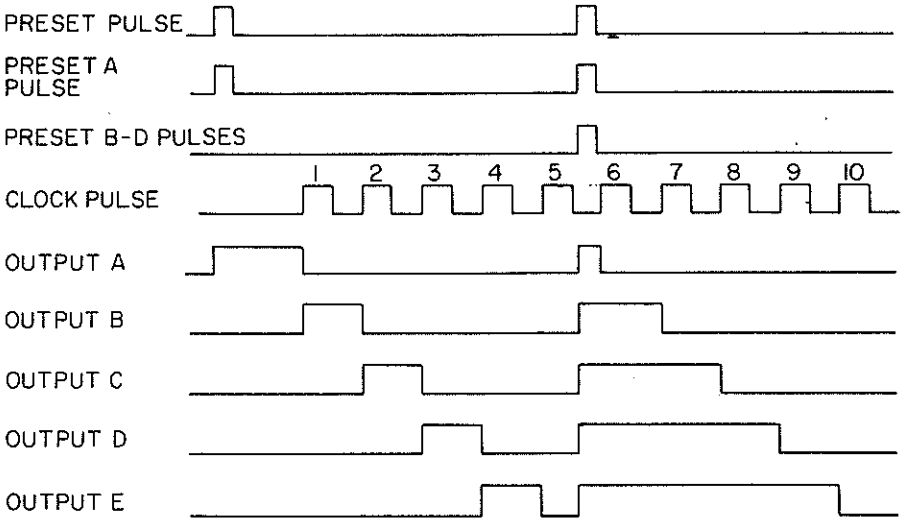


FIGURE 2. Truth tables and timing diagrams - Continued.

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ASYNCHRONOUS OPERATION

DEVICE TYPE 02

TYPICAL INPUT/OUTPUT VOLTAGE WAVEFORMS

NOTE:
INPUTS NOT SHOWN ARE HELD AT LOGIC LEVEL "L".

Device type 03

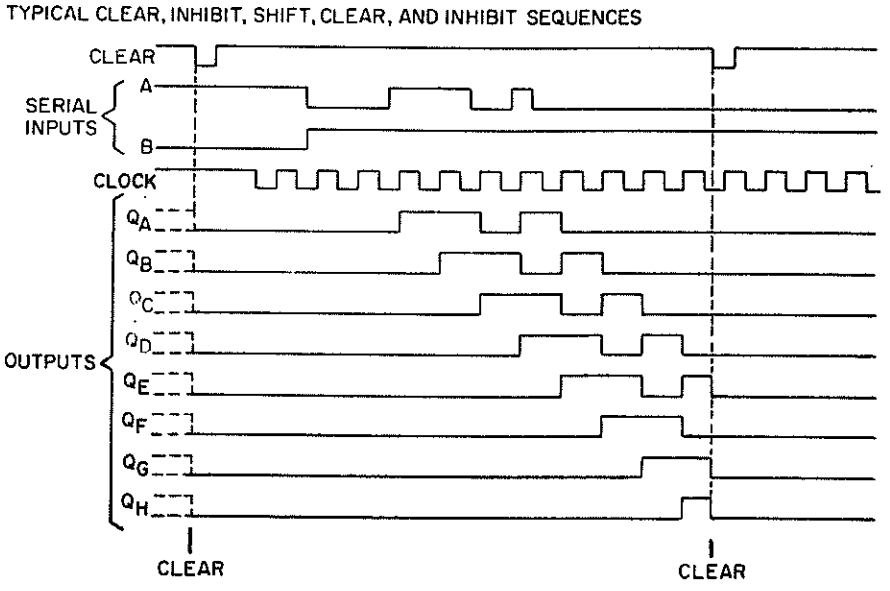
SERIAL INPUTS A AND B

INPUTS AT t_n		OUTPUT AT t_{n+1}
A	B	Q_A
H	H	H
L	H	L
H	L	L
L	L	L

Positive logic: t_n = bit time before clock pulse.
t_{n+1} = bit time after clock pulse.
Data at the serial inputs may be changed while the clock is high, but only information meeting the setup requirements will be entered. Clocking occurs on the low-to-high level transition of the clock input.
The clear input is asynchronous. Low-level at clear input sets all outputs to logical low.

FIGURE 2. Truth table and timing diagrams - continued.

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Device Type 04

Positive logic: Transfer of information to the output occurs when the clock input goes from a logical L to a logical H.

Clocking is accomplished through a 2-input positive-NOR gate, permitting one input to be used as a clock-inhibit function. Holding either of the clock inputs high inhibits clocking, and holding either clock input low with the load input high enables the other clock input. The clock-inhibit should be changed to the high level only while the clock input is high. Parallel loading is inhibited as long as the load input is high. When taken low, data at the parallel inputs are loaded directly into the register independently of the state of the clock.

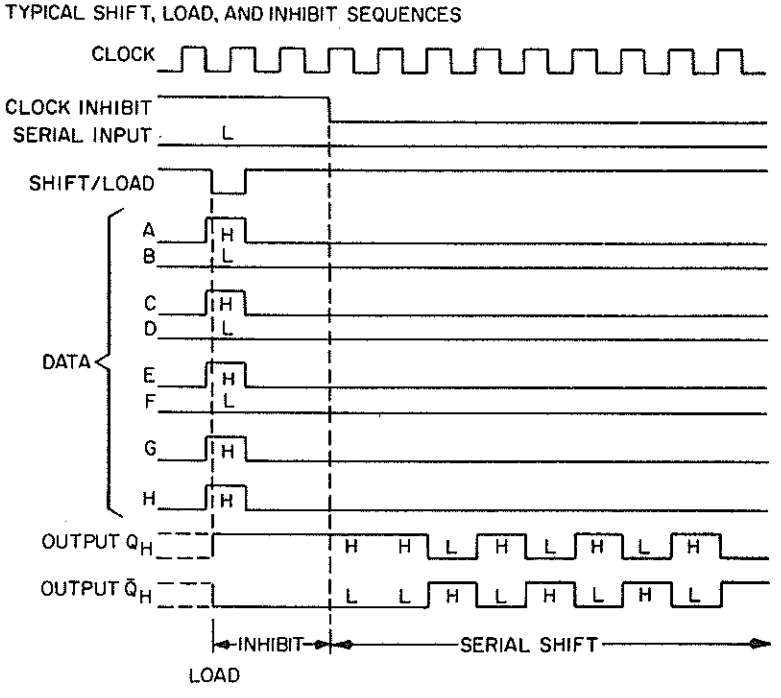


FIGURE 2. Truth tables and timing diagrams - continued.

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Device Type 05

Positive logic:
The register has four distinct modes of operation, namely:

	MODE CONTROL	
	S1	SO
Parallel (Broadside) Load	H	H
Shift Right (in the direction Q_A toward Q_D)	L	H
Shift Left (In the direction Q_D toward Q_A)	H	L
Inhibit Clock (Do nothing)	L	L

In the parallel-load mode, data is loaded into the associated flip-flop and appears at the output after the positive transition of the clock input. During loading, serial data flow is inhibited. Shift right is accomplished synchronously with the rising edge of the clock pulse when SO is high and S1 is low. Serial data for this mode is entered at the shift-right data input. When SO is low and S1 is high, data shifts left synchronously and new data is entered at the shift-left serial input. Clocking of the flip-flops is inhibited when both mode-control inputs are low. The mode controls should be changed only while the clock input is high.

typical clear, load, right-shift, left-shift, inhibit, and clear sequences

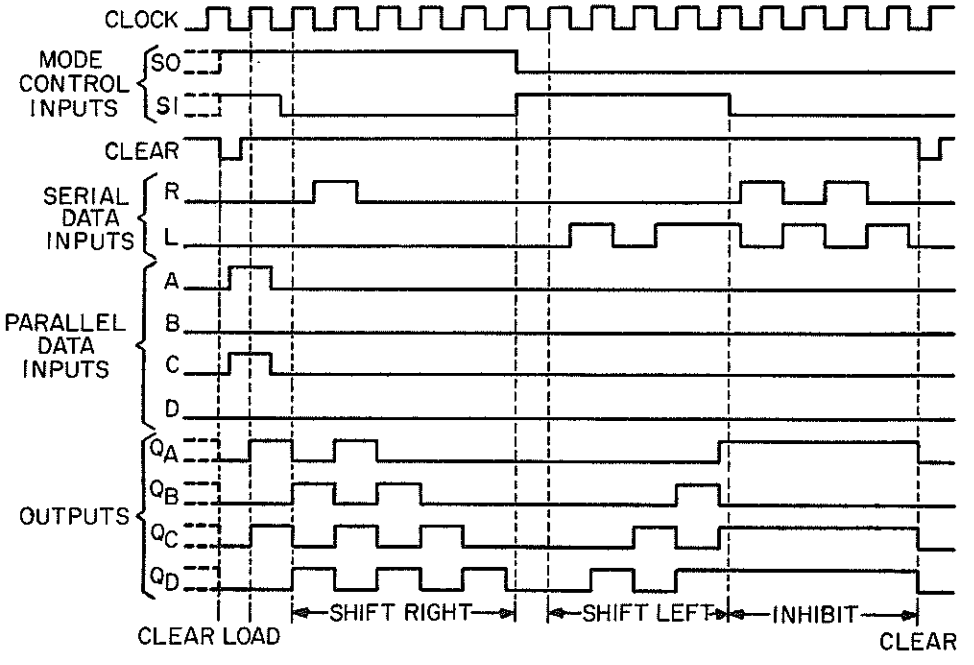


FIGURE 2. Truth tables and timing diagrams - continued.

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Device type 06

Positive logic:

The registers have two modes of operation:

Parallel (broadside) load

Shift (in direction Q_A toward Q_D)

Parallel loading is accomplished by applying the four bits of data and taking the shift/load control input low. The data is loaded into the associated flip-flop and appears at the outputs after the positive transition of the clock input. During loading, serial data flow is inhibited.

Shifting is accomplished synchronously when the shift/load control input is high. Serial data for this mode is entered at the J-$\overline{K}$ inputs. These inputs permit the first stage to perform as a J-$\overline{K}$, D-, or T-type flip-flop as shown in the truth table.

TRUTH TABLE

Inputs at t_n		Outputs at t_{n+1}				
J	$\overline{K}$	Q_A	Q_B	Q_C	Q_D	$\overline{Q}_D$
L	H	Q_{An}	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
L	L	L	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
H	H	H	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$
H	L	$\overline{Q}_{An}$	Q_{An}	Q_{Bn}	Q_{Cn}	$\overline{Q}_{Cn}$

H = high level, L = low level

NOTES: 1. t_n = bit time before clock pulse
2. t_{n+1} = bit time after clock pulse
3. Q_{An} = state of Q_{An} at t_n.

typical clear, shift, and load sequences

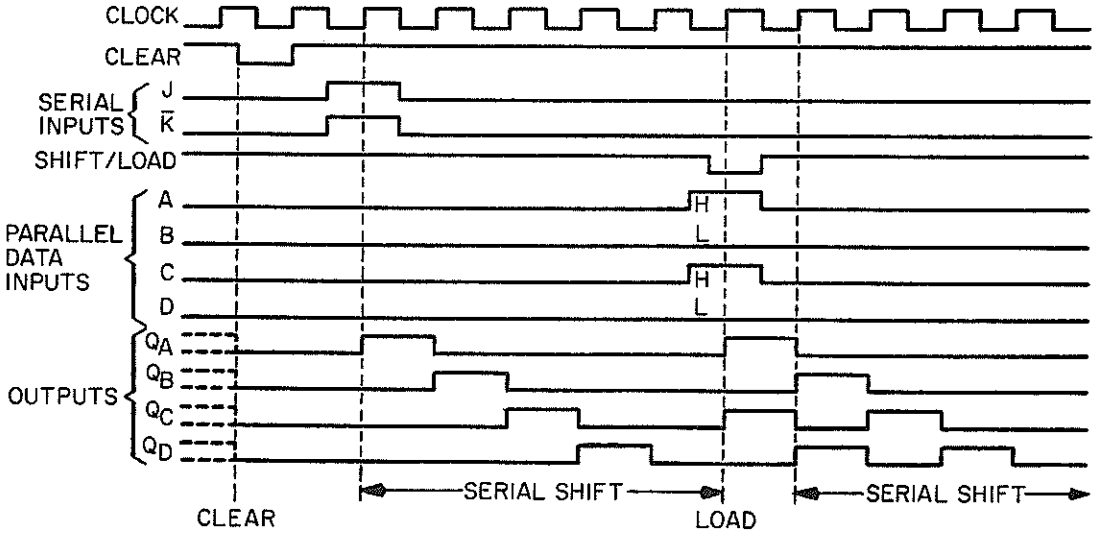


FIGURE 2. Truth tables and timing diagrams - continued.

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Device type 01

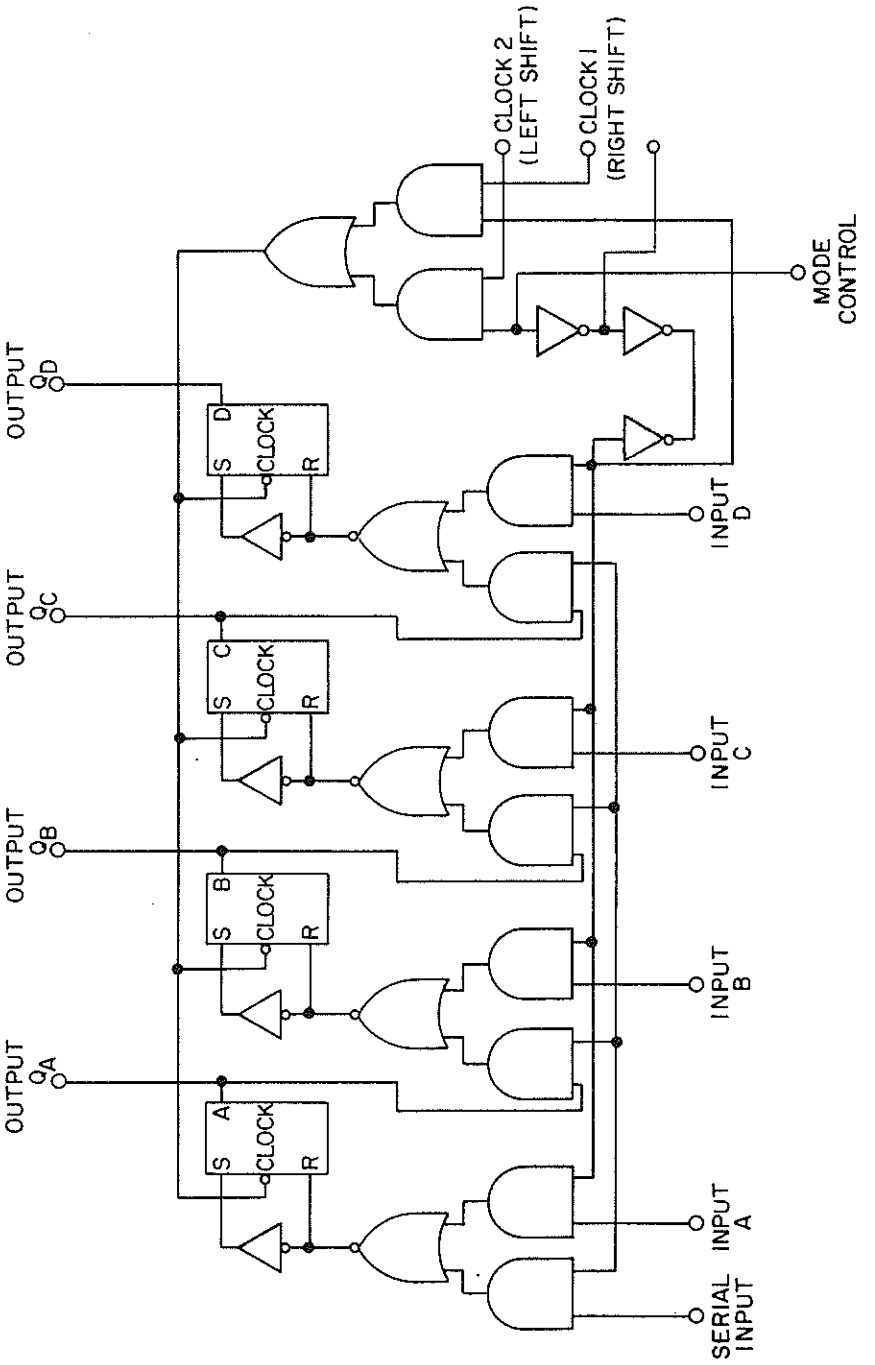


FIGURE 3. Logic diagrams.

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Device type 02

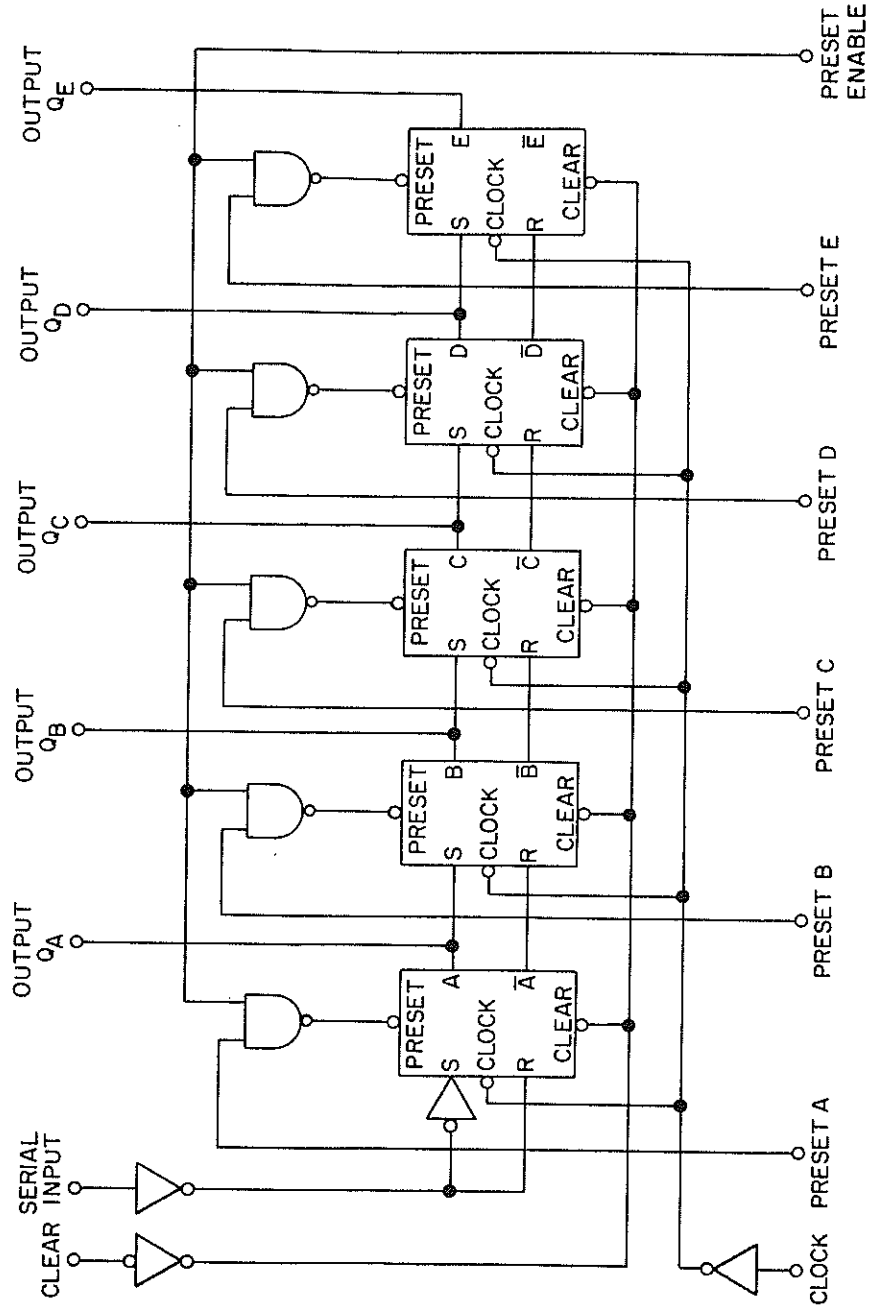
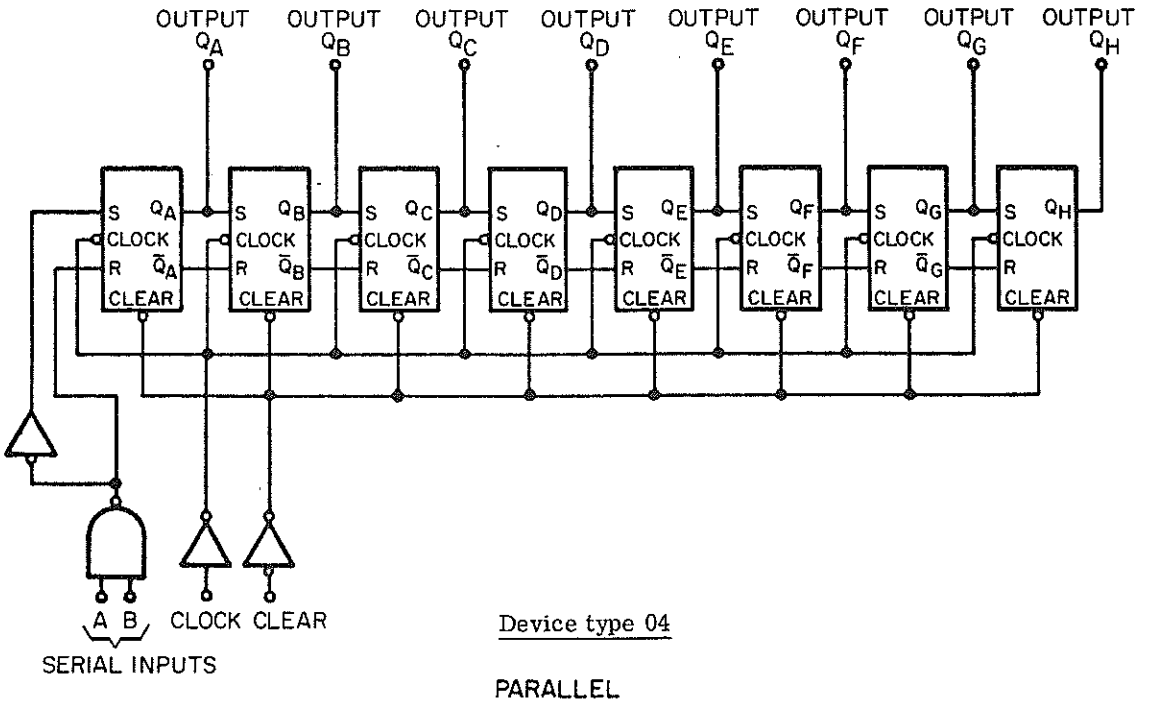


FIGURE 3. Logic diagram - Continued.

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Device type 03



Device type 04

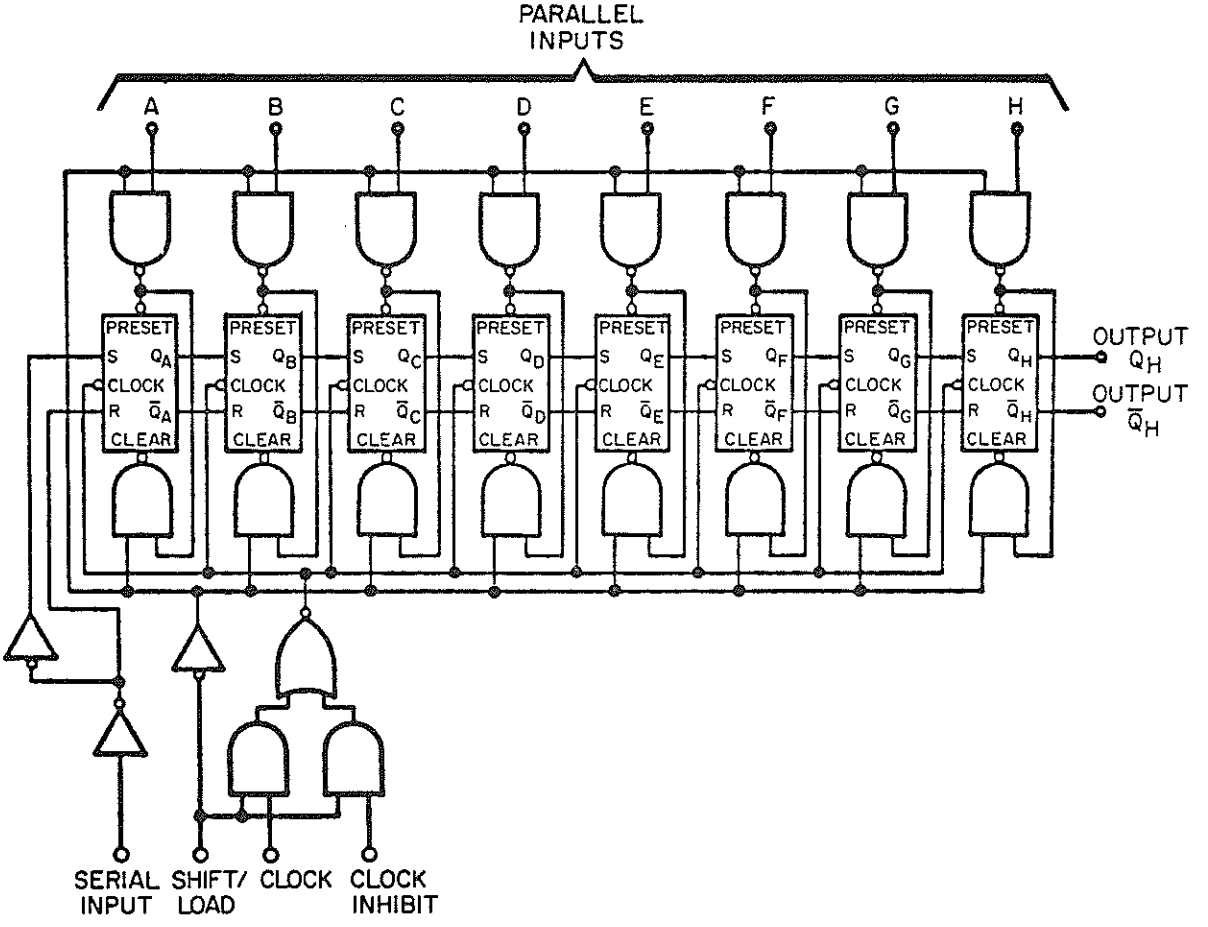
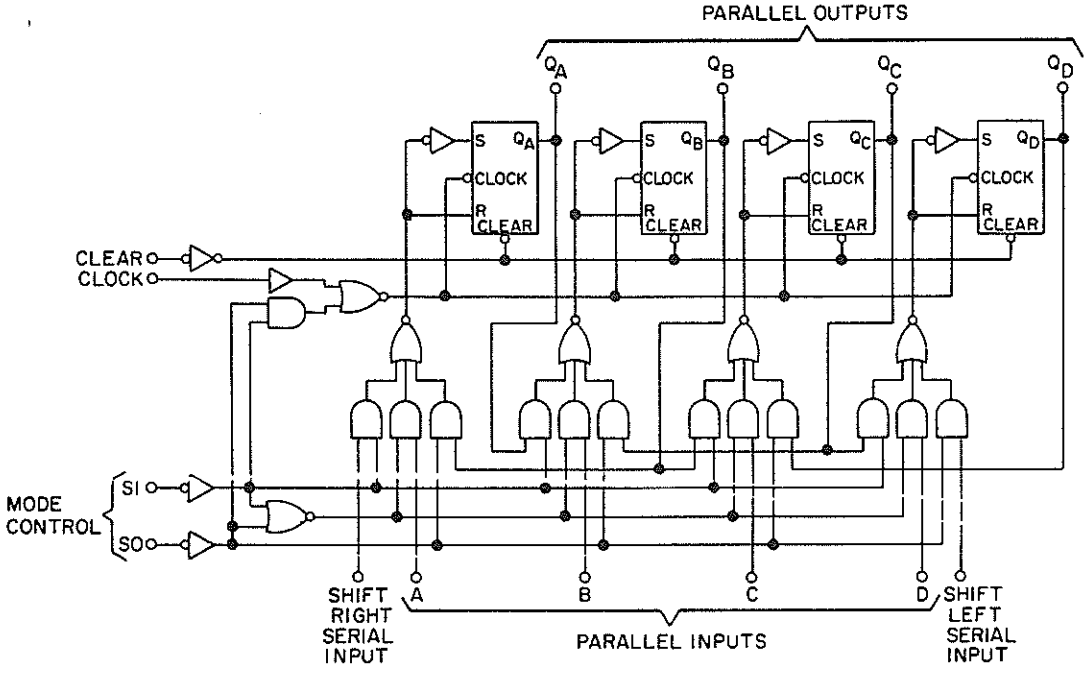


FIGURE 3. Logic diagrams - continued.

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Device type 05



Device type 06

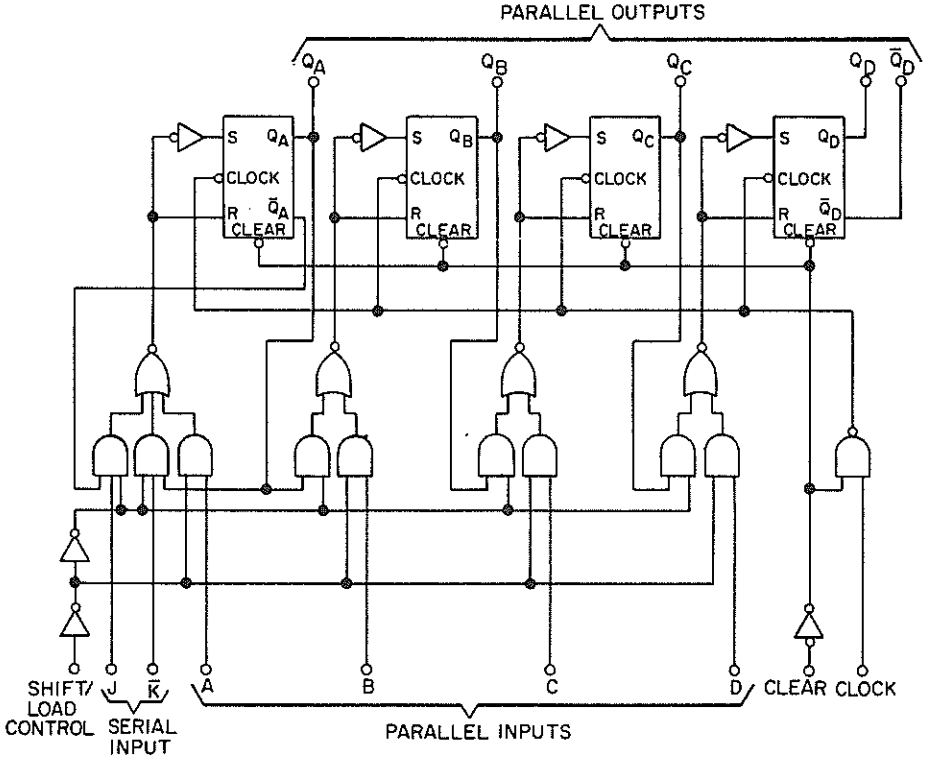


FIGURE 3. Logic diagrams - continued.

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Device type 01, circuit A

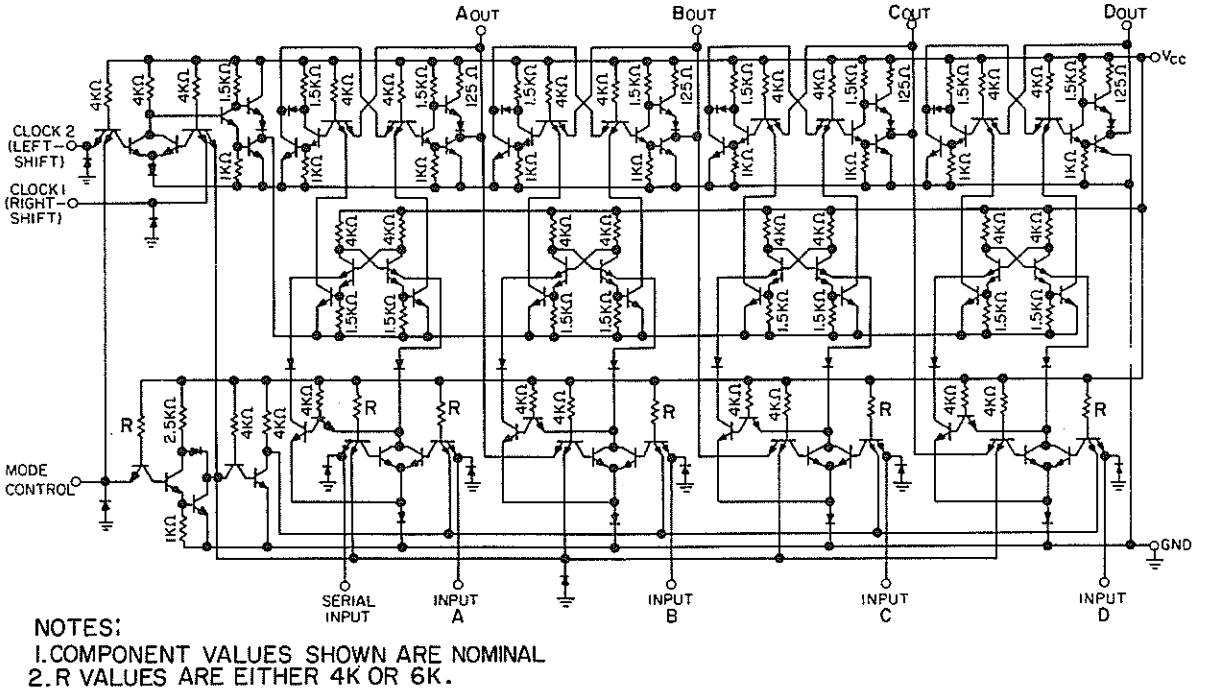
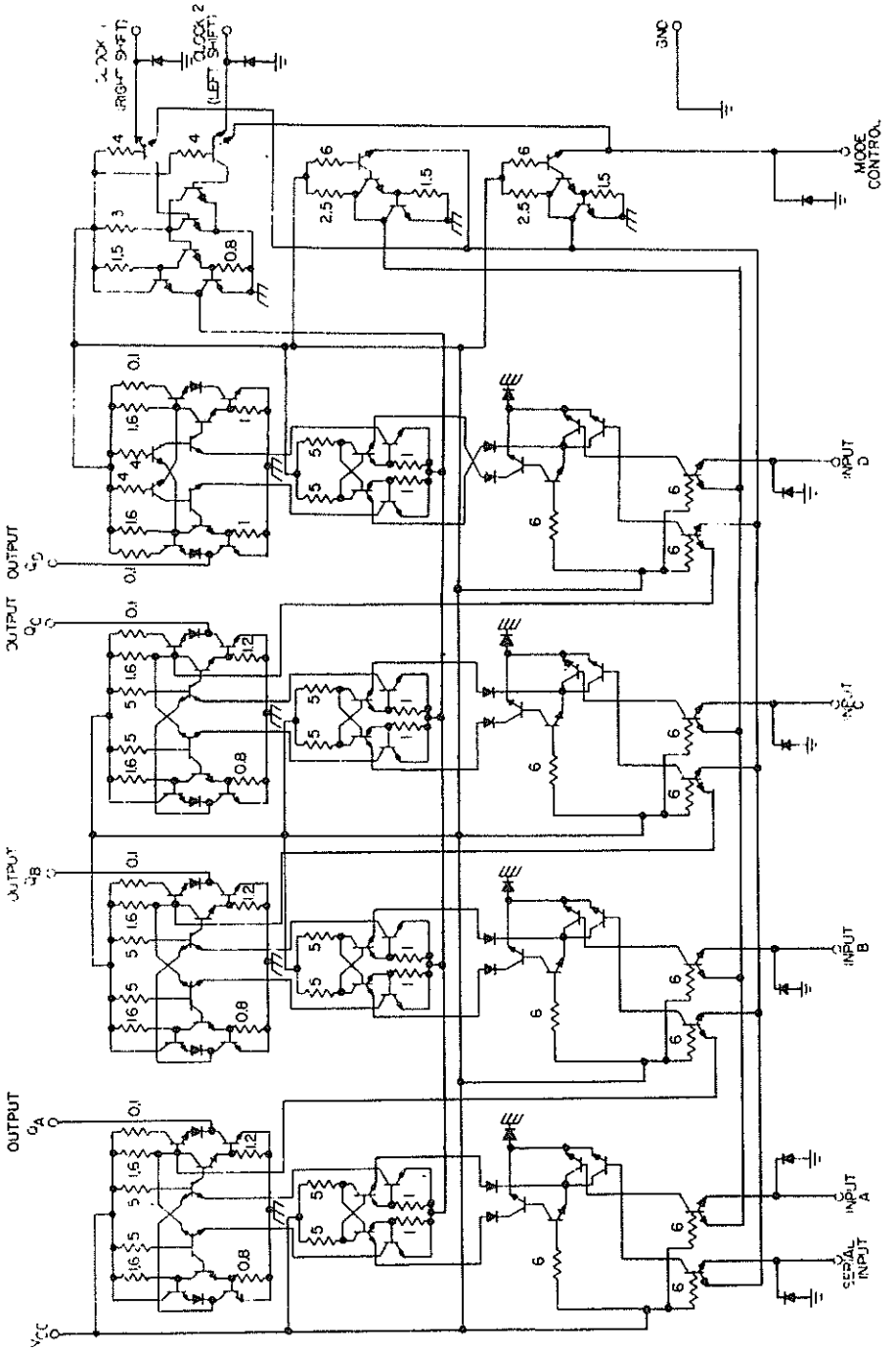


FIGURE 4. Schematic circuits.

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Device type 01, circuit B



NOTES:
1. Resistor values are in kilohms.
2. Component values shown are nominal.

FIGURE 4. Schematic circuits - Continued.

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Device type 02, circuit A

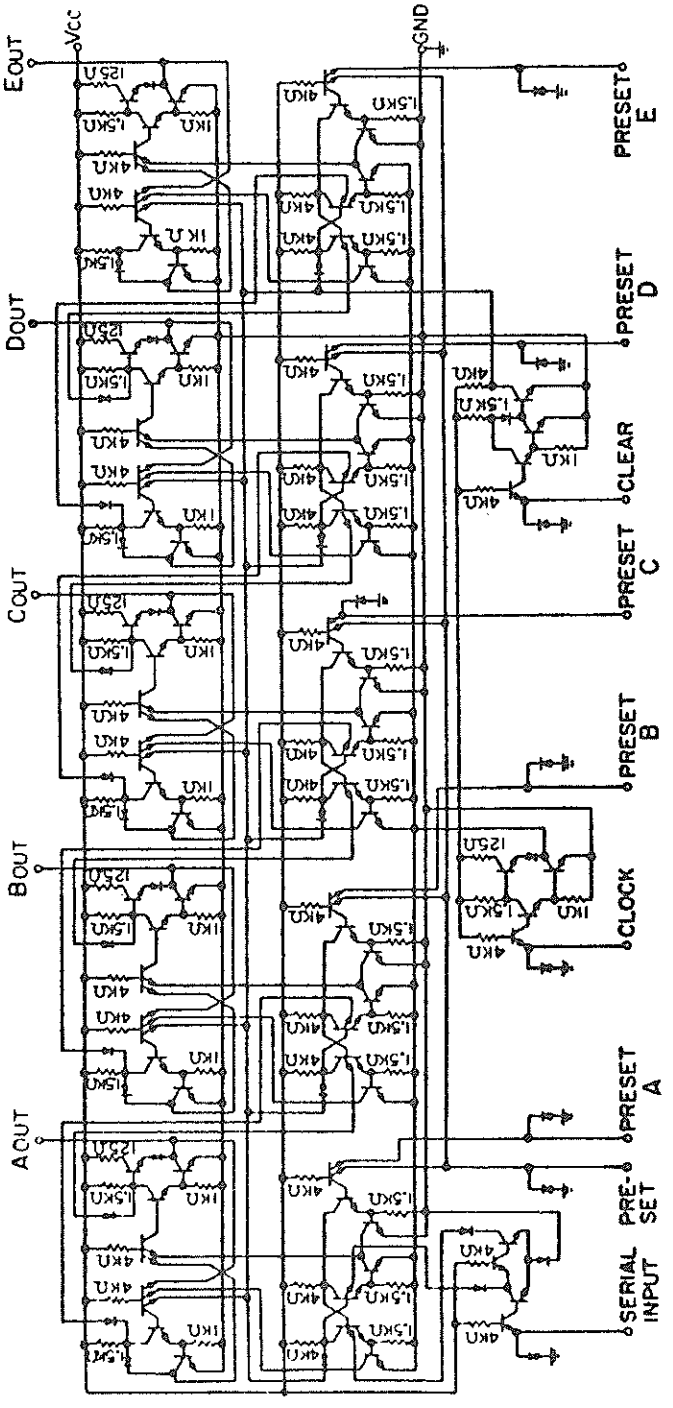
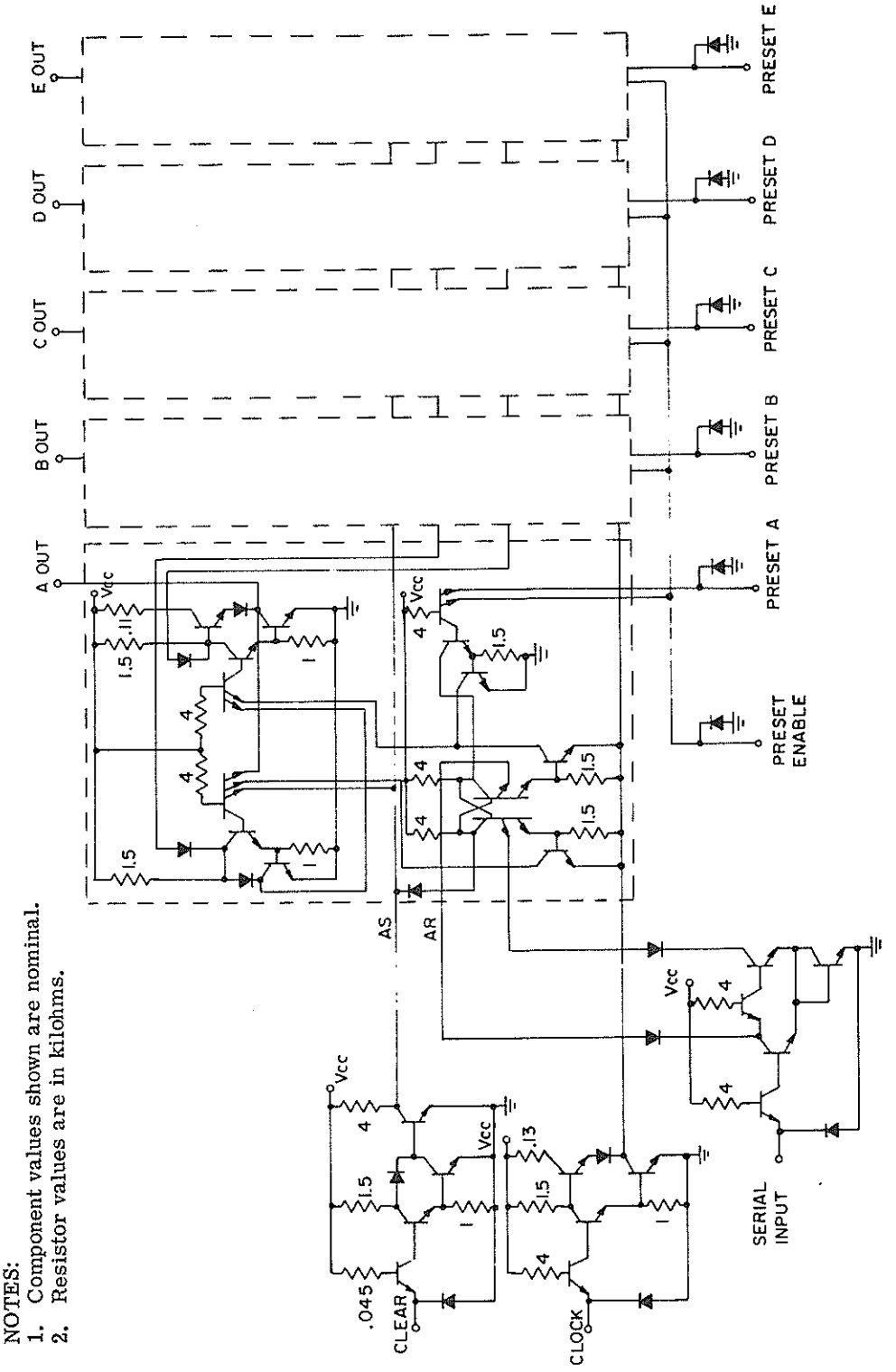


FIGURE 4. Schematic circuits - Continued.

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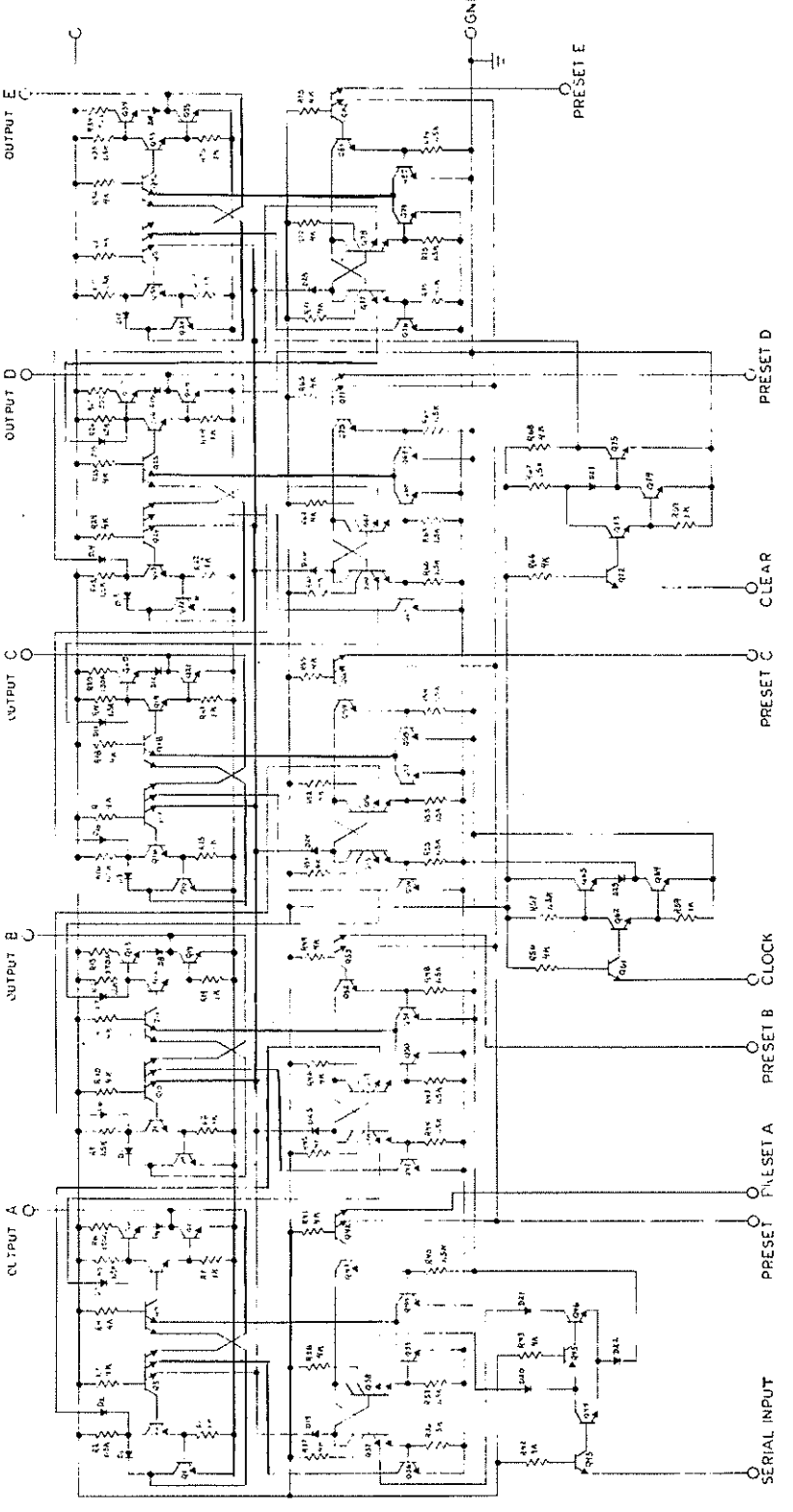
Device type 02, circuit B



NOTES:
1. Component values shown are nominal.
2. Resistor values are in kilohms.

FIGURE 4. Schematic circuits - Continued.

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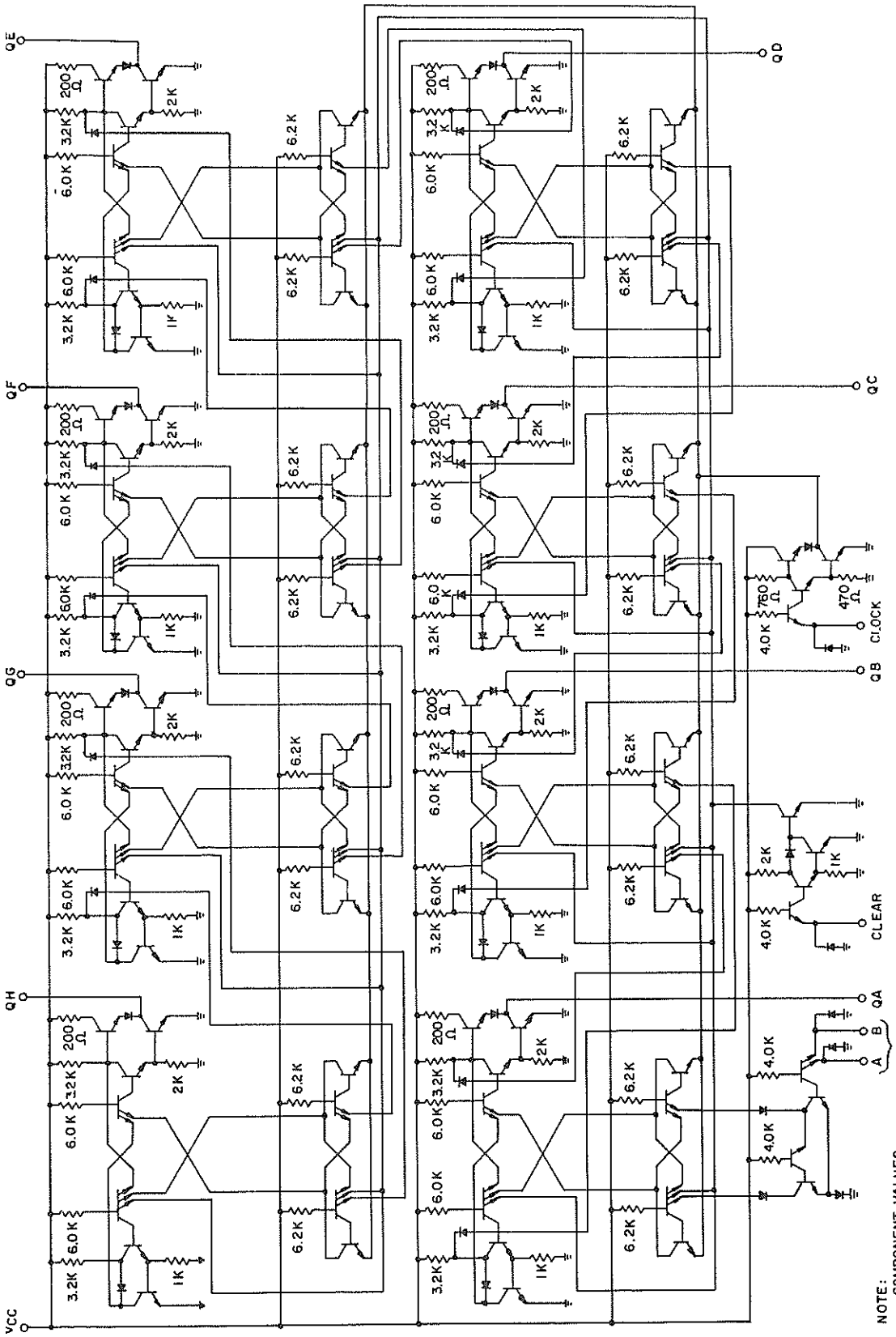
Device type 02

Circuit C

NOTE: Component values shown are nominal.

FIGURE 4. Schematic circuits - Continued.

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Device type 03, circuit A

FIGURE 4. Schematic circuits - Continued.

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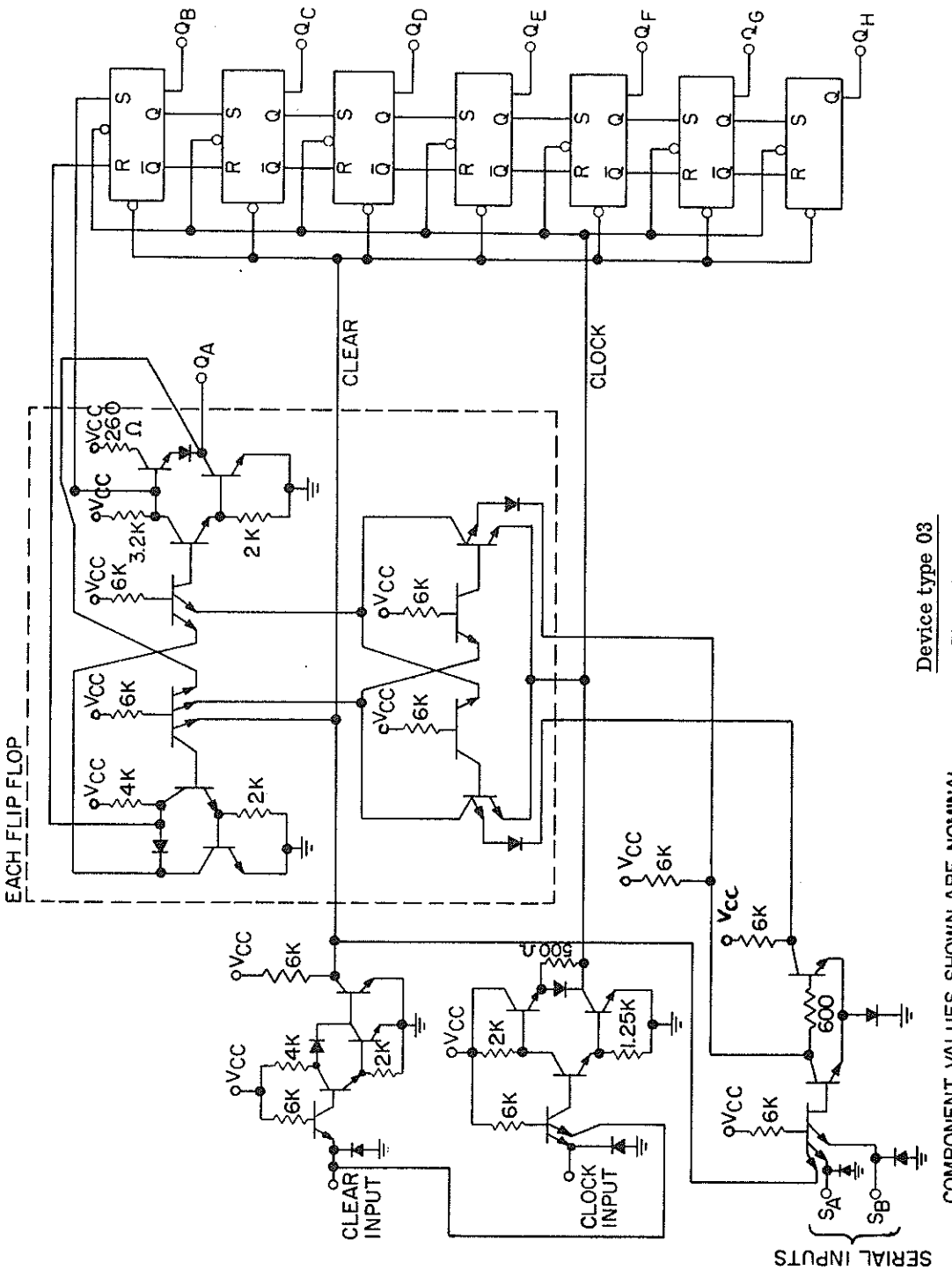
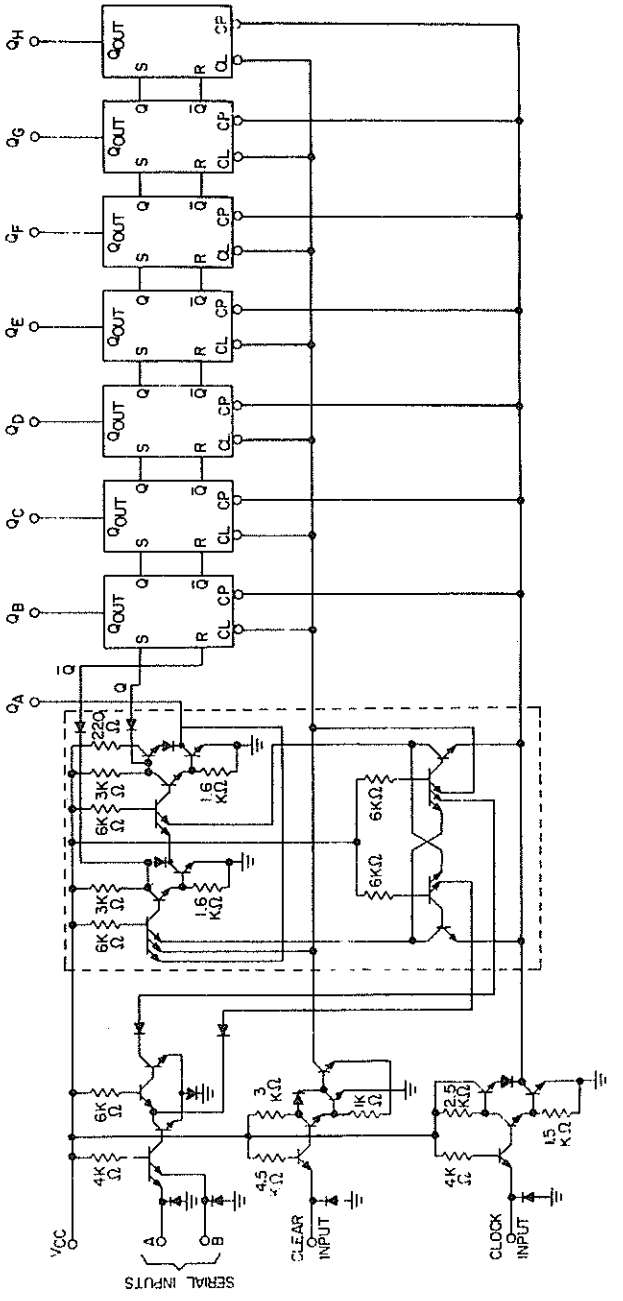


FIGURE 4. Schematic circuits - Continued.

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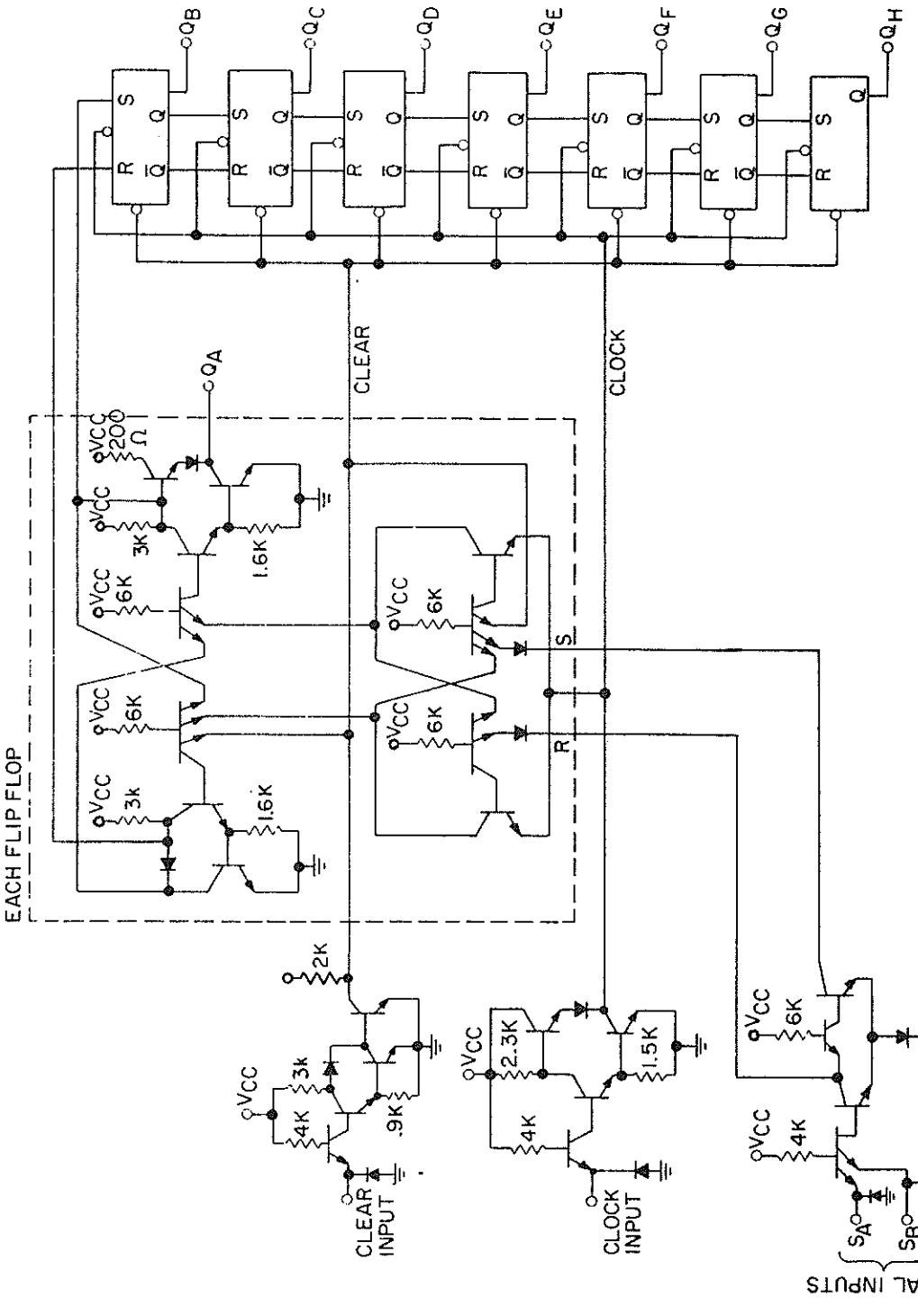
Device type 03, circuit C



Component values shown are nominal.

FIGURE 4. Schematic circuits - Continued.

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COMPONENT VALUES SHOWN ARE NOMINAL

FIGURE 4. Schematic circuits - Continued.

Device type 03, circuit D

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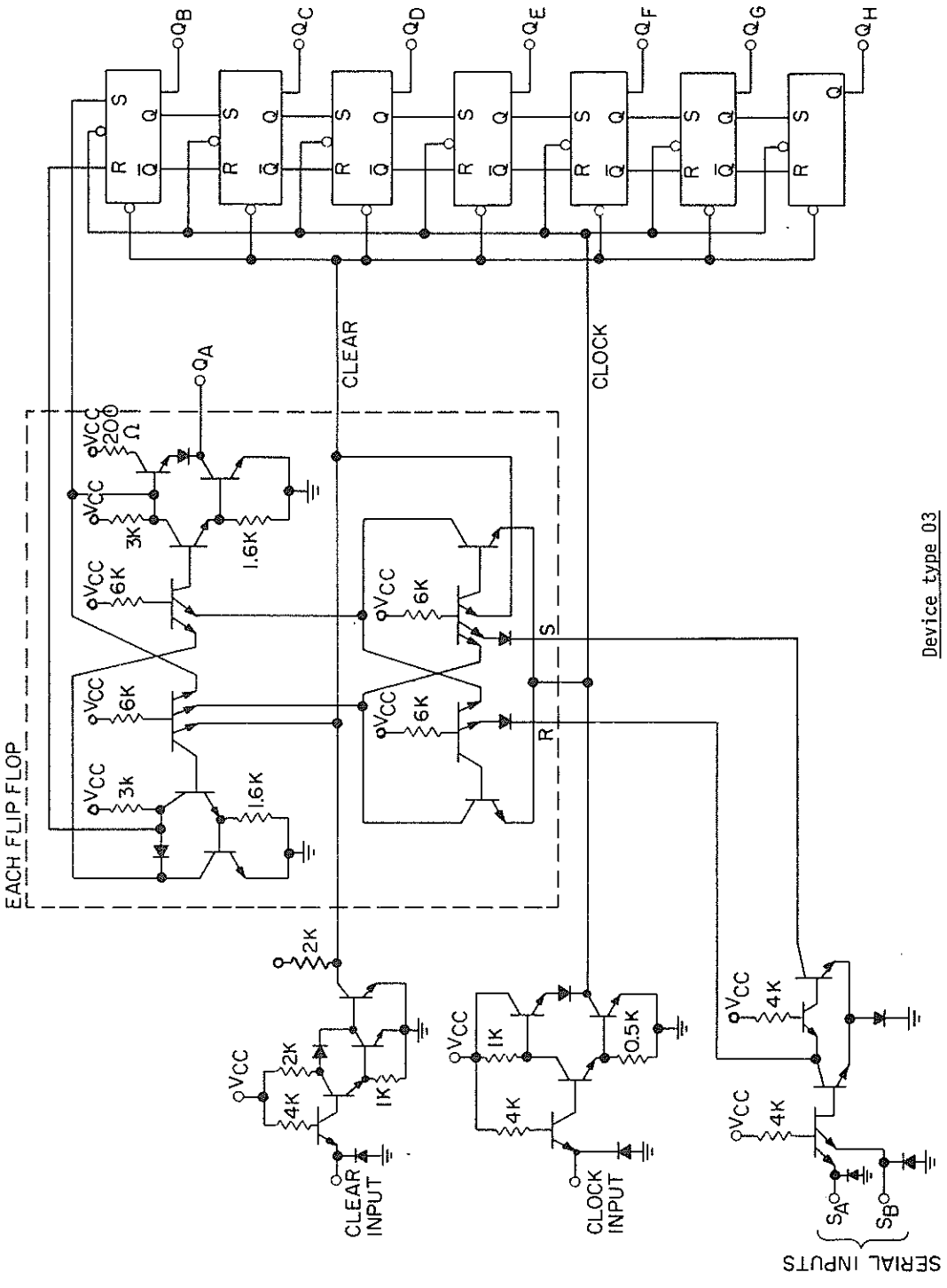
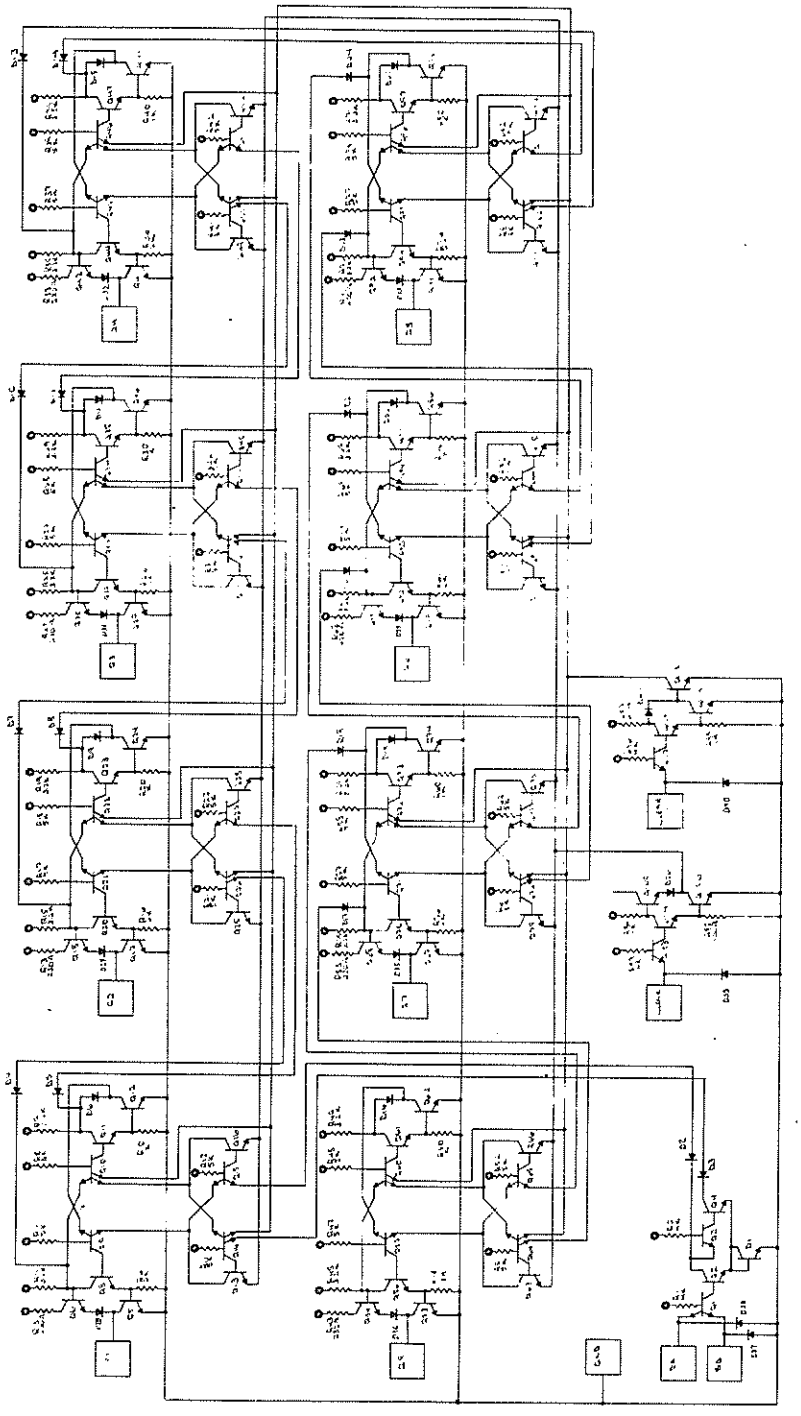


FIGURE 4. Schematic circuits - Continued.

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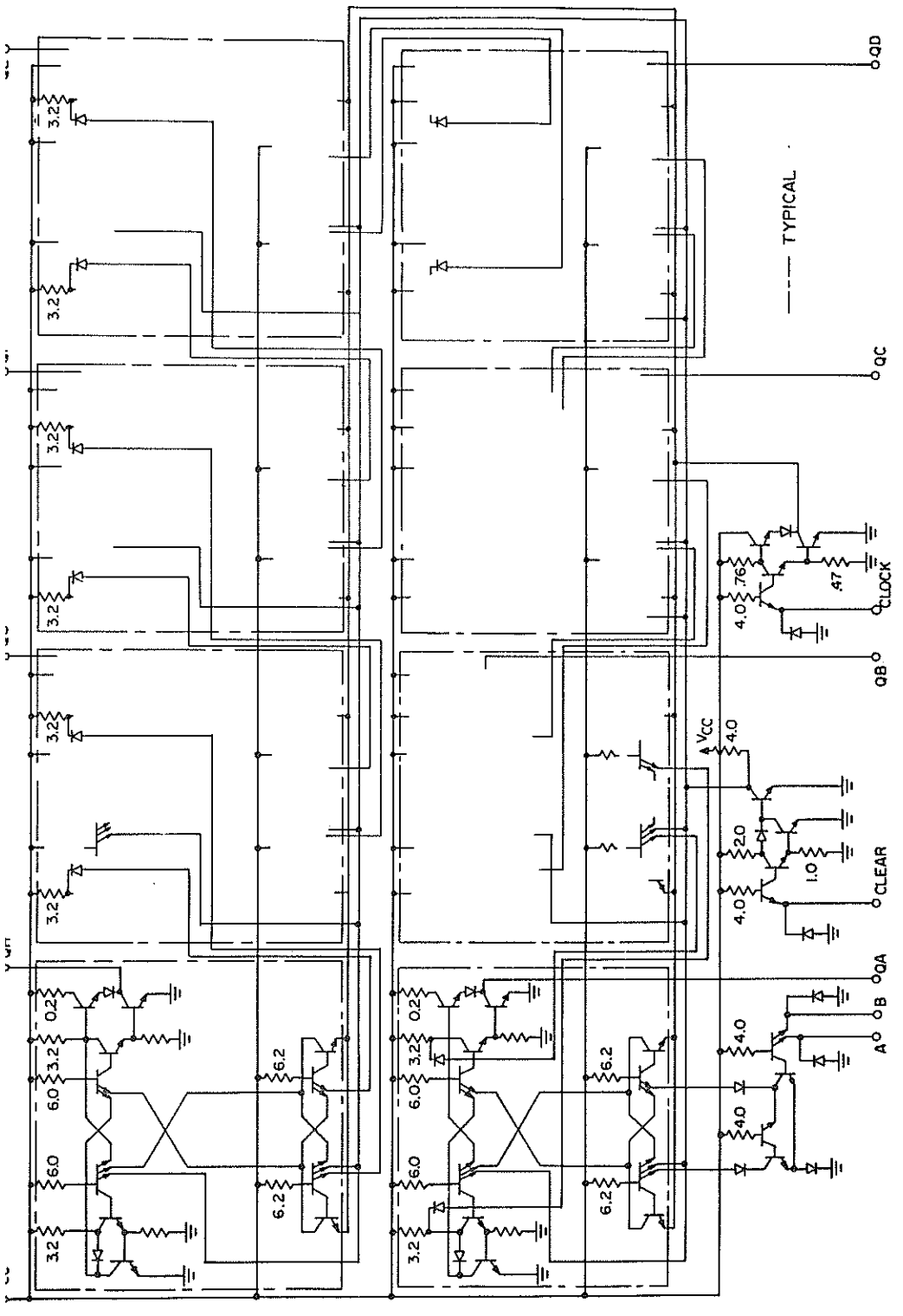
Device type 03

Circuit F

NOTES:
1. Component values shown are nominal.
2. $V_{CC} = 5$

FIGURE 4. Schematic circuits - Continued.

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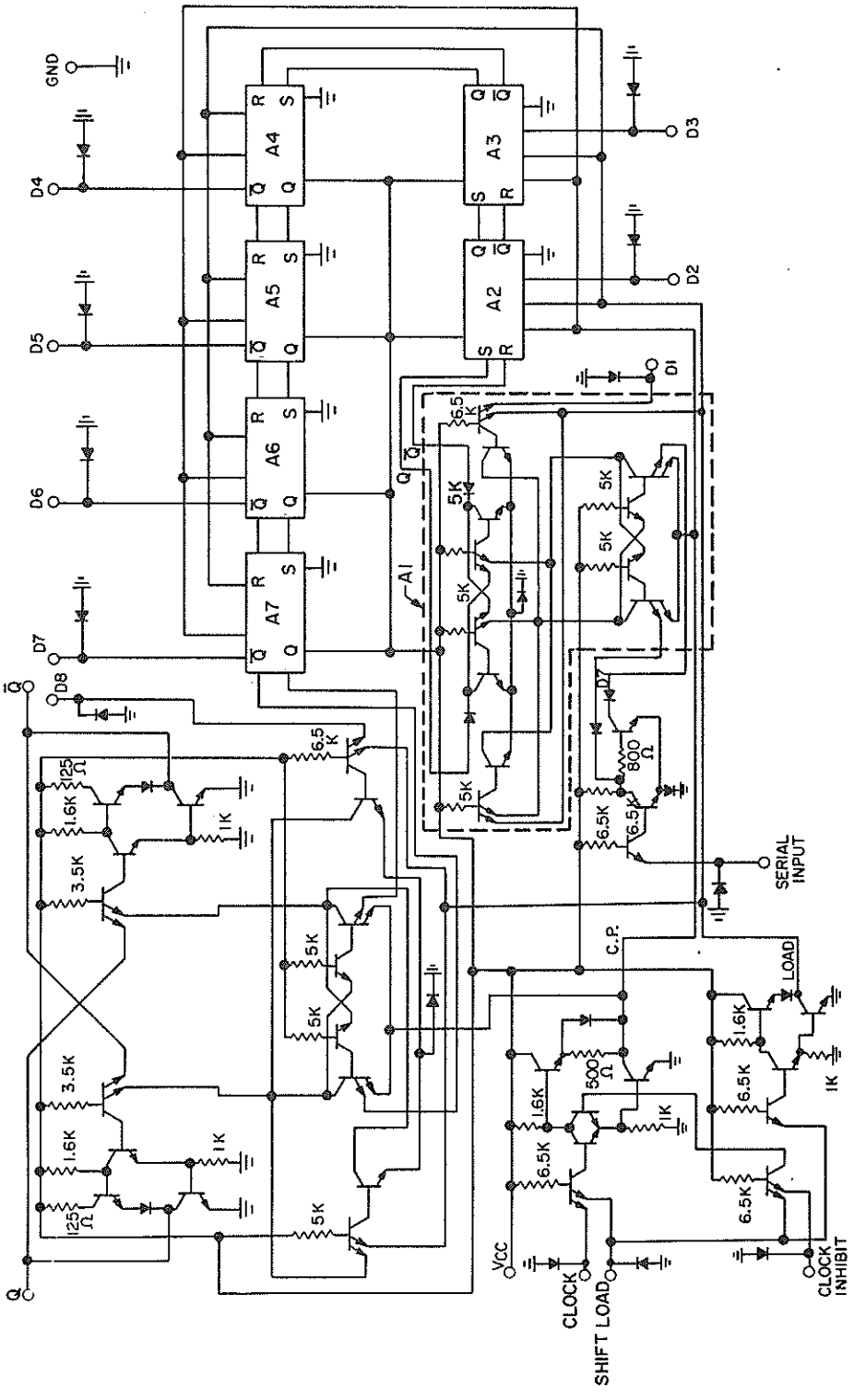
Device type 03

Circuit G

All resistor values are in kilohms and are nominal.

FIGURE 4. Schematic circuits - Continued.

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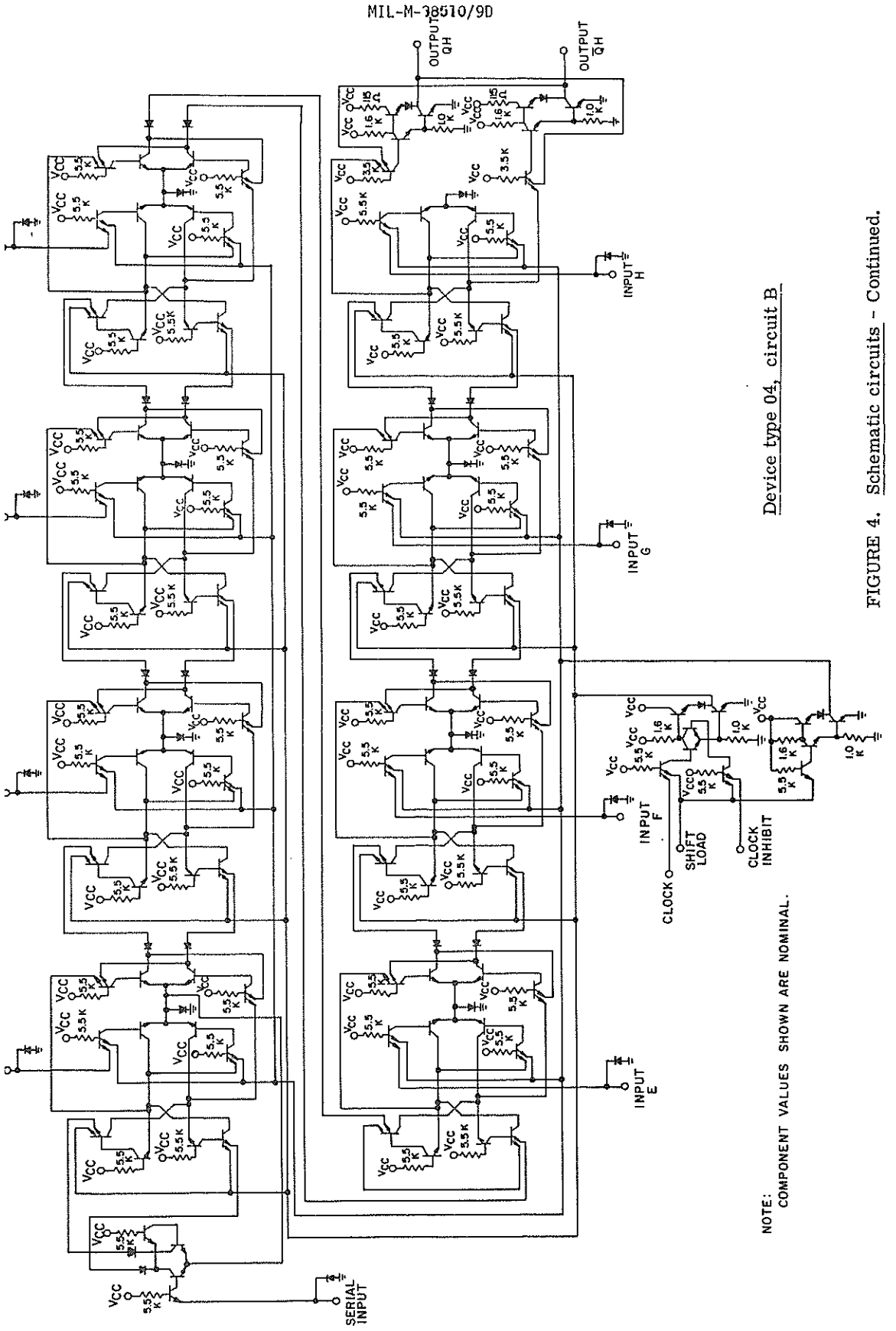


COMPONENT VALUES SHOWN ARE NOMINAL

Device type 04, circuit A

FIGURE 4. Schematic circuits - continued.

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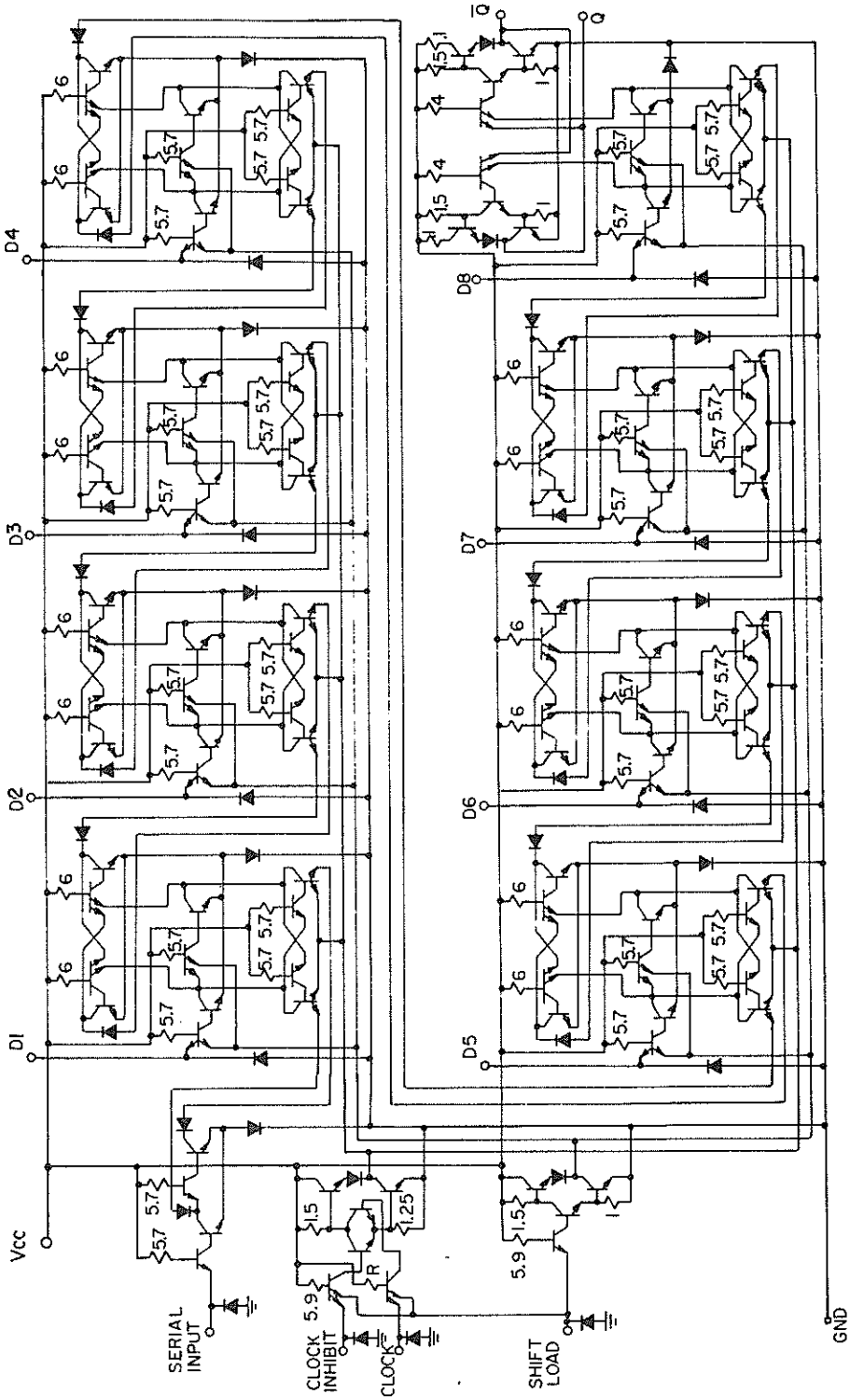


Device type 04, circuit B

FIGURE 4. Schematic circuits - Continued.

NOTE: COMPONENT VALUES SHOWN ARE NOMINAL.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
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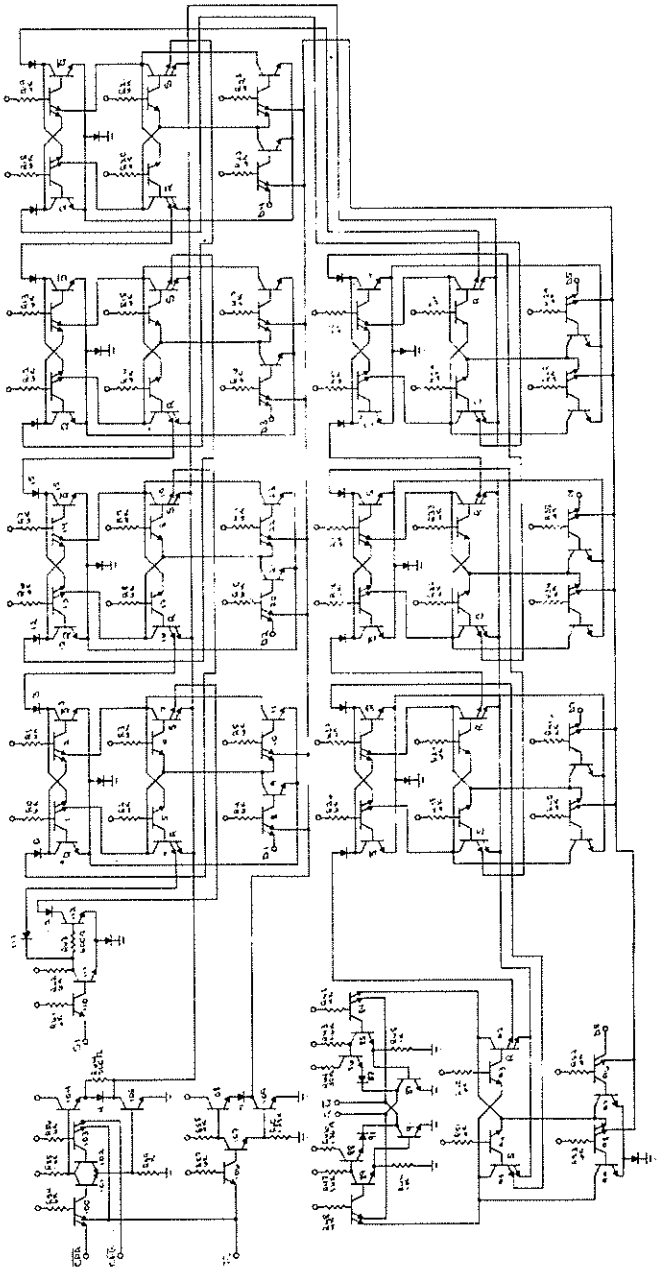


Device type 04, circuit C

NOTES:
1. Component values shown are nominal.
2. R value is either 4 kΩ or 6 kΩ.
3. Resistor values are in kilohms.

FIGURE 4. Schematic circuits - Continued.

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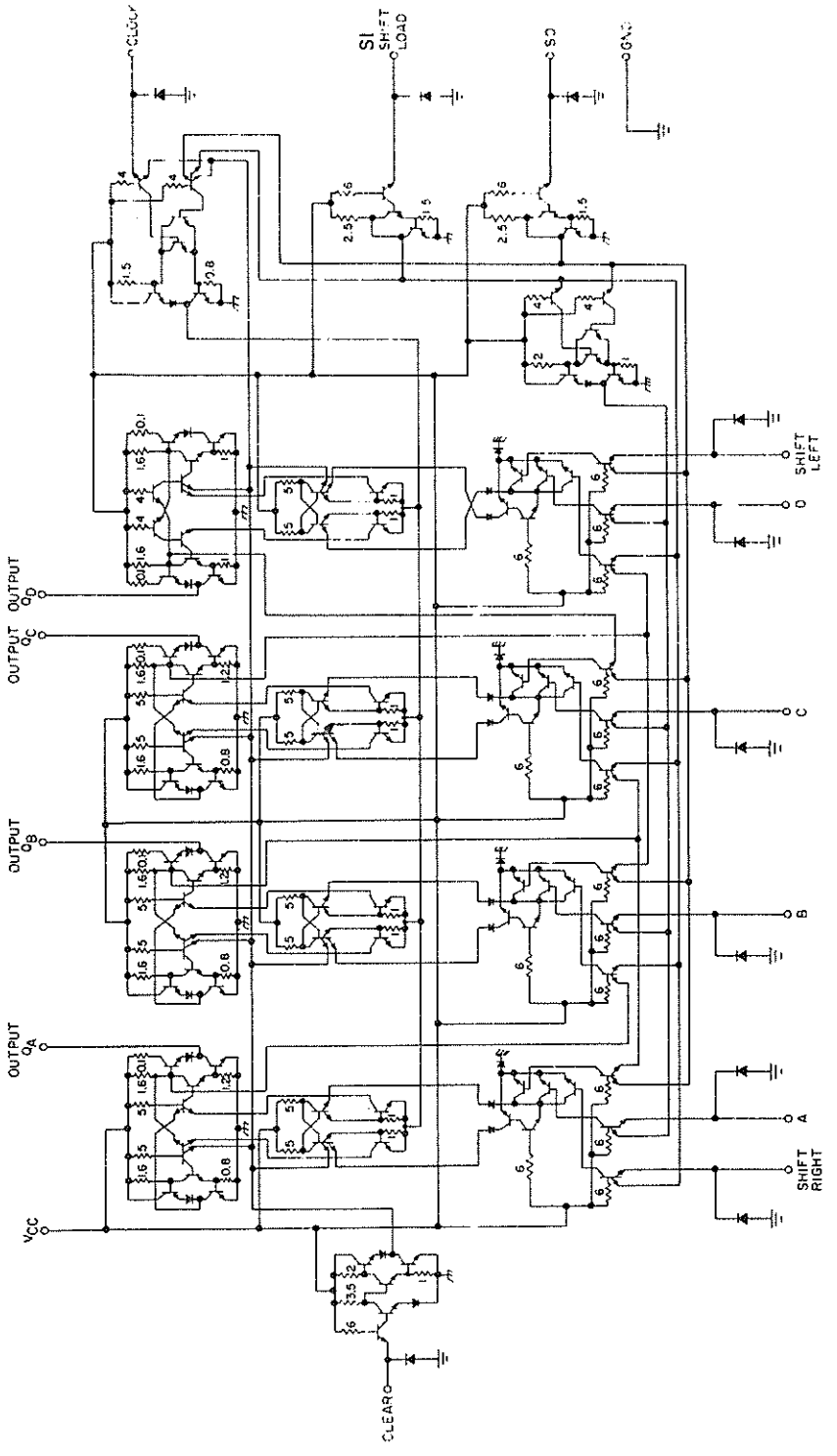
Device type 04

Circuit D

NOTES:
1. Component values shown are nominal.
2. V_{CC} = 9

FIGURE 4. Schematic circuits - Continued.

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Device type 05, circuit A

FIGURE 4. Schematic circuits - Continued.

NOTES:
1. Resistor values are in kilohms.
2. Component values shown are nominal.

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Device type 05, circuit B

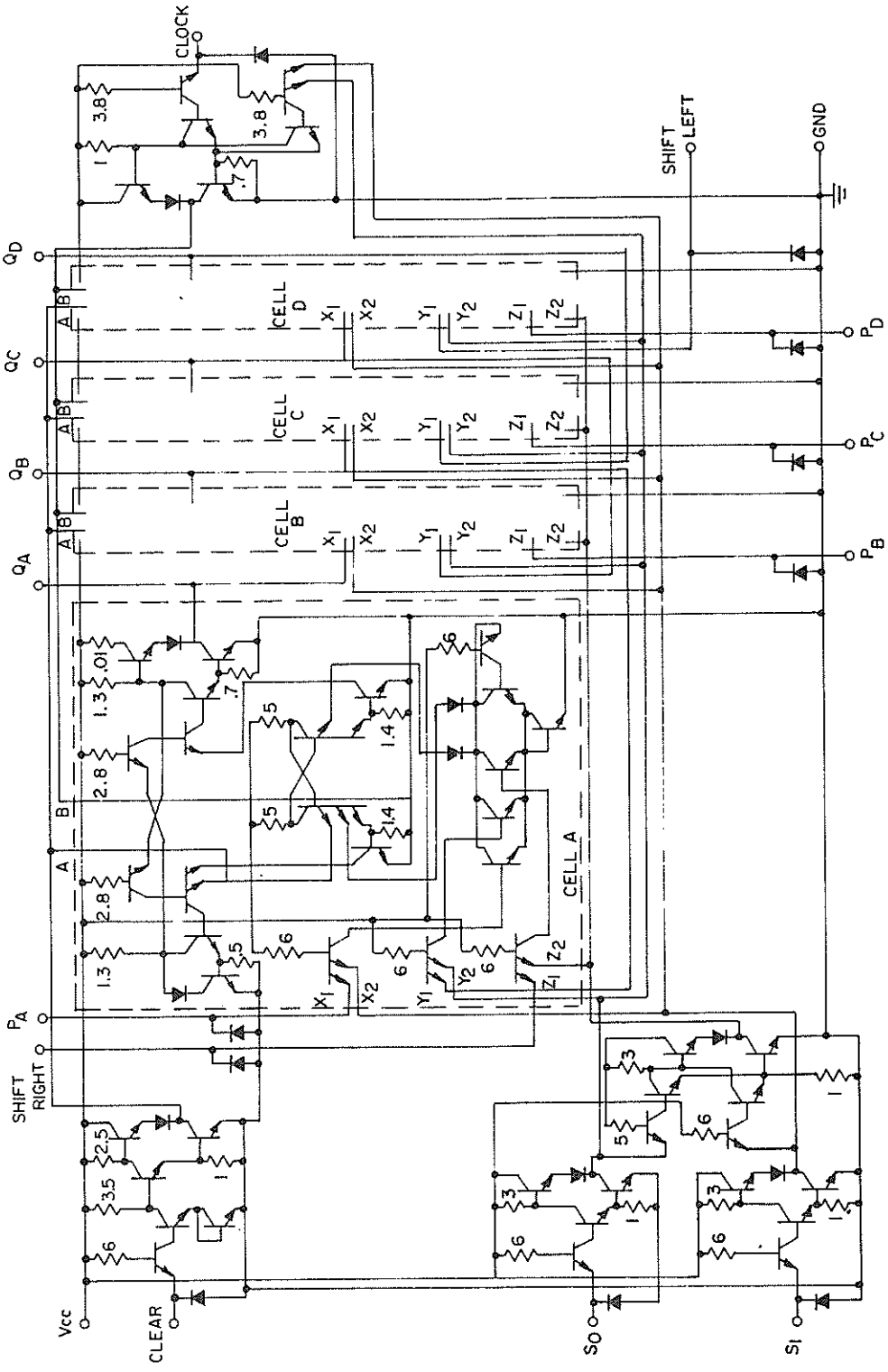
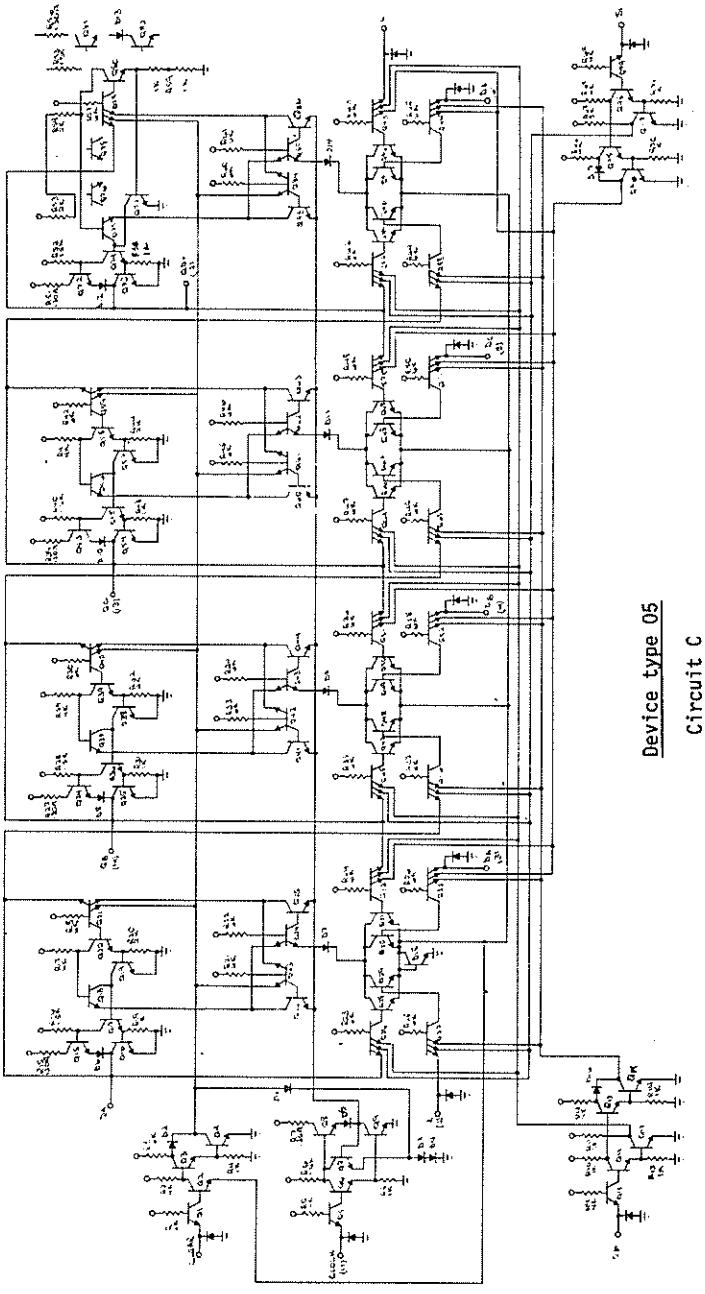


FIGURE 4. Schematic circuits - Continued.

NOTES:
1. Component values shown are nominal.
2. Resistor values are in kilohms.

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Device type 05

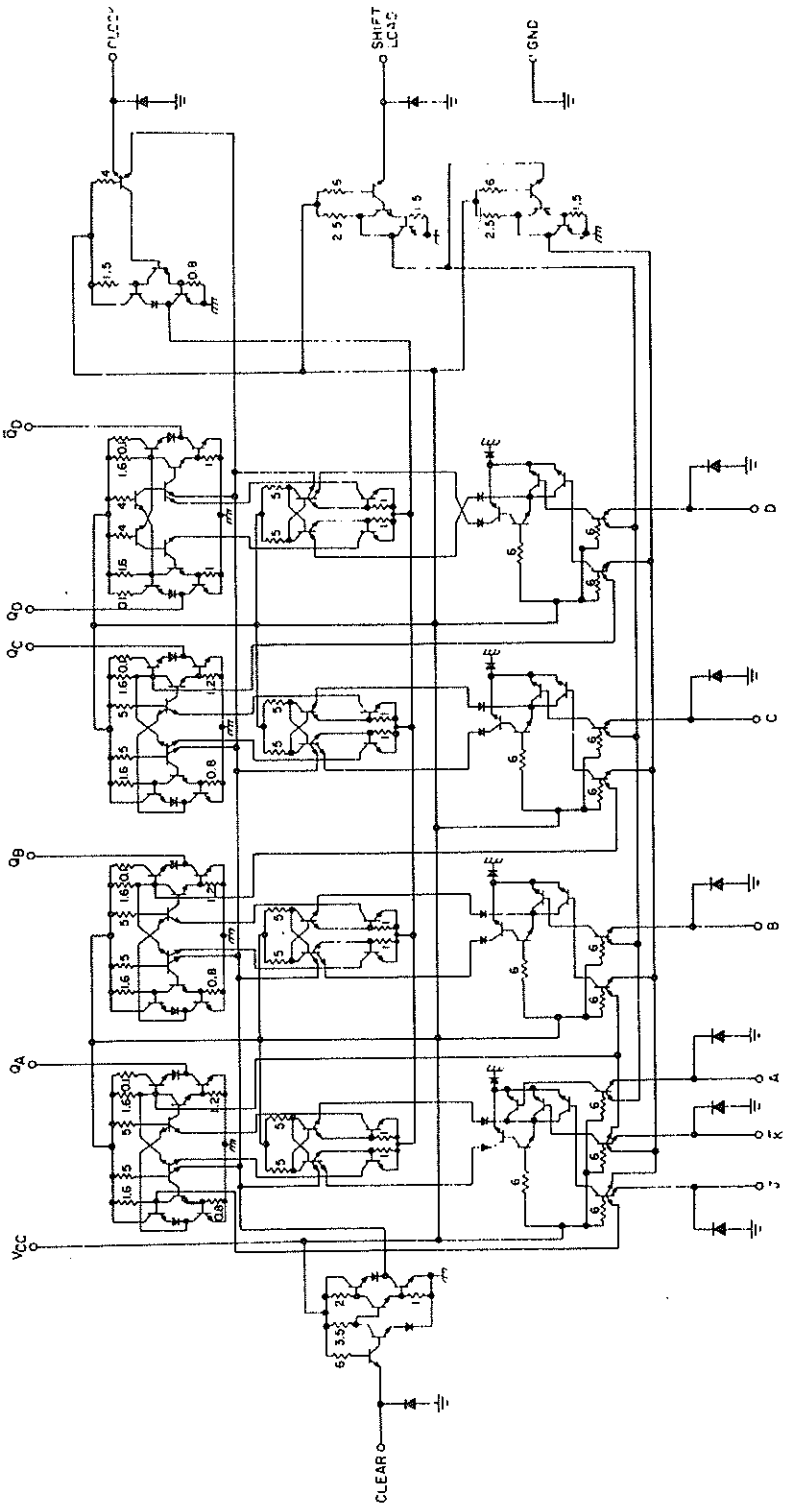
Circuit C

NOTES:
1. Component values shown are nominal.
2. V_{CC} = ?

FIGURE 4. Schematic circuits - Continued.

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Check the source to verify that this is the current version before use.

Device type 06, circuit A



NOTES:
1. Component values shown are nominal.
2. Resistor values are in kilohms.

FIGURE 4. Schematic circuits - Continued.

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Device type 06, circuit B

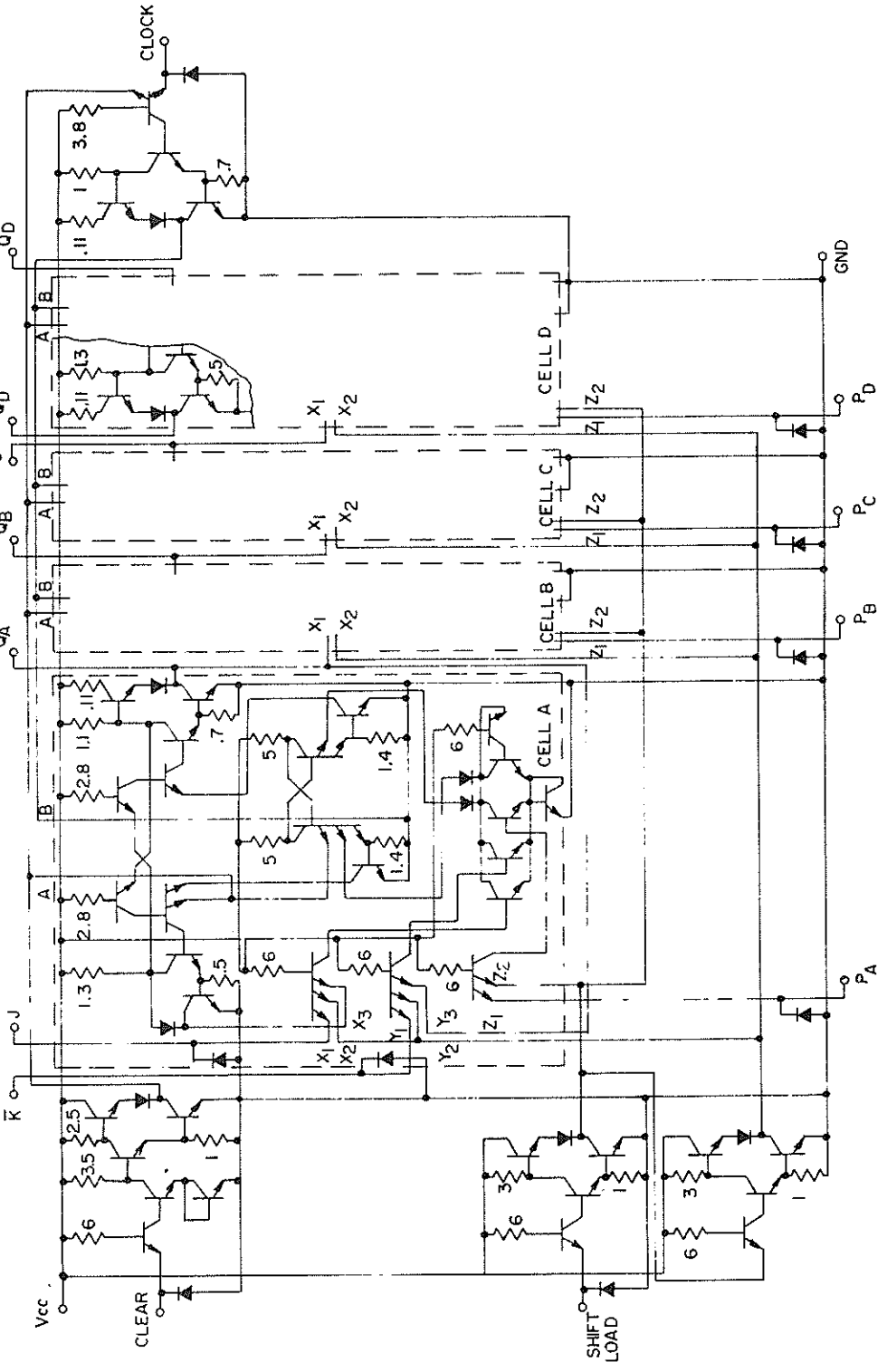
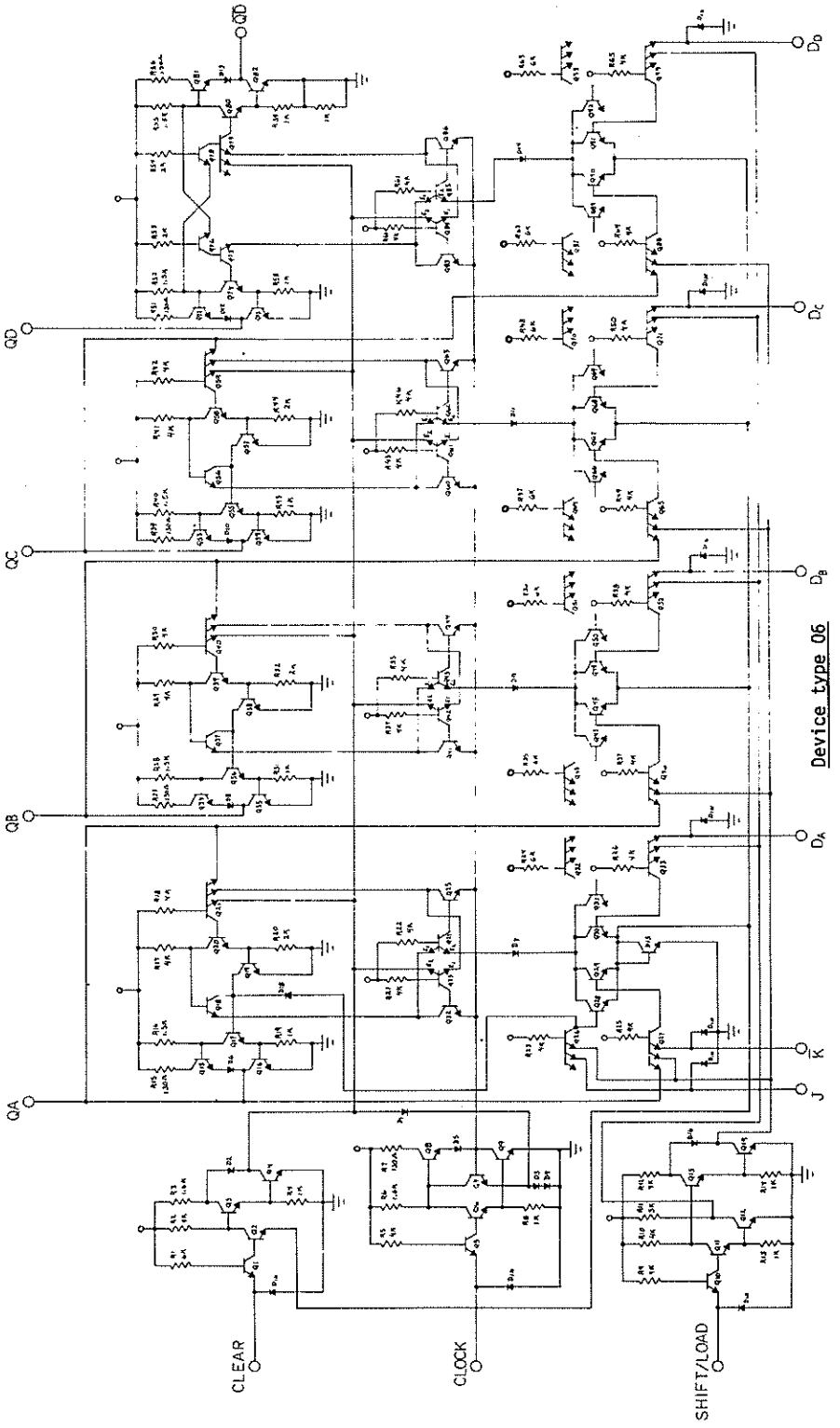


FIGURE 4. Schematic circuits - Continued.

NOTES:
1. Resistor values are in kilohms.
2. Component values shown are nominal.

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Device type 06

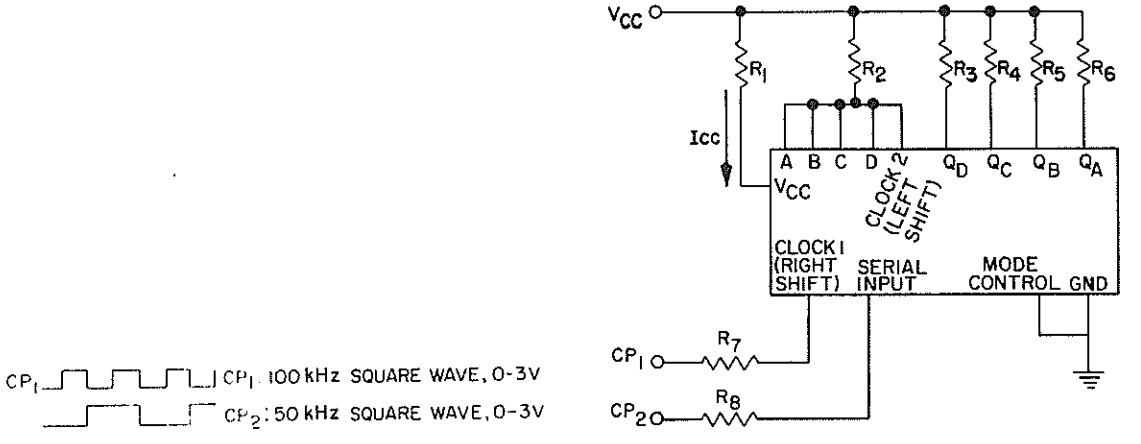
Circuit C

FIGURE 4. Schematic circuits - Continued.

NOTES:
1. Component values shown are nominal.
2. V_{CC} = 5

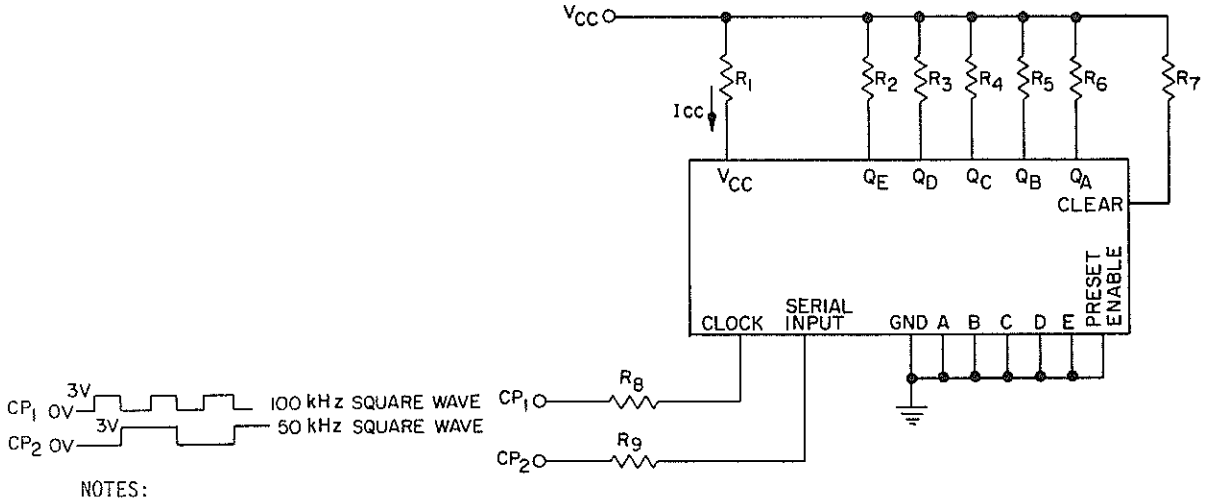
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

Device Type 01



NOTES:
R_1 = 10Ω maximum.
$R_2 = R_7 = R_8$ = 27Ω maximum.
$R_3 = R_4 = R_5 = R_6$ = 220Ω ±5%.
I_{cc} maximum = 72 mA.
V_{cc} shall be of such a value as to provide 5 V minimum at device terminals
CP_2 voltage transitions shall occur at least 5 ns after CP_1 transitions.

Device Type 02

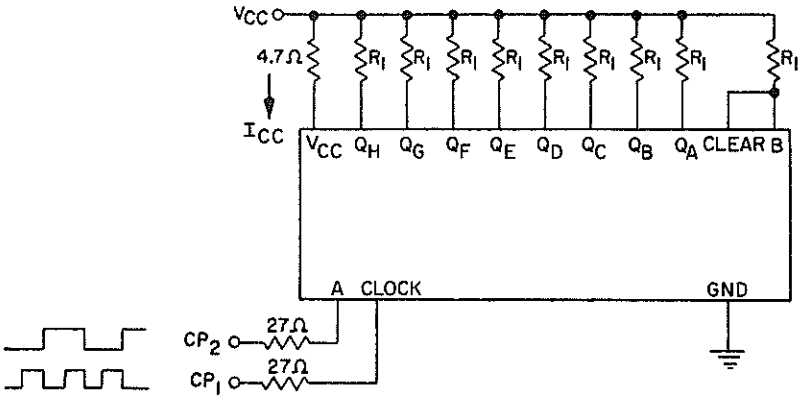


NOTES:
R_1 = 10Ω maximum.
$R_2 = R_3 = R_4 = R_5 = R_6 = R_7$ = 220Ω ±5%.
$R_8 = R_9$ = 27Ω maximum.
I_{cc} maximum = 68 mA.
V_{cc} shall be of such a value as to provide 5 V minimum at device terminals.
CP_2 voltage transitions shall occur at least 5 ns after CP_1 transitions.

FIGURE 5. Burn-in and life test circuits.

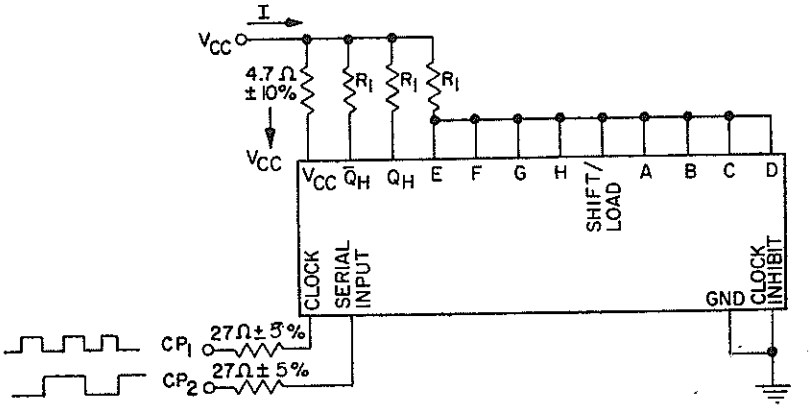
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

Device Type 03



NOTES:
ICCMAX = 54mA.
CP_1 = 100 kHz, 50% duty cycle, square wave, 0-3V
CP_2 = 50 kHz, 50% duty cycle, square wave, 0-3V
CP_2 = voltage transitions shall occur at least 5 ns after CP_1 transitions.
All resistors, ± 5%
R_1 = 220Ω
V_{cc} shall provide 5V min. at device terminals.

Device Type 04

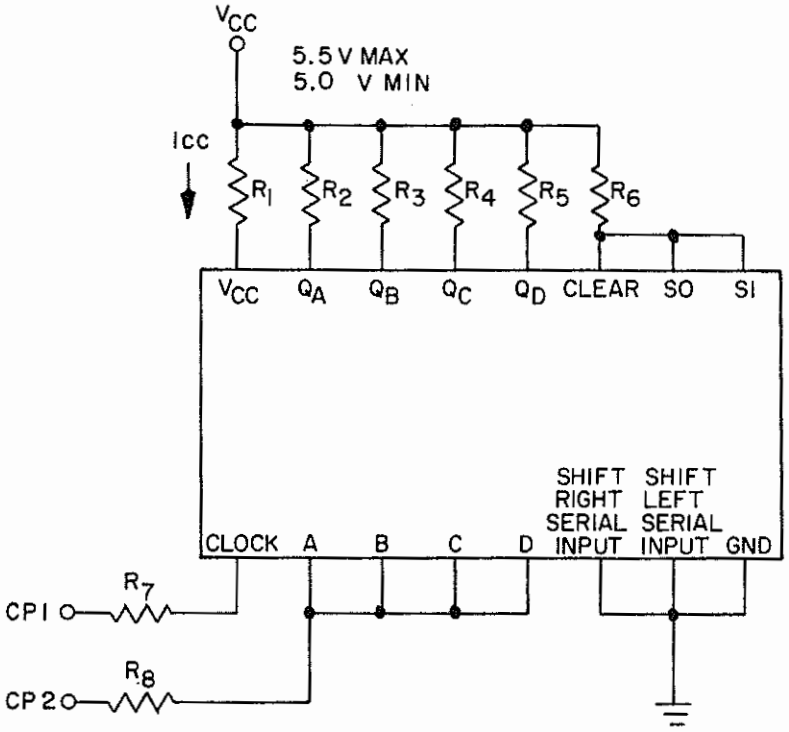


NOTES:
Iccmax - 63mA
CP_1 = 100 kHz, 50% duty cycle, square wave, 0-3V
CP_2 = 50kHz, 50% duty cycle, square wave, 0-3V
CP_2 voltage transitions shall occur at least 5 ns after CP_1 transitions.
R_1 = 220Ω ± 5%
V_{cc} shall provide 5V min. at device terminals.

FIGURE 5. Burn-in and life test circuits - Continued

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Device type 05

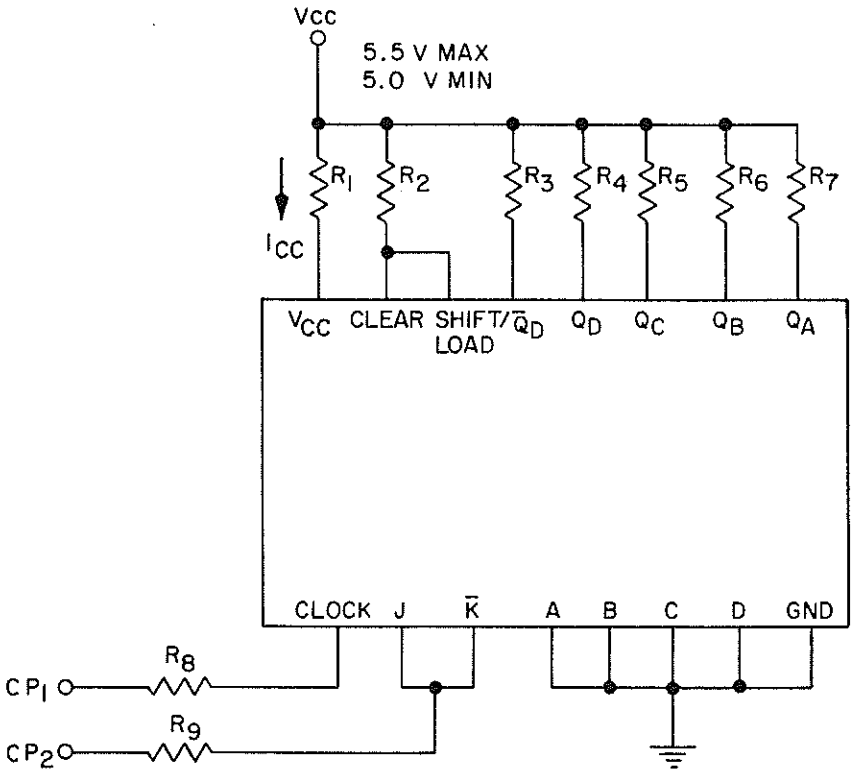


NOTES:
R_1 = 4.7 ohms ± 5 %
R_2 = R_6 = 220 ohms± 5 %
R_7 = R_8 = 27 ohms± 5 %
CP_1 = 100 kHz, 50% duty cycle, 0-3V
CP_2 = 50kHz, 50% duty cycle, 0-3V
I_{ccmax} = 63mA
V_{cc} shall provide 5V min. at device terminals.

FIGURE 5. Burn-in and lift test circuits - continued.

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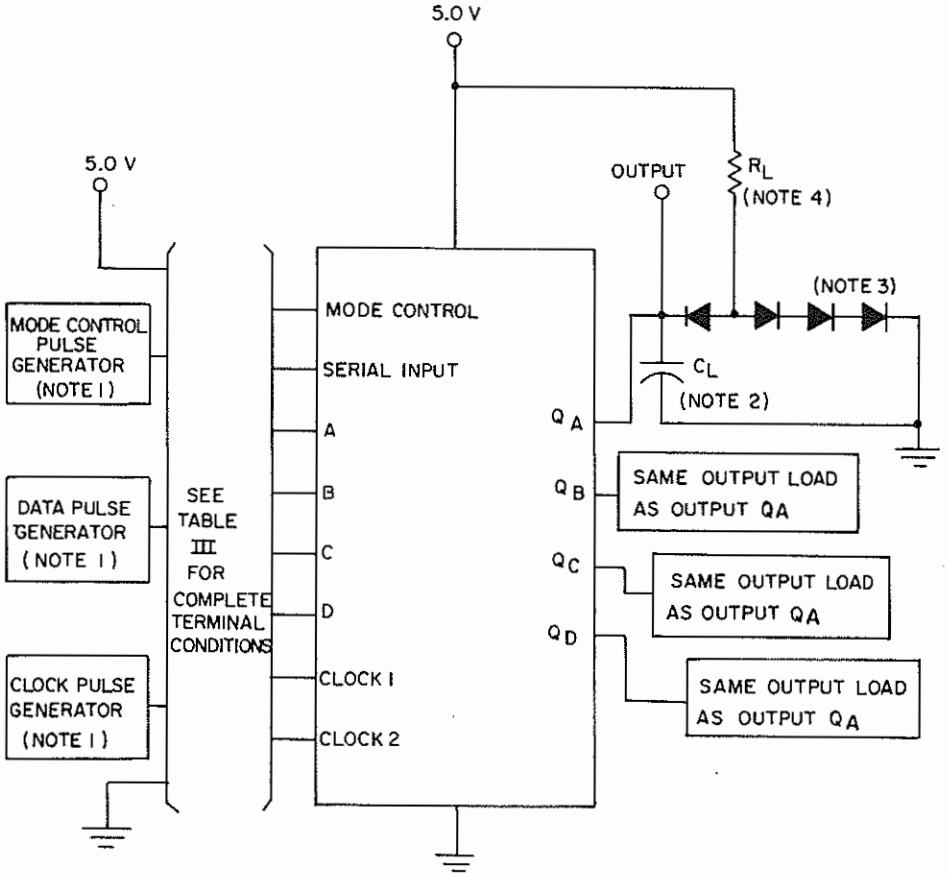
Device type 06



NOTES:
I_{ccmax} = 63mA.
CP_1 = 0 - 3.0V; 100kHz; 50% duty cycle
CP_2 = 0 - 3.0V; 50 kHz; 50% duty cycle
R_1 = 4.7Ω ±10%
$R_2 = R_3 = R_4 = R_5 = R_6 = R_7$ = 220Ω±10%
$R_8 = R_9$ = 27Ω±10%.
V_{cc} shall provide 5V min. at device terminals.

FIGURE 5. Burn-in and life test circuits - continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

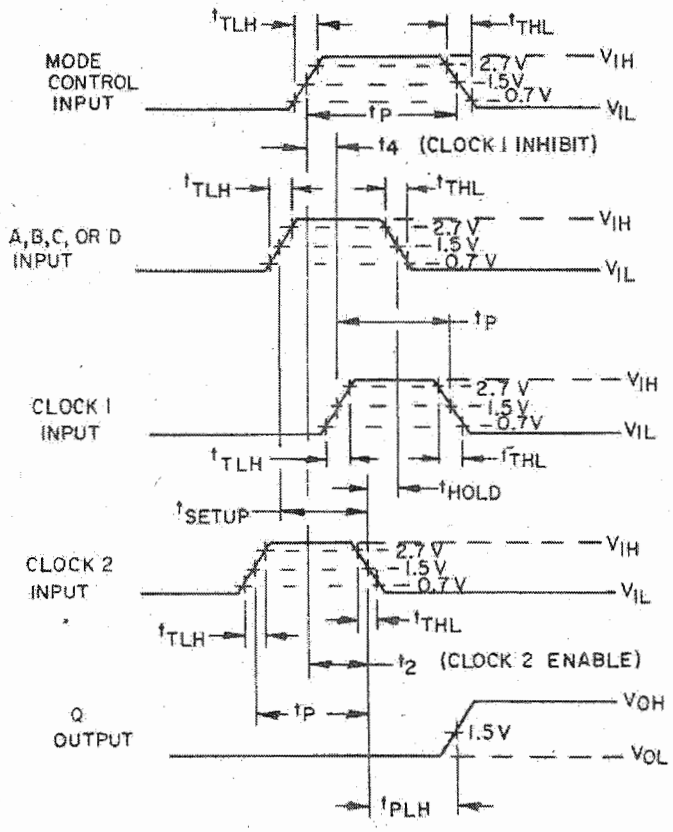


NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 10$ ns, $t_{THL} \leq 10$ ns, $V_{IH} = 3.0$ V minimum, $V_{IL} = 0$ V, $Z_{OUT} = 50\Omega$.
2. $C_L = 50$ pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064 or equivalent.
4. $R_L = 400\Omega \pm 5\%$.

FIGURE 6. Switching test circuits and waveforms for device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
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MAXIMUM SHIFT FREQUENCY (fMAX) and CLOCK TO OUTPUT (tPLH)

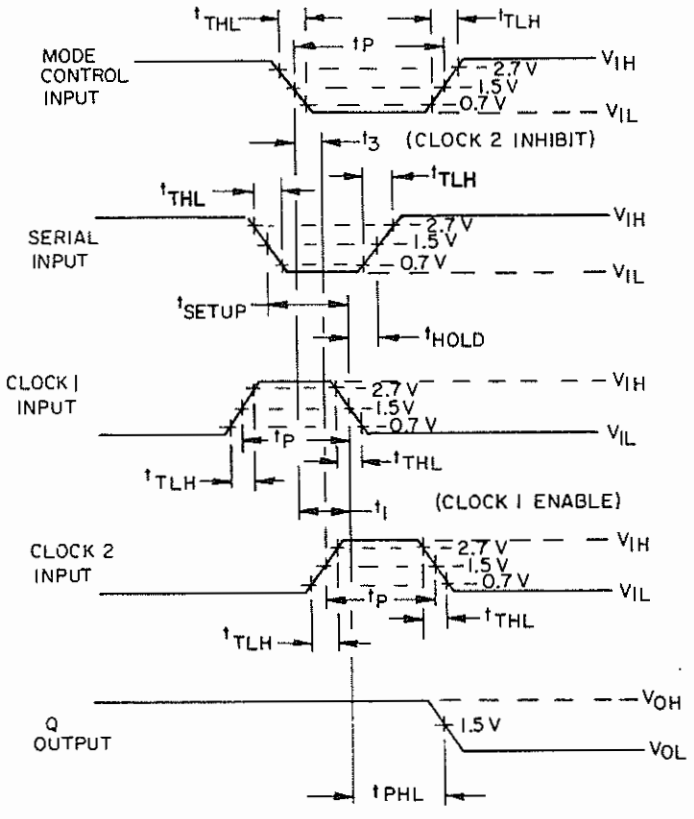


NOTES:
1. Mode control input characteristics: For f_{MAX}, PRR = 22 MHz at TA = 25°C and PRR = 16 MHz at -55°C ≤ TA ≤ 125°C. For t_{PLH}, PRR = 1 MHz, t_p = 35 ns, $t_{TLH} = t_{THL} \leq 10$ ns.
2. A, B, C, or D input characteristics: For f_{MAX}, PRR = 11 MHz at TA = 25°C and PRR = 8 MHz at -55°C ≤ TA ≤ 125°C. For t_{PLH}, PRR = 500 kHz, $t_p = t_{SETUP} + t_{HOLD}$. t_{SETUP} = 20 ns, t_{HOLD} = 5 ns, $t_{TLH} = t_{THL} \leq 10$ ns.
3. Clock 1 input characteristics: When testing f_{MAX}, PRR = 11 MHz at 25°C and PRR = 8 MHz at -55°C ≤ TA ≤ 125°C. For t_{PLH}, PRR = 500 kHz, t_p = 20 ns minimum, $t_{TLH} = t_{THL} \leq 10$ ns.
4. Clock 2 input characteristics: When testing f_{MAX}, PRR = 22 MHz at 25°C and PRR = 16 MHz at -55°C ≤ TA ≤ 125°C. For t_{PLH}, PRR = 1 MHz, t_p = 20 ns minimum, $t_{TLH} = t_{THL} \leq 10$ ns.
5. Serial input = GND.
6. Except for input under test, all other data inputs are open.

FIGURE 6. Switching test circuits and waveforms for device type 01 - Continued.

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Check the source to verify that this is the current version before use.

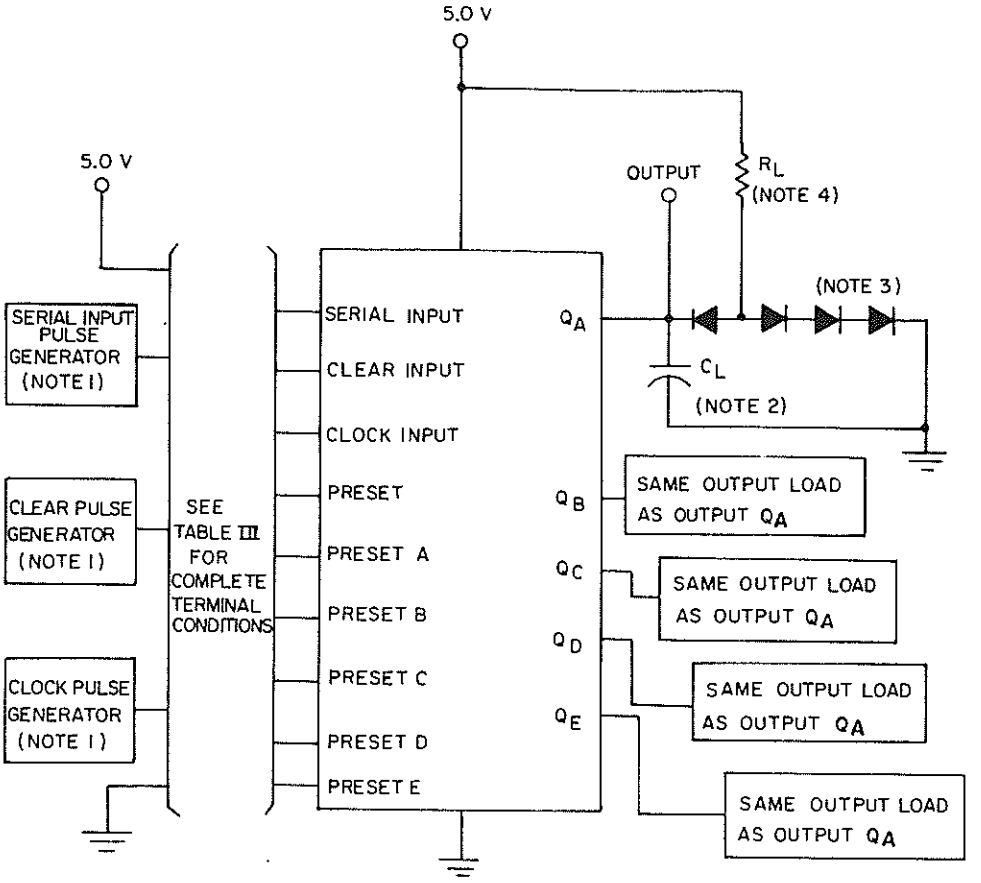
CLOCK TO OUTPUT (t_{PHL})



NOTES:
1. Mode control input characteristics: PRR = 1 MHz, t_p = 35 ns, $t_{TLH} = t_{THL} \leq 10$ ns.
2. Serial input characteristics: PRR = 500 kHz, $t_p = t_{SETUP} + t_{HOLD}$, t_{SETUP} = 20 ns minimum, t_{HOLD} = 5, $t_{TLH} = t_{THL} \leq 10$ ns.
3. Clock 1 input characteristics: PRR = 1 MHz, t_p = 20 ns minimum, $t_{TLH} = t_{THL} \leq 10$ ns.
4. Clock 2 input characteristics: PRR = 500 kHz, t_p = 20 ns minimum, $t_{TLH} = t_{THL} \leq 10$ ns.
5. Inputs A thru D = OPEN.

FIGURE 6. Switching test circuits and waveforms for device type 01 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

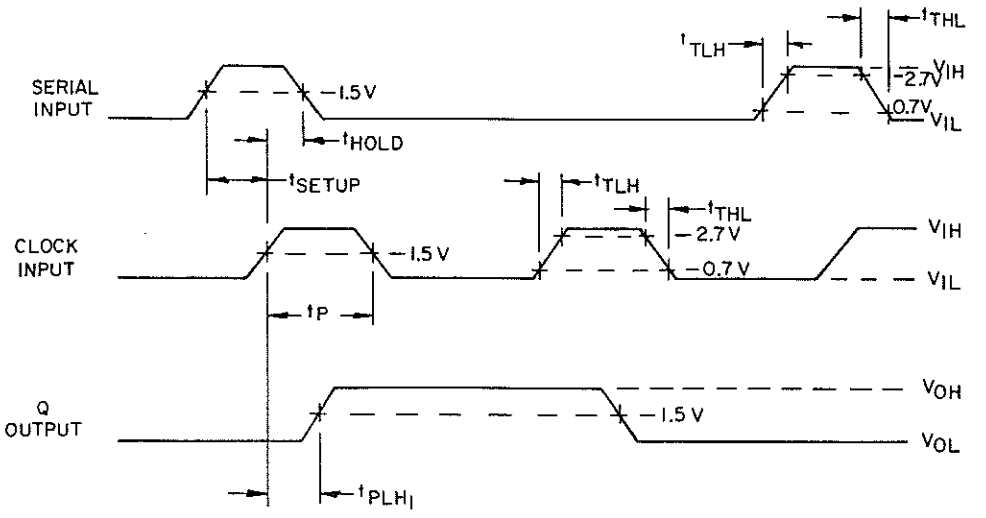


NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 10$ ns, $t_{THL} \leq 10$ ns, $V_{IH} = 3.0$ V minimum, $V_{IL} = 0$ V, $Z_{OUT} = 50\Omega$.
2. $C_L = 50$ pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064 or equivalent.
4. $R_L = 400\Omega \pm 5\%$.

FIGURE 7. Switching test circuits and waveforms for device type 02.

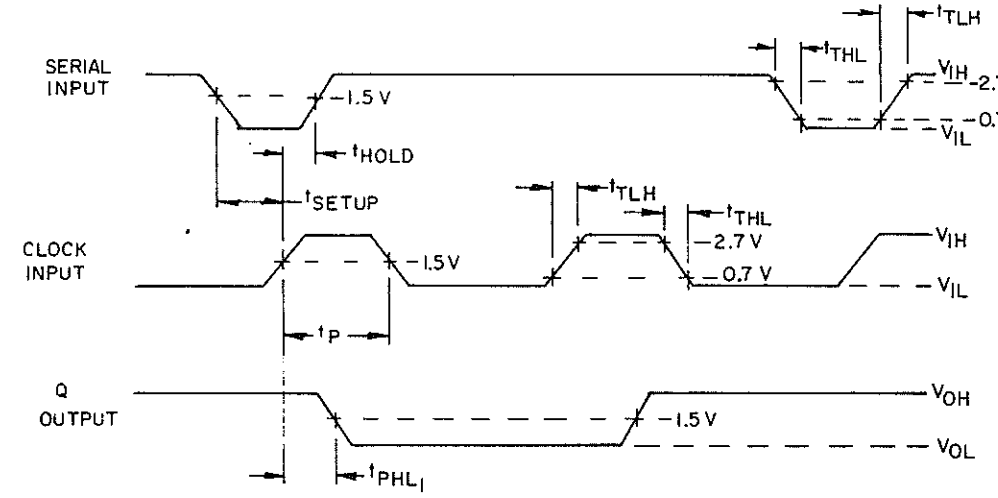
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

MAXIMUM CLOCK FREQUENCY (f_{MAX}) and CLOCK TO OUTPUT (t_{PLH1}).



NOTES:
1. Serial input characteristics: For f_{MAX}, PRR = 5 MHz at TA = 25°C, PRR = 3.5MHz at -55° ≤ TA ≤ 125°C. For t_{PLH1}, PRR = 500 kHz, t_p = t_{SETUP} + t_{HOLD}, t_{SETUP} = 30 ns, t_{HOLD} = 0 ns, t_{THL} = t_{TLH} ≤ 10 ns.
2. Clock input characteristics: For f_{MAX}, PRR = 10 MHz at TA = 25°C, PRR = 7 MHz at -55° ≤ TA ≤ 125°C. For t_{PLH1}, PRR = 1 MHz, t_p = 35 ns, t_{TLH} = t_{THL} ≤ 10 ns.
3. Clear = 4.5 V, preset enable = GND, preset A thru E = OPEN.

CLOCK TO OUTPUT (t_{PHL1})

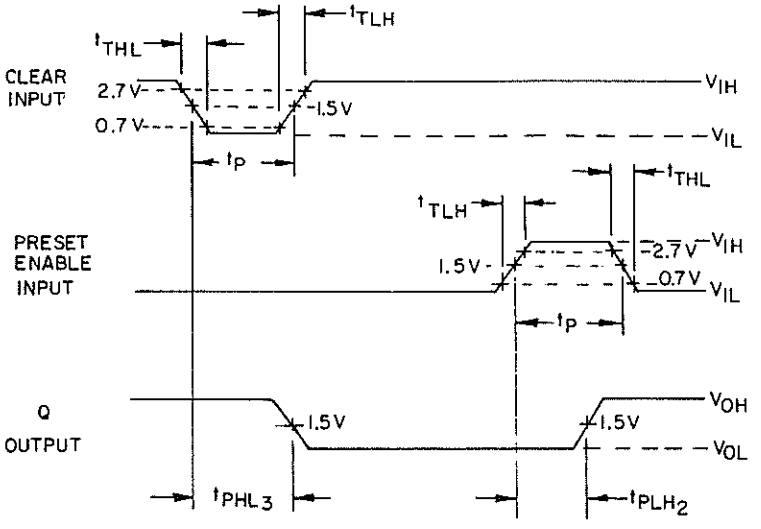


NOTES:
1. Serial input characteristics: PRR = 500 kHz, t_{THL} = t_{TLH} ≤ 10 ns, t_p = t_{SETUP} + t_{HOLD}, t_{SETUP} = 30 ns, t_{HOLD} = 0 ns.
2. Clock input characteristics: PRR = 1 MHz, t_{THL} = t_{TLH} ≤ 10 ns, t_p = 35 ns.
3. Clear = 4.5 V, preset enable = GND, preset A thru E = OPEN.

FIGURE 7. Switching test circuits and waveforms for device type 02 - Continued.

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Check the source to verify that this is the current version before use.

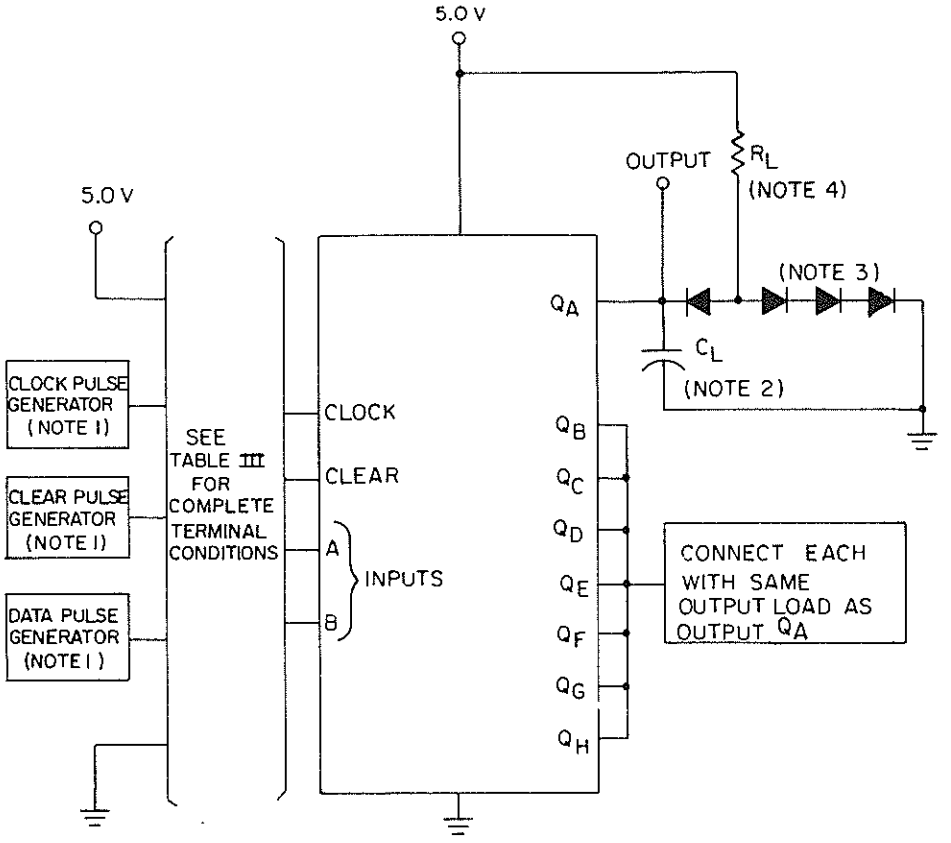
PRESET ENABLE TO OUTPUT (t_{PLH2}) and CLEAR TO OUTPUT (t_{PHL3})



NOTES:
1. Clear input characteristics: PRR = 1 MHz, $t_{THL} = t_{TLH} \leq 10$ ns, $t_p = 30$ ns.
2. Preset enable characteristics: PRR = 1 MHz, $t_{THL} = t_{TLH} \leq 10$ ns, $t_p = 30$ ns.
3. Preset A thru E = 4.5 V, clock = GND, serial = OPEN.

FIGURE 7. Switching test circuits and waveforms for device type 02 - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

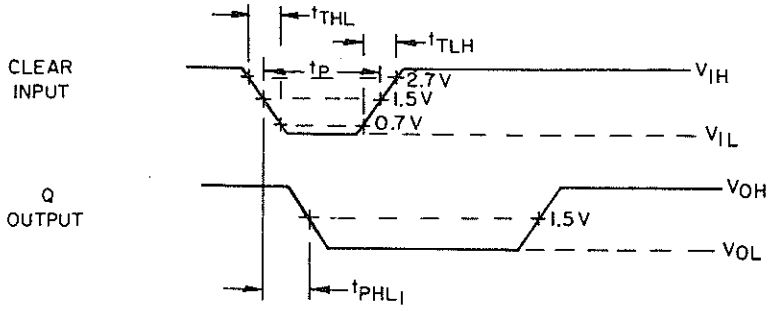


NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 10$ ns, $t_{THL} \leq 10$ ns, $V_{IH} = 3.0$ V minimum, $V_{IL} = 0$ V, $Z_{OUT} = 50\Omega$.
2. $C_L = 50$ pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064 or equivalent.
4. $R_L = 800\Omega \pm 5\%$.
5. Q_A outputs are illustrated in the individual waveforms. Relationship of serial input A and B data to other Q outputs is illustrated in the typical shift sequence.

FIGURE 8. Switching test circuit and waveforms for device type 03.

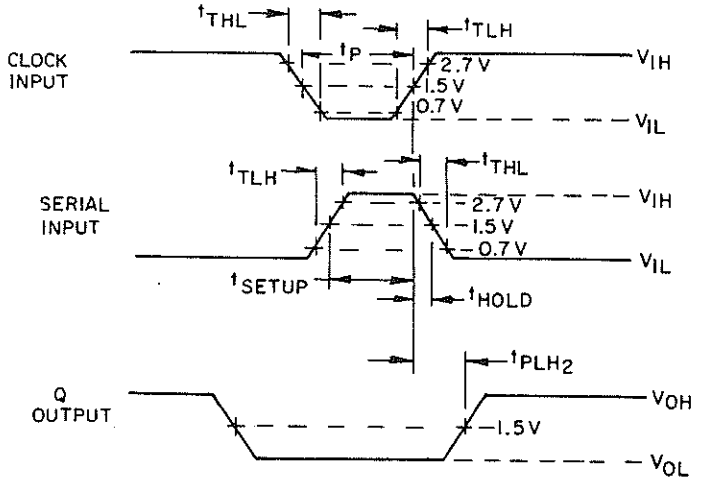
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

CLEAR INPUT TO Q OUTPUTS (t_{PHL1})



NOTES:
1. Clear input characteristics: PRR = 1 MHz, $t_{THL} = t_{TLH} \leq 10$ ns, $t_p = 50$ ns maximum.
2. Clock = GND, serial inputs A and B = OPEN.

MAXIMUM CLOCK FREQUENCY, (f_{MAX}) and CLOCK INPUT TO Q OUTPUT (t_{PLH2}).

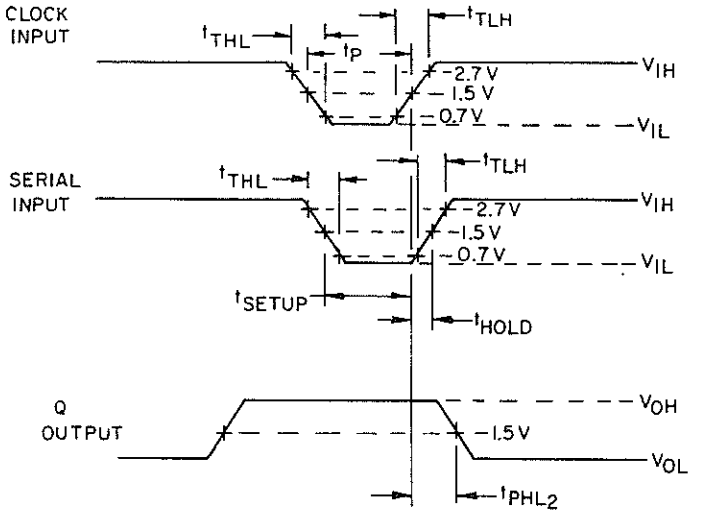


NOTES:
1. Clock input characteristics: For f_{MAX}, PRR = 22 MHz at $T_A = 25°C$, PRR = 18 MHz at $-55°C \leq T_A \leq 125°C$. For t_{PLH2}, PRR = 1 MHz, $t_p = 30$ ns maximum, $t_{THL} = t_{TLH} \leq 10$ ns.
2. Serial input characteristics: For f_{MAX}, PRR = 11 MHz at 25°C, PRR = 9 MHz at $-55° \leq T_A \leq 125°C$. For t_{PLH2}, PRR = 500 kHz, $t_p = t_{SETUP} + t_{HOLD}$, $t_{SETUP} = 15$ ns minimum, $t_{HOLD} = 10$ ns maximum, $t_{THL} = t_{TLH} \leq 10$ ns.
3. Clear = 4.5 V.

FIGURE 8. Switching test circuit and waveforms for device type 03 - Continued.

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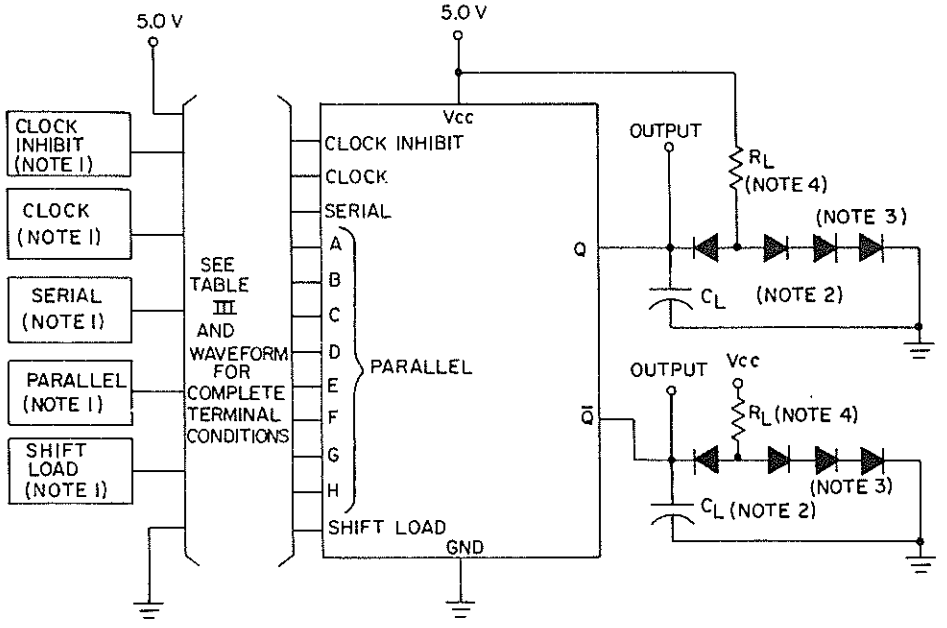
CLOCK INPUT TO Q OUTPUT (t_{PHL2})



NOTES:
1. Clock input characteristics: PRR = 1 MHz, $t_{THL} = t_{TLH} \leq 10$ ns, $t_p = 30$ ns maximum.
2. Serial input characteristics: PRR = 500 kHz, $t_p = t_{SETUP} + t_{HOLD}$, $t_{SETUP} = 15$ ns minimum, $t_{HOLD} = 10$ ns maximum, $t_{THL} = t_{TLH} \leq 10$ ns.
3. Clear = 4.5 V.

FIGURE 8. Switching test circuit and waveforms for device type 03 - Continued.

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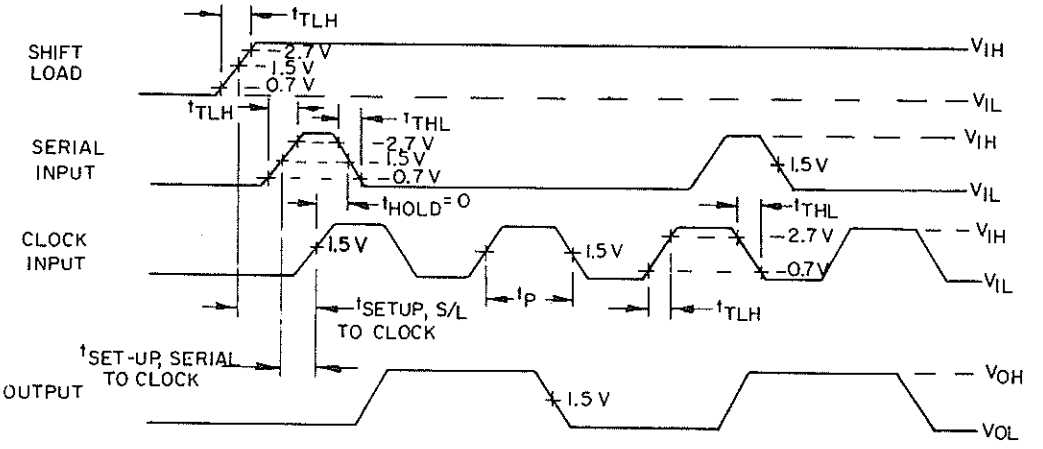


NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 10$ ns, $t_{THL} \leq 10$ ns, $V_{IH} = 3.0$ V minimum, $V_{IL} = 0$ V, $Z_{OUT} \approx 50\Omega$.
2. $C_L = 50$ pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064 or equivalent.
4. $R_L = 400\Omega \pm 5\%$.

FIGURE 9. Switching test circuit and waveforms for device type 04.

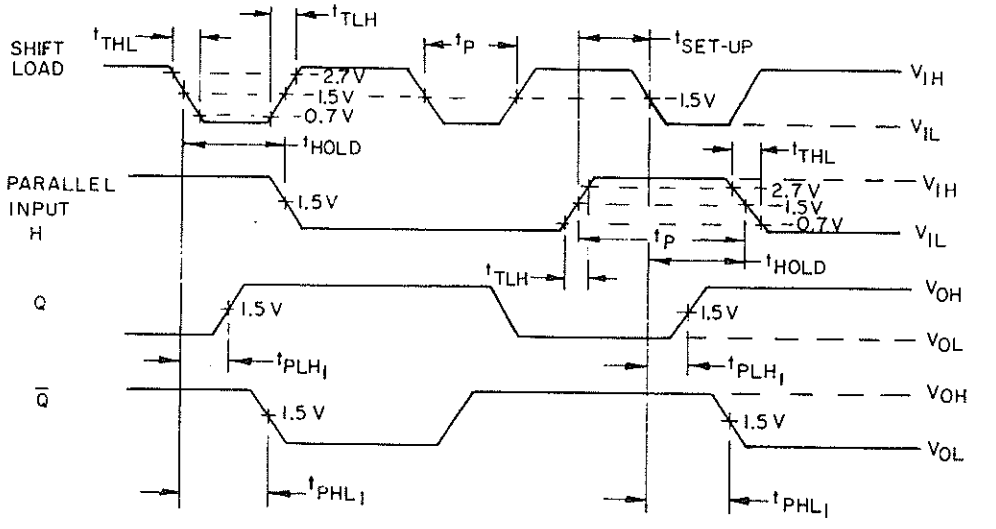
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
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MAXIMUM CLOCK FREQUENCY, f_{MAX}



NOTES:
1. Clock pulse characteristics: PRR = 18 MHz at TA = 25°C, PRR = 14 MHz at -55°C $\leq$ TA $\leq$ 125°C, $t_{TLH} = t_{THL} \leq 10$ ns, $t_p = 20$ ns minimum.
2. Serial pulse characteristics: PRR = 9 MHz at TA = 25°C, 7 MHz at -55° $\leq$ TA $\leq$ 125°C $t_p = t_{SETUP} + t_{HOLD}$, t_{SETUP} = 35 ns minimum, t_{HOLD} = 0, $t_{TLH} = t_{THL} \leq 5$ ns.
3. Shift load characteristics: $t_{TLH} \leq 10$ ns, t_{SETUP} = 45 ns.
4. Clock inhibit = GND, A thru H = GND.

SHIFT LOAD TO OUTPUT (t_{PHL1}, t_{PLH1})

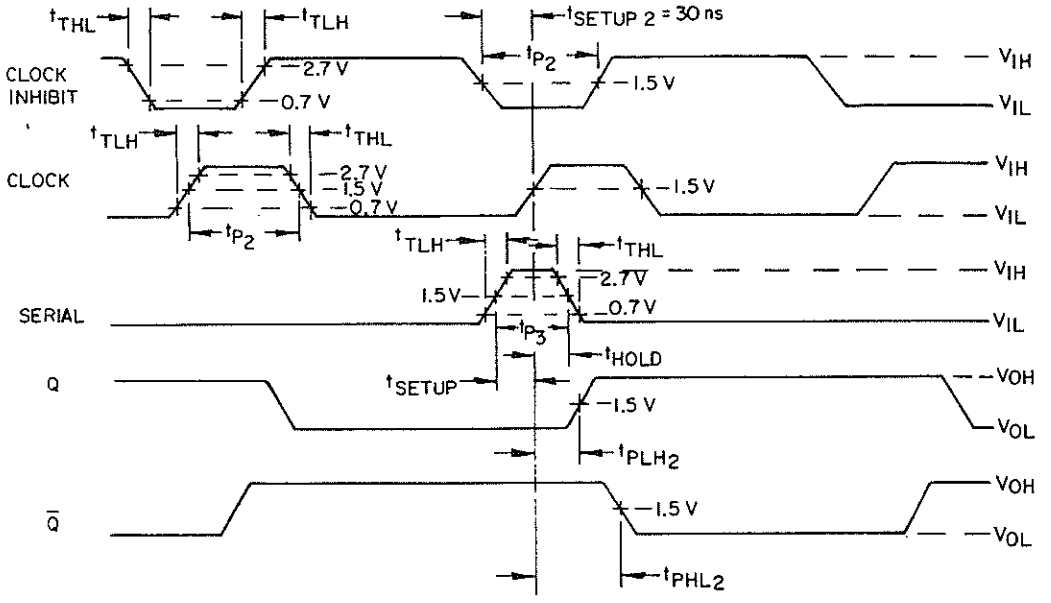


NOTES:
1. Shift load characteristics: PRR = 1 MHz, t_p = 25 ns, $t_{THL} = t_{TLH} \leq 10$ ns.
2. Parallel input characteristics: PRR = 500 kHz, $t_p = t_{SETUP} + t_{HOLD}$ = 40 ns, t_{SETUP} = 10 ns, t_{HOLD} = 30 ns, $t_{THL} = t_{TLH} \leq 10$ ns.
3. Clock = clock inhibit = GND, A thru G = GND, serial = OPEN.

FIGURE 9. Switching test circuit and waveforms for device type 04 - Continued.

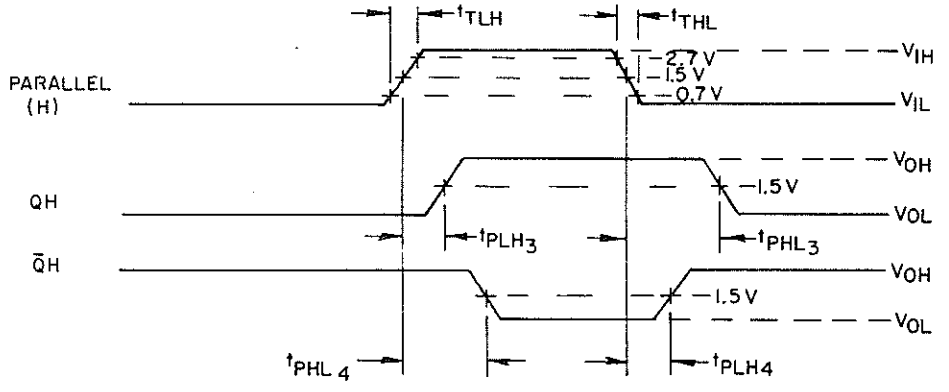
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

CLOCK TO OUTPUT (t_{PHL2} and t_{PLH2})



NOTES:
1. Clock inhibit characteristics: PRR = 1 MHz, t_{p2} = 50 ns, $t_{THL} = t_{TLH} \leq 10$ ns t_{SETUP2} = 30 ns.
2. Clock pulse characteristics: PRR = 1 MHz, t_{p1} = 25 ns, $t_{THL} = t_{TLH} \leq 10$ ns.
3. Serial pulse characteristics: PRR = 500 kHz, $t_{p3} = t_{SETUP} + t_{HOLD}$, t_{SETUP} = 35 ns t_{HOLD} = 0, $t_{TLH} = t_{THL} \leq 5$ ns.
4. Shift/load = 5.0 V.

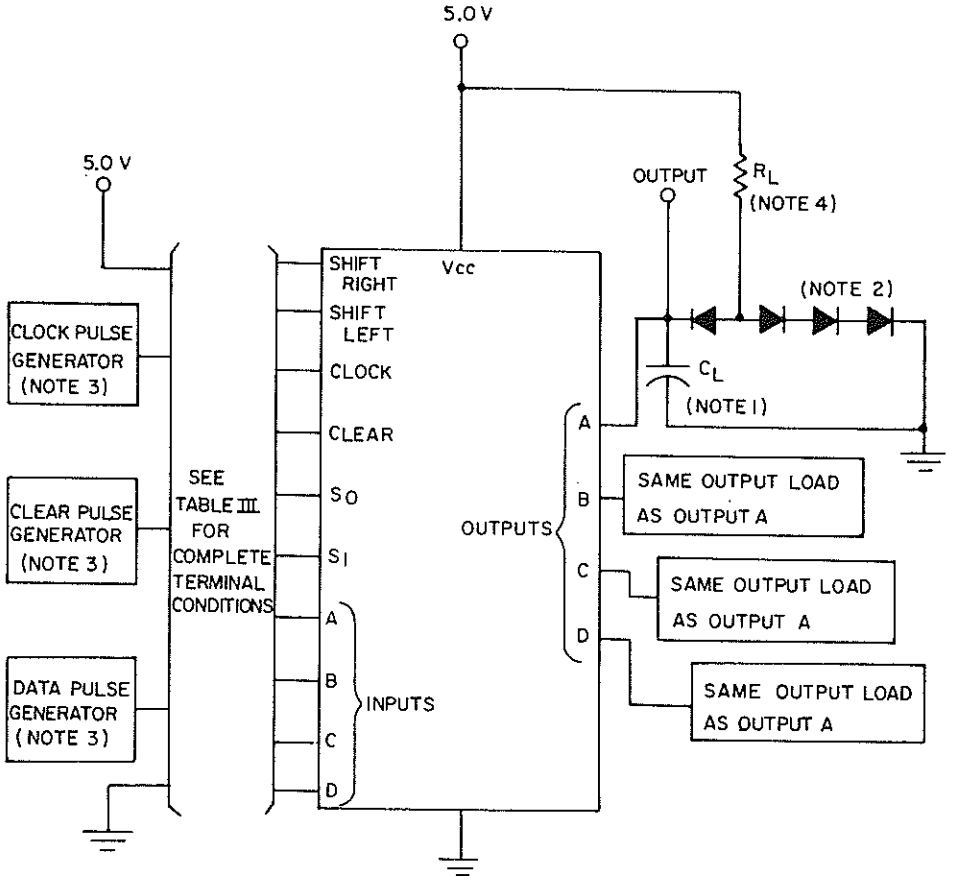
(H) INPUT TO QH and $\overline{QH}$ OUTPUTS (t_{PHL3} and t_{PLH3}) (t_{THL4} and t_{PLH4})



NOTES:
1. (H) input characteristics: PRR = 1 MHz, 50% duty cycle, $t_{TLH} = t_{THL} \leq 10$ ns.
2. Shift/load = GND, clock inhibit = GND, serial = GND, A thru G = GND, clock = GND.

FIGURE 9. Switching test circuit and waveforms for device type 04 - Continued.

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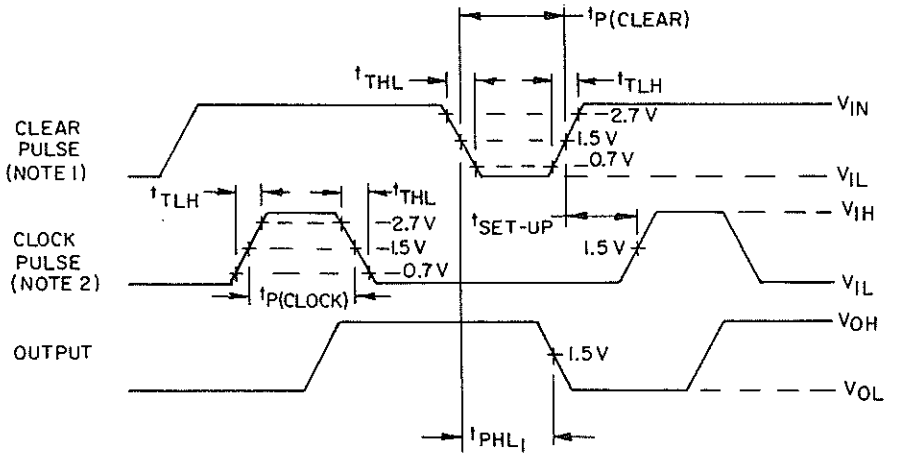


NOTES:
1. C_L = 50 pF minimum including probe and jig capacitance.
2. All diodes are 1N3064, or equivalent.
3. Unless otherwise specified in the notes associated with the individual tests, all pulse generators have the following characteristics: $Z_{OUT} \approx 50\Omega$, $t_{TLH} \leq 7$ ns, $t_{THL} \leq 7$ ns, V_{IH} = 3.0 V minimum, V_{IL} = 0.
4. $R_L = 400\Omega \pm 5\%$.

FIGURE 10. Switching test circuit and waveforms for device type 05.

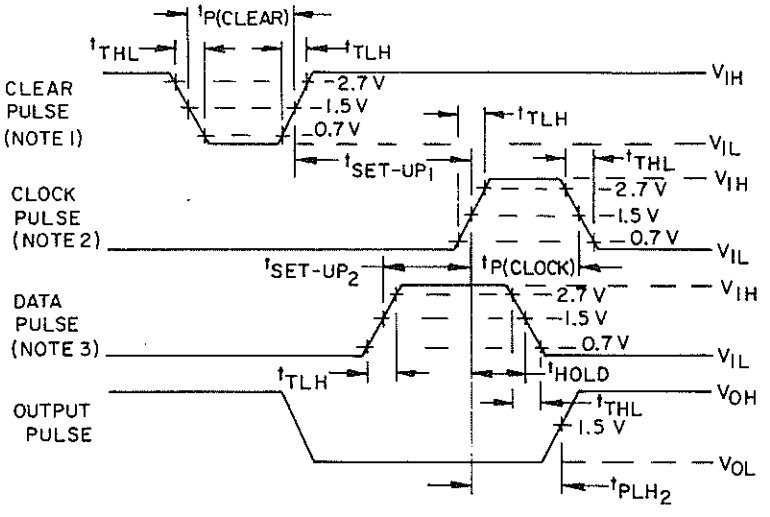
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CLEAR TO OUTPUT



NOTES:
1. The clear pulse has the following characteristics: $t_{P(CLEAR)}$ = 20 ns, t_{SETUP} = 25 ns, PRR = 1 MHz.
2. The clock pulse has the following characteristics: $t_{P(CLOCK)}$ = 20 ns, PRR = 1 MHz.

CLOCK TO OUTPUT, DATA PULSE TO PARALLEL INPUT (t_{PLH2})

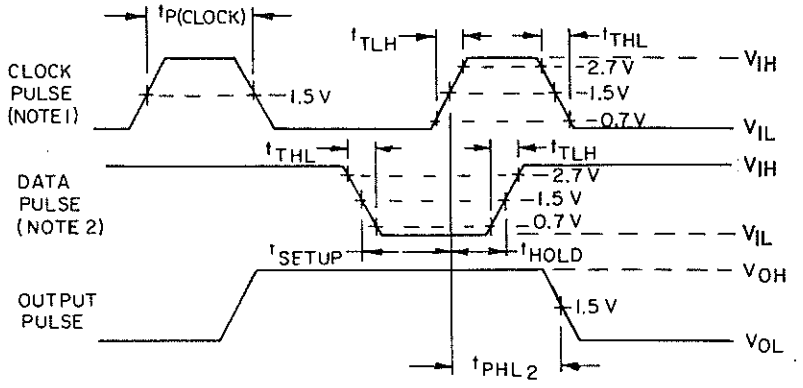


NOTES:
1. The clear pulse is a momentary ground, then V_{IH} is applied to the input. $t_{P(CLEAR)} \leq 75$ ns, $t_{THL} \leq 15$ ns and $t_{TLH} \leq 15$ ns, t_{SETUP} = 25 ns.
2. Clock pulse characteristics: $t_{P(CLOCK)}$ = 20 ns, PRR = 2 MHz.
3. Data pulse characteristics: $t_{P(DATA)} = t_{(SETUP2)} + t_{HOLD}$, t_{SETUP2} = 20 ns, t_{HOLD} = 7, PRR = 1 MHz.

FIGURE 10. Switching test circuit and waveforms for device type 05 - Continued.

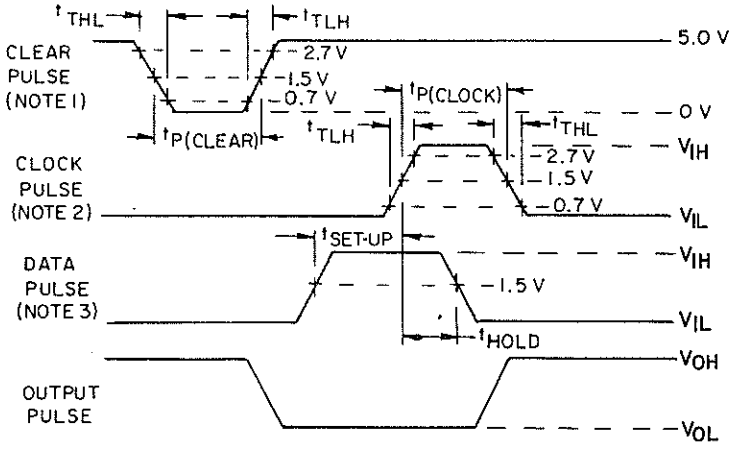
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

CLOCK TO OUTPUT, DATA PULSE TO PARALLEL INPUT (t_{PHL2})



NOTES:
1. Clock pulse characteristics: tP(CLOCK) = 20 ns, PRR = 2 MHz.
2. Data pulse characteristics: tP(DATA) = tSETUP = 20 ns, PRR = 1 MHz.

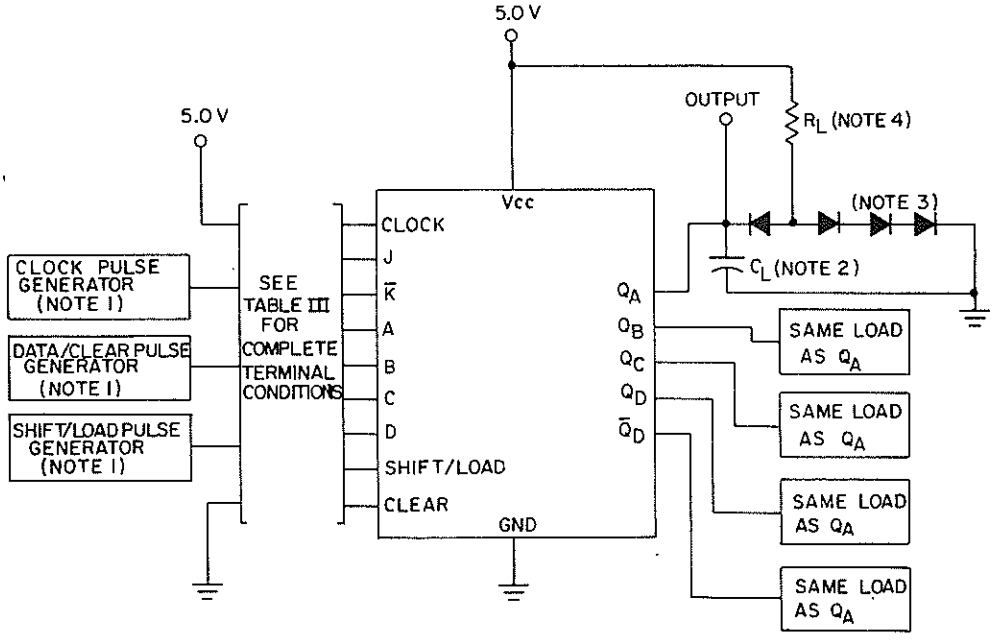
MAXIMUM CLOCK FREQUENCY (f_{MAX})



NOTES:
1. The clear pulse is a momentary GND, then V_{IH} is applied to the input, $t_{P(CLEAR)} \leq 20$ ns, $t_{THL} \leq 15$ ns, $t_{TLH} \leq 15$ ns.
2. Clock pulse characteristics: tP(CLOCK) = 20 ns, PRR = 18 MHz at $-55°C \leq T_A \leq 125°C$ (22 MHz at $T_A = 25°C$).
3. Data pulse characteristics: tP(DATA) = tSETUP = 20 ns, PRR = 9 MHz at $-55°C \leq T_A \leq 125°C$ (11 MHz at $T_A = 25°C$).

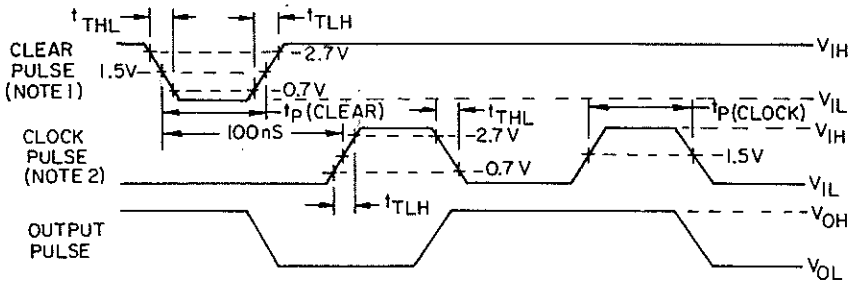
FIGURE 10. Switching test circuit and waveforms for device type 05 - Continued.

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NOTES:
1. Unless otherwise specified in the notes with the individual waveforms, all pulse generators shall have the following characteristics: $t_{TLH} \leq 7$ ns, $t_{THL} \leq 7$ ns, V_{IH} = 3.0 V minimum, V_{IL} = 0, $Z_{OUT} \approx 50\Omega$.
2. C_L = 50 pF minimum, including jig and probe capacitance.
3. All diodes are 1N3064 or equivalent.
4. $R_L = 400\Omega \pm 5\%$.

MAXIMUM CLOCK FREQUENCY, fMAX

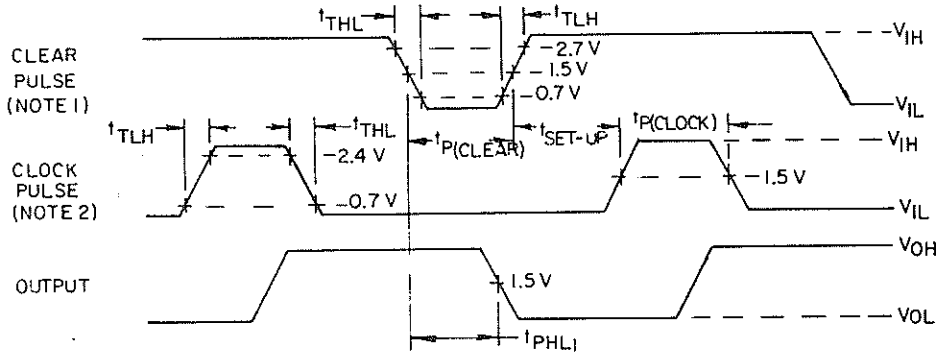


NOTES:
1. The clear pulse is a momentary GND, then V_{IH} is applied to the input. $t_{THL} \leq 15$ ns, $t_{TLH} \leq 15$ ns, $t_{P(CLEAR)} \leq 75$ ns.
2. Clock pulse characteristics: $t_{P(CLOCK)}$ = 16 ns, PRR = 24 MHz at $-55°C \leq T_A \leq 125°C$ (30 MHz at T_A = 25°C), V_{IH} = 3.0 V minimum, V_{IL} = GND.

FIGURE 11. Switching test circuit and waveforms for device type 06.

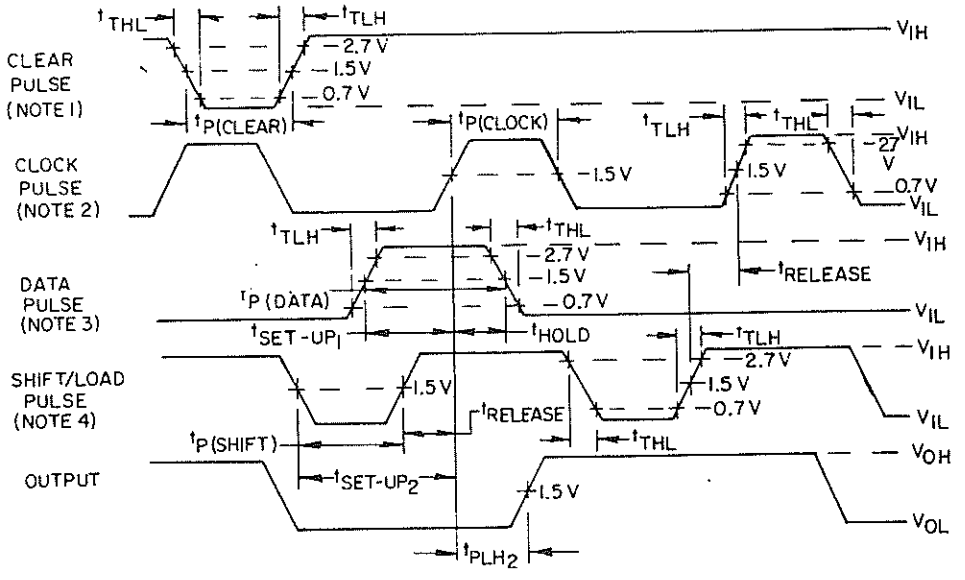
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:49Z
Check the source to verify that this is the current version before use.

CLEAR TO OUTPUT (tPHL1)



NOTES:
1. Clear pulse characteristics: tP(CLEAR) = 12 ns, tSETUP = 25 ns, PRR = 1 MHz.
2. Clock pulse characteristics: tP(CLOCK) = 16 ns, PRR = 1 MHz.

CLOCK TO OUTPUT (tPLH2)

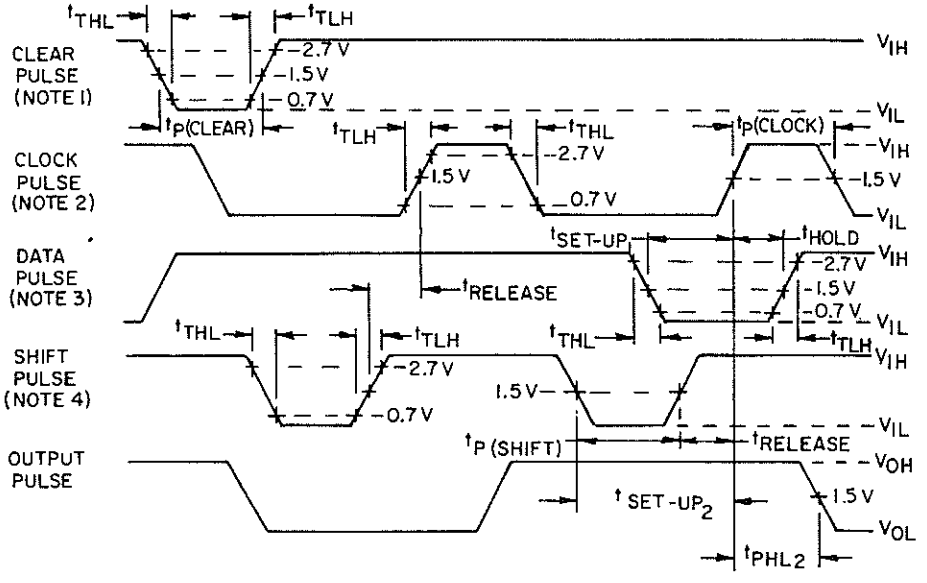


NOTES:
1. The clear pulse is a momentary GND, then VIH is applied to the input. tTHL $\leq$ 15 ns, tTLH $\leq$ 15 ns, tP(CLEAR) $\leq$ 100 ns.
2. Clock pulse characteristics: tP(CLOCK) = 16 ns, PRR = 2 MHz.
3. Data pulse characteristics: tP(DATA) = 25 ns, tSETUP1 = 25 ns, tHOLD = 0, PRR = 1 MHz.
4. Shift/load pulse characteristics: tP(SHIFT) = 17 ns, tRELEASE = 10 ns, tSETUP2 = 27 ns, PRR = 2 MHz.

FIGURE 11. Switching test circuit and waveforms for device type 06 - Continued.

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CLOCK TO OUTPUT (t_{PHL2})



NOTES:
1. The clear pulse is a momentary GND, then VIH is applied to the input. $t_{THL} \leq 15$ ns, $t_{TLH} \leq 15$ ns, $t_{P(CLEAR)} \leq 100$ ns.
2. Clock pulse characteristics: $t_{P(CLOCK)} = 16$ ns, PRR = 2 MHz.
3. Data pulse characteristics: $t_{P(DATA)} = t_{SETUP} + t_{HOLD} = 20$ ns, $t_{SETUP1} = 20$ ns, $t_{HOLD} = 0$, PRR = 1 MHz.
4. Shift/load pulse characteristics: $t_{P(SHIFT)} = 22$ ns, $t_{RELEASE} = 10$ ns, $t_{SETUP2} = 32$ ns, PRR = 2 MHz.

FIGURE 11. Switching test circuit and waveforms for device type 06 - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Input serial	2 Input A	3 Input B	4 Input C	5 Input D	6 Mode control	7 GND	8 Clock 2	9 Clock 1	10 Q_D	11 Q_C	12 Q_B	13 Q_A	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 $T_A = 25°C$	V_{OH}	3006	1 2 3 4		2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	A 1/		-.8 mA	-.8mA	-.8 mA	-.8 mA	4.5 V	Q_A Q_B Q_C Q_D	2.4		V
	V_{OL}	3007	5 6 7 8		0.8 V	0.8 V	0.8 V	0.8 V					16 mA	16 mA	16 mA	16 mA		Q_A Q_B Q_C Q_D		0.4	
	V_{IC}		9 10 11 12 13 14 15 16	-12 mA	-12mA	-12 mA	-12mA	-12 mA	-12 mA		-12 mA	-12 mA						Input serial Input A Input B Input C Input D Mode control Clock 2 Clock 1		-1.5	
	I_{IL1}	3009	17 18 19 20 21 22 23	0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	GND 4.5 V GND		0.4 V	0.4 V					5.5 V	Input serial Input A Input B Input C Input D Clock 2 Clock 1	2/		
	I_{IL2}		24						0.4 V		4.5 V							Mode control	3/		
	I_{IH1}	3010	25 26 27 28 29 30 31	2.4 V	2.4 V	2.4 V	2.4 V	2.4 V	4.5 V GND 4.5 V		2.4 V	2.4 V						Input serial Input A Input B Input C Input D Clock 2 Clock 1		40	µA

See footnotes at end of device type 01.

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Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Input serial	2 Input A	3 Input B	4 Input C	5 Input D	6 Mode control	7 GND	8 Clock 2	9 Clock 1	10 Q_D	11 Q_C	12 Q_B	13 Q_A	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	I_{IH2}	3010	32	5.5 V					4.5 V	GND							5.5 V	Input serial		100	µA
			33		5.5 V				GND									Input A			
			34			5.5 V												Input B			
			35				5.5 V											Input C			
			36					5.5 V										Input D			
			37								5.5 V							Clock 2			
			38						4.5 V			5.5 V						Clock 1			
	I_{IH3}		39						2.4 V		GND							Mode control		80	
	I_{IH4}		40						5.5 V		GND							Mode control		200	
	I_{OS}	3011	41		4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		A							Q_A	-18	-57	mA
			42											GND	GND	GND		Q_B			
			43															Q_C			
			44										GND					Q_D			
	I_{CC}	3005	45		GND	GND	GND	GND				GND						V_{CC}		72	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
7 T_A= 25°C 4/ 7/	Truth table test 5/	3014	46	B	B	B	B	B	A	GND	B	B	X	X	X	X	4.5 V		6/		
			47						A		A		X	X	X	X					
			48						A		B		L	L	L	L					
			49						B												
			50	A																	
			51	A								A									
			52	A								B				H					
			53	B								B				H					
			54	B								A				H					
			55	B								B			H	L					
			56	A								B			H	L					
			57	A								A			H	L					
			58	A								B		H	L	H					
			59	B								B		H	L	H					
			60	B								A		H	L	H					
			61	B								B	H	L	H	L					
			62	A								B	H	L	H	L					
			63	A								A	H	L	H	L					
			64	A								B	L	H	L	H					
			65	B								B	L	H	L	H					
			66									A	L	H	L	H					
			67									B	L	H	L	H					
			68									A	H	L	H	L					
			69									B	L	H	L	H					
			70									A	L	H	L	H					
			71									B	L	H	L	L					
			72									A	H	L	H	L					
			73									B	L	H	L	L					
			74		A		A					B	L								

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Input serial	2 Input A	3 Input B	4 Input C	5 Input D	6 Mode control	7 GND	8 Clock 2	9 Clock 1	10 Q_D	11 Q_C	12 Q_B	13 Q_A	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_A = 25°C 4/ 7/	Truth table test 5/	3014	75	B	A	B	A	B	A	GND	B	B	L	L	L	L	4.5 V			6/	
			76		A	B	A	B			A			L		L					
			77		A	B	A	B			B			H		H					
			78		B	A	B	A			B			H		H					
			79		B		B				A		H	H	H	H					
			80		B		B				B			L		L					
			81		A		A				B		L	L		L					
			82		A		A				A			L		L					
			83		A		A				B			H		H					
			84		B		B				B			H		H					
			85		B		B				A		H	H	H	H					
			86		B		B				B		L	L	L	L					
8	Repeat subgroup 7 at T_A = +125°C and T_A = -55°C.																				
9 T_A = 25°C	f_{MAX}	(Fig 6)	87	GND				-IN	IN	GND	IN	IN	OUT				5.0 V	Q_D	11		MHz
	t_{PLH}	3003 (Fig 6)	88		IN											OUT		Q_A	10	30	ns
			89			IN									OUT			Q_B			
			90				IN							OUT				Q_C			
			91					IN					OUT					Q_D			
	t_{PHL}		92	IN												OUT				35	
			93												OUT			Q_A			
			94											OUT				Q_B			
			95										OUT					Q_C			
																		Q_D			
10 T_A=125°C	f_{MAX}	(Fig 6)	96	GND				IN					OUT					Q_D	8		MHz
	t_{PLH}	3003 (Fig 6)	97		IN											OUT		Q_A	10	42	ns
			98			IN									OUT			Q_B			
			99				IN							OUT				Q_C			
			100					IN					OUT					Q_D			
	t_{PHL}		101	IN												OUT				49	
			102												OUT			Q_A			
			103											OUT				Q_B			
			104										OUT					Q_C			
																		Q_D			
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55°C.																				

1/ A = normal clock pulse, except for subgroups 7 and 8 (see 4/).
2/ For device type 01, with schematics incorporating a 4 kΩ base resistor, the minimum and maximum limits shall be -0.7 and -1.6 mA, respectively. For schematics incorporating a 6 kΩ base resistor, the minimum and maximum limits shall be -0.4 and -1.3 mA, respectively.
3/ For device type 01, with schematics incorporating a 4 kΩ base resistor in the mode control input circuit, the minimum and maximum limits shall be -1.4 and -3.2 mA, respectively. For schematics incorporating a 6 kΩ base resistor in the mode control input circuit, the minimum and maximum limits shall be -0.8 and -2.6 mA, respectively.
4/ For subgroups 7 and 8, A = V_{CC}, B = GND, and X = indeterminate.
5/ The tests in subgroups 7 and 8 shall be performed in the sequence specified.
6/ Output voltages shall be either:
(a) H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator, or
(b) H ≥ 1.5 V and L < 1.5 V when using a high speed checker single comparator.
7/ Only a summary of attributes data is required.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clock	2 Preset A	3 Preset B	4 Preset C	5 V_{CC}	6 Preset D	7 Preset E	8 Preset enable	9 Input serial	10 Q_E	11 Q_D	12 GND	13 Q_C	14 Q_B	15 Q_A	16 Clear	Meas. terminal	Test limits Min	Max	Unit
1 $T_A = 25°C$	V_{OH}	3006	1 2 3 4 5		2.0 V	2.0 V	2.0 V	4.5 V	2.0 V	2.0 V	2.0 V		-.4 mA	-.4 mA	GND	-.4 mA	-.4 mA	-.4 mA	2.0 V	Q_A Q_B Q_C Q_D Q_E	2.4		V
	V_{OL}	3007	6 7 8 9 10								0.8 V		16 mA	16 mA		16 mA	16 mA	16 mA	0.8 V	Q_A Q_B Q_C Q_D Q_E		0.4	
	V_{IC}		11 12 13 14 15 16 17 18 19	-12 mA	-12 mA	-12 mA	-12 mA		-12 mA	-12 mA	-12 mA	-12 mA							-12 mA	Clock Preset A Preset B Preset C Preset D Preset E Preset enable Serial input Clear		-1.5	
	I_{IL1}	3009	20 21 22 23 24 25 26 27	0.4 V	0.4 V	0.4 V	0.4 V	5.5 V	0.4 V	0.4 V	4.5 V	0.4 V							0.4 V	Clock Preset A Preset B Preset C Preset D Preset E Serial input Clear	-0.7	-1.6	mA
	I_{IL2}		28		4.5 V	4.5 V	4.5 V		4.5 V	4.5 V	0.4 V									Preset enable	-3.0	-8.0	
	I_{IH1}	3010	29 30 31 32 33 34 35 36	2.4 V	2.4 V	2.4 V	2.4 V		2.4 V	2.4 V	GND	2.4 V							2.4 V	Clock Preset A Preset B Preset C Preset D Preset E Serial input Clear		40	µA
	I_{IH2}		37 38 39 40 41 42 43 44	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V	GND	5.5 V							5.5 V	Clock Preset A Preset B Preset C Preset D Preset E Serial input Clear		100	

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	Clock	Preset A	Preset B	Preset C	V_{CC}	Preset D	Preset E	Preset enable	Input serial	Q_E	Q_D	GND	Q_C	Q_B	Q_A	Clear		Min	Max	Unit
1 T_A = 25°C	I_{IH3}	3010	45		GND	GND	GND	5.5 V	GND	GND	2.4 V				GND					Preset enable		200	µA
	I_{IH4}	3010	46		GND	GND	GND		GND	GND	5.5 V									Preset enable		500	µA
	I_{OS}	3011	47 48 49 50 51		4.5 V	4.5 V	4.5 V		4.5 V	4.5 V	4.5 V		GND	GND		GND	GND	GND	4.5 V	Q_A Q_B Q_C Q_D Q_E	-20	-57	mA
	I_{CCH}	3005	52	4.5 V								GND							5.5 V	V_{CC}		68	
	I_{CCL}	3005	53	GND							GND	GND							GND	V_{CC}		68	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																						
7 T_A = 25°C 1/ 4/	Truth table test 3/	3014	54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87	B A B A B A	B A B A	B B A B A B	B B B A B B B A B	4.5 V	B A B B A B	B A B	B A B B B A B B B A B B B A B B B A B B B A B B B A B B B A	B A B B A A B A B A A B A A B A A B A A B A A B A A B A A B A	L H L L H L L H L	L H L H L L H L L H L H L L H L H L L H L H L L H L H L L H	GND	L L L H L H L H H L H H L H H L H H L H H L H H L H H L H H	L L H L H L L H H L L H H L L H H L L H H L L H H L L H H	L H L H L H L H H L L H H L L H H L L H H L L H H L L H H	B A B A B A B A B A A B A A B A A B A A B A A B A A B A A B A		2/		
See footnotes at end of device type 02.																							

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clock	2 Preset A	3 Preset B	4 Preset C	5 V_{CC}	6 Preset D	7 Preset E	8 Preset enable	9 Input serial	10 Q_E	11 Q_D	12 GND	13 Q_C	14 Q_B	15 Q_A	16 Clear	Meas. terminal	Test limits Min	Max	Unit
7 T_A= 25°C 1/ 4/	Truth table test 3/	3014	88	B	B	B	B	4.5 V	B	B	B	A	L	H	GND	H	H	H	A				
			89	A		B					B	A	H					H					
			90	B		A					A	B						H					
			91	A		A												L					
			92	B		B											L						
			93	A					A							L							
			94	B					A														
			95	A					B	A													
			96	B						A													
			97	A						B	B		L										
			98	B																			
			99	A	A							A	L										
			100	B								A											
			101	A							A	B						H					
			102	B													H						
			103	A		A																	
			104	B	B											H		L					
			105	A		A	A										L			2/			
			106	B		B								H									
			107	A			A						H					L					
			108	B			B		A							L							
			109	A					A	A			H										
			110	B					B					L									
			111	A						A				L									
			112	B						B	B			L									
			113	A							B		L										
			114	B		A			A		A	A	L	H		H	H						
			115	A		A			A				H	H			H	H					
			116	B	A	B	A		B	A				H			L	L					
			117	A								B		L			L	H	B				
			118	B										L				H	A				
			119	A		A								H				H					
			120	B					A					H				H					
			121	A										H				H					
			122	B							B	A	L	L		L	L	L	B				
			123	A							B	A	L	L		L	L	L	B				
8	Repeat subgroup 7 at T_A = +125°C and T_A = -55°C.																						
9 T_A = 25°C	f_{MAX}	(Fig 7)	124	IN				5.0 V			GND	IN	OUT		GND				4.5 V	Q_E	5		MHz
	t_{PLH1}	3003 (Fig 7)	125															OUT		Q_A	12	40	ns
			126																	Q_B			
			127																	Q_C			
			128											OUT		OUT	OUT			Q_D			
			129										OUT							Q_E			
	t_{PHL1}		130															OUT		Q_A			
			131																	Q_B			
			132																	Q_C			
			133											OUT		OUT	OUT			Q_D			
			134										OUT							Q_E			

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	Clock	Preset A	Preset B	Preset C	V_{CC}	Preset D	Preset E	Preset enable	Input serial	Q_E	Q_D	GND	Q_C	Q_B	Q_A	Clear		Min	Max	Unit
9 T_A = 25°C	t_{PLH2}	3003 (Fig 7)	135 136 137 138 139	GND	4.5 V	4.5 V	4.5 V	5.0 V	4.5 V	4.5 V	IN		OUT	OUT	GND	OUT	OUT	OUT	IN	Q_A Q_B Q_C Q_D Q_E	12	42	ns
	t_{PHL3}		140 141 142 143 144										OUT	OUT		OUT	OUT	OUT		Q_A Q_B Q_C Q_D Q_E		55	
10 T_A = 125°C	f_{MAX}	(Fig 7)	145	IN							GND	IN	OUT						4.5 V	Q_E	3.5		MHz
	t_{PLH1}	3003 (Fig 7)	146 147 148 149 150										OUT	OUT		OUT	OUT	OUT		Q_A Q_B Q_C Q_D Q_E	12	56	ns
	t_{PHL1}		151 152 153 154 155										OUT	OUT		OUT	OUT	OUT		Q_A Q_B Q_C Q_D Q_E			
	t_{PLH2}		156 157 158 159 160	GND	4.5 V	4.5 V	4.5 V		4.5 V	4.5 V	IN		OUT	OUT		OUT	OUT	OUT	IN	Q_A Q_B Q_C Q_D Q_E		59	
	t_{PHL3}		161 162 163 164 165										OUT	OUT		OUT	OUT	OUT		Q_A Q_B Q_C Q_D Q_E		77	
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_A = -55°C.																						

1/ For subgroups 7 and 8, A = V_{CC} and B = GND.
2/ Output voltages shall be either:
(a) H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 V and L $<$ 1.5 V when using a high speed checker single comparator.
3/ The tests in subgroups 7 and 8 shall be performed in the sequence specified.
4/ Only a summary of attributes data is required.

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Input serial A	2 Input serial B	3 Q_A	4 Q_B	5 Q_C	6 Q_D	7 GND	8 Clock	9 Clear	10 Q_E	11 Q_F	12 Q_G	13 Q_H	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	2.0 V	2.0 V	-0.4 mA	-0.4 mA	-0.4 mA	-0.4 mA	GND	A 1/	4.5 V	-0.4 mA	-0.4 mA	-0.4 mA	-0.4 mA	4.5 V	Q_A Q_B Q_C Q_D Q_E Q_F Q_G Q_H	2.4		V
	V_{OL}	3007	9 10 11 12 13 14 15 16	0.8 V	0.8 V	8 mA	8 mA	8 mA	8 mA				8 mA	8 mA	8 mA	8 mA		Q_A Q_B Q_C Q_D Q_E Q_F Q_G Q_H		0.4	
	V_{IC}		17 18 19 20	-12 mA	-12 mA						-12 mA	-12 mA						Input serial A Input serial B Clock Clear		-1.5	
	I_{IL1}	3009	21 22 23	0.4 V 5.5 V	5.5 V 0.4 V						0.4 V						5.5 V	Input serial A Input serial B Clock		7/	
	I_{IL2}		24									0.4 V						Clear		8/	
	I_{IH1}	3010	25 26 27	2.4 V GND	GND 2.4 V						2.4 V							Input serial A Input serial B Clock		40	μA
	I_{IH2}		28 29 30	5.5 V GND	GND 5.5 V						5.5 V							Input serial A Input serial B Clock		100	
	I_{IH3}		31									2.4 V						Clear		[illegible]	
	I_{IH4}		32									5.5 V						Clear		10	

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Input serial A	2 Input serial B	3 Q_A	4 Q_B	5 Q_C	6 Q_D	7 GND	8 Clock	9 Clear	10 Q_E	11 Q_F	12 Q_G	13 Q_H	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	I_{OS}	3011	33 34 35 36 37 38 39 40	4.5 V	4.5 V	GND	GND	GND	GND	GND	A	4.5 V	GND	GND	GND	GND	5.5 V	Q_A Q_B Q_C Q_D Q_E Q_F Q_G Q_H	-10	-27.5	mA
	I_{CC1} I_{CC2}	3005 3005	41 42	GND GND	GND GND						0.4 V 2.4 V	B 2/ B						VCC VCC		44 54	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																				
7 T_A= 25°C 3/ 6/	Truth table test 5/	3014	43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78	A B A	A	L L L H L	L H L	L H L	L H L	GND	A A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A B A A	B A	L H L	L H L	L H L	L H L	4.5 V		4/		

See footnotes at end of device type 03.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Input serial A	2 Input serial B	3 Q_A	4 Q_B	5 Q_C	6 Q_D	7 GND	8 Clock	9 Clear	10 Q_E	11 Q_F	12 Q_G	13 Q_H	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_A= 25°C 3/ 6/	Truth table test 5/	3014	79	A	A	L	L	L	L	GND	B	A	L	L	L	L	4.5 V				
			80			H	L				A										
			81				L				B										
			82				H				A										
			83					H			B										
			84								A										
			85								B										
			86						H		A										
			87								B										
			88								A		H								
			89								B										
			90								A			H							
			91								B										
			92								A				H						
			93								B										
			94								A					H					
			95		B						A										
			96								B										
			97			L					A										
			98								B										
			99				L				A										
			100								B										
			101					L			A										
			102								B										
			103						L		A										
			104								B										
			105								A		L								
			106								B										
			107								A			L							
			108								B										
			109								A				L						
			110								B										
			111								A					L					
			112		A						A										
			113								B										
			114			H					A										
			115								B										
			116				H				A										
			117					H			B										
			118								A										
			119								B										
			120						H		A										
			121								B										
			122								A		H								
			123								B										
			124								A			H							
			125								B										
			126								A				H						
			127								B				H						
			128								A				H	H					

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Input serial A	2 Input serial B	3 Q_A	4 Q_B	5 Q_C	6 Q_D	7 GND	8 Clock	9 Clear	10 Q_E	11 Q_F	12 Q_G	13 Q_H	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_A= 25°C 3/ 6/	Truth table test 5/	3014	129	B	B	H	H	H	H	GND	A	A	H	H	H	H	4.5 V				
			130			H					B										
			131			L					A										
			132								B										
			133				L				A										
			134								B										
			135					L			A										
			136								B										
			137						L		A										
			138								B										
			139								A		L								
			140								B										
			141								A			L							
			142								B										
			143								A				L						
			144								B										
			145								A					L					
			146	A	A						A							4/			
			147			H					A										
			148								B										
			149								A										
			150				H				B										
			151								A										
			152					H			A										
			153								B										
			154						H		A										
			155								B										
			156								A		H								
			157								B										
			158								A			H							
			159								B										
			160								A				H						
			161								B				H						
			162								A				H						
			163			L	L	L	L		A	B	L	L	L	L					
8	Repeat subgroup 7 at T_A = 125°C and T_A = -55°C.																				
9 T_A = 25°C	f_{MAX}	(Fig 8)	164	IN	IN					GND	IN	4.5 V				OUT	5.0 V	Q_H	11		MHz
	t_{PHL1}	3003 (Fig 8)	165			OUT	OUT	OUT	OUT		GND	IN	OUT	OUT	OUT			Q_A	12	49	ns
			166															Q_B			
			167															Q_C			
			168															Q_D			
			169															Q_E			
			170															Q_F			
			171															Q_G			
			172													OUT		Q_H			

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Subgroup	Symbol	MIL-STD-883 method	Case A,B,C,D Test No.	1 Input serial A	2 Input serial B	3 Q_A	4 Q_B	5 Q_C	6 Q_D	7 GND	8 Clock	9 Clear	10 Q_E	11 Q_F	12 Q_G	13 Q_H	14 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_A = 25°C	t_{PLH2}	3003 (Fig 8)	173 174 175 176 177 178 179 180	IN	IN	OUT	OUT	OUT	OUT	GND	IN	4.5 V	OUT	OUT	OUT	OUT	5.0 V	Q_A Q_B Q_C Q_D Q_E Q_F Q_G Q_H	10	30	ns
	t_{PHL2}		181 182 183 184 185 186 187 188			OUT	OUT	OUT	OUT				OUT	OUT	OUT	OUT		Q_A Q_B Q_C Q_D Q_E Q_F Q_G Q_H		37	
10 T_A=125°C	f_{MAX}	(Fig 8)	189															Q_H	9		MHz
	t_{PHL1}	3003 (Fig 8)	190 191 192 193 194 195 196 197			OUT	OUT	OUT	OUT		GND	IN	OUT	OUT	OUT	OUT		Q_A Q_B Q_C Q_D Q_E Q_F Q_G Q_H	12	63	ns
	t_{PLH2}		198 199 200 201 202 203 204 205	IN	IN	OUT	OUT	OUT	OUT		IN	4.5 V	OUT	OUT	OUT	OUT		Q_A Q_B Q_C Q_D Q_E Q_F Q_G Q_H	10	42	
	t_{PHL2}		206 207 208 209 210 211 212 213			OUT	OUT	OUT	OUT				OUT	OUT	OUT	OUT		Q_A Q_B Q_C Q_D Q_E Q_F Q_G Q_H		52	
11	Same tests, terminal conditions and limits as for subgroup 10, except T_A = -55°C.																				

1/ A = normal clock pulse, except for subgroups 7 and 8 (see 3/).
2/ B = momentary GND, then 4.5 V to clear register prior to test, except for subgroups 7 and 8 (see 3/).
3/ For subgroups 7 and 8, A = V_{CC} and B = GND.
4/ Output voltages shall be either:
(a) H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 V and L $<$ 1.5 V when using a high speed checker single comparator.
5/ The tests in subgroups 7 and 8 shall be performed in the sequence specified.
6/ Only a summary of attributes data is required.
7/ For device type 03, with schematics incorporating 4 kΩ base resistors, the minimum and maximum limits shall be -0.7 and -1.6 mA, respectively. For schematics incorporating 6 kΩ base resistors, the minimum and maximum limits shall be -0.4 and -1.3 mA, respectively.
8/ For device type 03, schematic circuits A, C, D, E, and F, the minimum and maximum limits shall be -0.7 and -1.6 mA, respectively. For schematic circuit B, the minimum and maximum limits shall be -0.8 and -2.6 mA, respectively.
9/ For device type 03, schematic circuits A, C, D, E, and F, the maximum limit shall be 40 µA. For schematic circuit B, the maximum limit shall be 80 µA.
10/ For device type 03, schematic circuits A, C, D, E, and F, the maximum limit shall be 100 µA. For schematic circuit B, the maximum limit shall be 200 µA.

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TABLE III. Group A inspection for device type 04.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Shift load	2 Clock	3 E	4 F	5 G	6 H	7 $\overline{Q}_H$	8 GND	9 Q_H	10 Serial input	11 A	12 B	13 C	14 D	15 Clock inhibit	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 $T_A = 25°C$	V_{OH}	3006	1	0.8 V					2.0 V		GND	-.8 mA							4.5 V	Q_H	2.4		V
	V_{OH}	3006	2	↓					0.8 V	-.8 mA	↓								↓	$\overline{Q}_H$	2.4		↓
	V_{OL}	3007	3						2.0 V	16mA										$\overline{Q}_H$		0.4	
	V_{OL}	3007	4						0.8 V	↓		16mA								Q_H		0.4	
	V_{IC}		5	-12mA																Shift load		-1.5	
	↓		6		-12 mA															Clock		↓	
			7										-12 mA							Serial input			
			8															-12 mA		Clock inhibit			
			9											-12 mA						A			
			10												-12 mA					B			
			11													-12 mA				C			
			12														-12 mA			D			
			13			-12 mA														E			
			14				-12 mA													F			
			15					-12 mA												G			
			16						-12 mA										↓	H	↓	↓	↓
	I_{IL1}	3009	17	0.4 V	4.5 V													4.5 V	5.5 V	Shift load	-1.2	-3.9	mA
	I_{IL2}		18										0.4 V							Serial input	-0.4	-1.3	
	↓		19	4.5 V														0.4 V		Clock inhibit	↓	↓	
			20	GND										0.4 V						A			
			21												0.4 V					B			
			22													0.4 V				C			
			23														0.4 V			D			
			24			0.4 V														E			
			25				0.4 V													F			
			26					0.4 V												G			
			27	↓					0.4 V											H			
	I_{IL3}	↓	28	4.5 V	0.4 V						↓								↓	Clock	5/		↓

See footnotes at end of device type 04.

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Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Shift load	2 Clock	3 E	4 F	5 G	6 H	7 $\overline{Q}_H$	8 GND	9 Q_H	10 Serial input	11 A	12 B	13 C	14 D	15 Clock inhibit	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 T_A = 25°C	I_{IH1}	3010	29								GND		2.4 V						5.5 V	Serial input		40	µA
			30	GND														2.4 V		Clock inhibit			
			31	5.5 V										2.4 V						A			
			32												2.4 V					B			
			33													2.4 V				C			
			34														2.4 V			D			
			35			2.4 V														E			
			36				2.4 V													F			
			37					2.4 V												G			
			38						2.4 V											H			
			39	GND	2.4 V															Clock			
	I_{IH2}		40										5.5 V							Serial input		100	
			41	GND														5.5 V		Clock inhibit			
			42	5.5 V										5.5 V						A			
			43												5.5 V					B			
			44													5.5 V				C			
			45														5.5 V			D			
			46			5.5 V														E			
			47				5.5 V													F			
			48					5.5 V												G			
			49						5.5 V											H			
			50	GND	5.5 V															Clock			
	I_{IH3}		51	2.4 V	GND													GND		Shift load		120	
	I_{IH4}		52	5.5 V	GND													GND		Shift load		300	
	I_{OS}	3011	53	C 1/	4.5 V				4.5 V									4.5 V		Q_H	-20	-55	mA
		3011	54	C $\overline{1}$/	4.5 V				GND	GND		GND						4.5 V		$\overline{Q}_H$	-20	-55	mA
	I_{CC}	3005	55	C 1/	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V				GND	4.5 V	4.5 V	4.5 V	4.5 V	4.5 V		V_{CC}		63	mA
2	Same tests, terminal conditions and limits as for subgroup 1, except T_A = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																						
7 T_A= 25°C 2/ 5/	Truth table test 4/	3014	56	B	B	B	B	B	B	H	GND	L	A	B	B	B	B	B	4.5 V	3/			
			57	A	B	B	B	B	B					B	B	B	B	B					
			58		A	A	A	A	A					A	A	A	A	A					
			59		B																		
			60		A																		
			61		B																		
			62		A																		
			63		B																		
			64		A																		
			65		B																		
			66		A																		
			67		B																		
			68		A																		
			69		B																		

See footnotes at end of device type 04.

MIL-M-38510/9D

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	Shift load	Clock	E	F	G	H	$\overline{Q}_H$	GND	Q_H	Serial input	A	B	C	D	Clock inhibit	V_{CC}		Min	Max	Unit
7 T_A = +25°C 2/ 5/	Truth table test 4/	3014	70	A	A	A	A	A	A	H	GND	L	A	A	A	A	B	B	4.5 V				
			71		B					H		L	A										
			72		A					L		H	A										
			73		B								B										
			74		A								B										
			75		B								A										
			76		A								A										
			77		B								B										
			78		A								B										
			79		B								A										
			80		A								A										
			81		B								B										
			82		A								B										
			83		B								A										
			84		A								A										
			85		B								B										
			86		A								B										
			87		B								A										
			88		A					H		L											
			89		B																		
			90		A													A					
			91		B																		
			92		A																		
			93		B																		
			94		A																		
			95		B								B					B		3/			
			96		A					L		H	B										
			97		B					L		H	A										
			98		A					H		L	A										
			99		B					H		L	B										
			100		A					L		H	B										
			101		B					L		H	A										
			102		A					H		L	A										
			103		B					H		L	B										
			104		A					L		H	B										
			105		B					L		H	A										
			106		A					H		L	A										
			107		B					H		L	B										
			108		A					L		H	B										
			109		B		B		B	H		L	A		B		B						
			110		A																		
			111		B																		
			112		A																		
			113		B																		
			114		A													A					
			115		A													A					
			116		B								B					B					
			117		A		A		A	L		H	B		A		A						
			118		B					L		H	A										
			119		A					H		L	A										
			120		B					H		L	B										

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.

Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	Shift load	Clock	E	F	G	H	$\overline{Q}_H$	GND	Q_H	Serial input	A	B	C	D	Clock inhibit	V_{CC}		Min	Max	Unit
7 T_A= 25°C 2/ 5/	Truth table test 4/	3014	121	A	A	A	A	A	A	L	GND	H	B	A	A	A	A	B	4.5 V	3/			
			122		B					L		H	A										
			123		A					H		L	A										
			124		B					H		L	B										
			125		A					L		H	B										
			126		B					L		H	A										
			127		A					H		L	A										
			128		B					H		L	B										
			129		A					L		H	B										
			130		B					L		H	A										
			131		A					H		L											
			132	B	B	B		B		L		H		B		B							
			133		A																		
			134		B																		
			135		A																		
			136		B																		
			137		A													A					
			138	A	A								B					A					
			139		B	B		B		H		L		B		B		B					
			140		A	A		A		H		L		A		A							
			141		B					H		L											
			142		A					L		H											
			143		B					L		H											
			144		A					H		L											
			145		B					H		L											
			146		A					L		H											
			147		B					L		H											
			148		A					H		L											
			149		B					H		L											
			150		A					L		H											
			151		B					L		H											
			152		A					H		L											
			153		B																		
			154		A																		
			155		B																		
			156		A																		
			157		B																		
			158		A																		
			159		B																		
			160		A																		
			161		B																		
			162		A																		
			163		B																		
			164		A																		
			165		B																		
			166		A																		
			167		B																		
			168		A																		
			169	B	B					L		H											
			170	A	B					L		H											
			171	A	A					L		H											

See footnotes at end of device type 04.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Shift load	2 Clock	3 E	4 F	5 G	6 H	7 $\overline{Q}_H$	8 GND	9 Q_H	10 Serial input	11 A	12 B	13 C	14 D	15 Clock inhibit	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_A= 25°C 2/ 5/	Truth table test 4/	3014	172	A	B	A	A	A	A	L	GND	H	B	A	A	A	A	B	4.5 V	3/			
			173		A																		
			174		B																		
			175		A																		
			176		B																		
			177		A																		
			178		B																		
			179		A																		
			180		B																		
			181		A																		
			182		B																		
			183		A																		
			184	B					B	H		L											
			185	A					B	H		L											
			186	B				B	A	L		H											
			187	A				B	A	L		H											
8	Repeat subgroup 7 at T_A = +125°C and T_A = -55°C.																						
9 T_A = 25°C	f_{MAX}	(Fig 9)	188	IN	IN	GND	GND	GND	GND		GND	OUT	IN	GND	GND	GND	GND	GND	5.0 V	Q_H	18		MHz
	t_{PLH1}	3003	189	IN	GND	GND	GND	GND	IN			OUT		GND	GND	GND	GND	GND		Shift-Q_H	6	35	ns
	t_{PHL1}	(Fig 9)	190	IN	GND	GND	GND	GND	IN	OUT				GND	GND	GND	GND	GND		Shift-$\overline{Q}_H$	7	44	
	t_{PLH2}		191	5.0 V	IN							OUT	IN					IN		Clock-Q_H	5	28	
	t_{PHL2}		192	5.0 V	IN					OUT			IN					IN		Clock-$\overline{Q}_H$	6	35	
	t_{PLH3}		193	GND	GND	GND	GND	GND	IN			OUT	GND	GND	GND	GND	GND	GND		H - Q_H	5	21	
	t_{PHL3}		194									OUT								H - $\overline{Q}_H$	7	40	
	t_{PLH4}		195							OUT										H - Q_H	6	31	
	t_{PHL4}		196							OUT										H - $\overline{Q}_H$	6	31	
10 T_A= 125°C	f_{MAX}	(Fig 9)	197	IN	IN				GND			OUT	IN							Q_H	14		MHz
	t_{PLH1}	3003	198	IN	GND				IN			OUT								Shift-Q_H	10	40	ns
	t_{PHL1}	(Fig 9)	199	IN	GND				IN	OUT										Shift-$\overline{Q}_H$	11	60	
	t_{PLH2}		200	5.0 V	IN							OUT	IN					IN		Clock-Q_H	6	37	
	t_{PHL2}		201	5.0 V	IN					OUT			IN					IN		Clock-$\overline{Q}_H$	10	47	
	t_{PLH3}		202	GND	GND	GND	GND	GND	IN			OUT	GND	GND	GND	GND	GND	GND		H - Q_H	5	27	
	t_{PHL3}		203									OUT								H - $\overline{Q}_H$	11	54	
	t_{PLH4}		204							OUT										H - Q_H	10	41	
	t_{PHL4}		205							OUT										H - $\overline{Q}_H$	10	41	
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_A = -55°C.																						

1/ C = after all other input conditions, but prior to measurement, apply momentary GND, then 4.5 V.
2/ For subgroups 7 and 8, A = V_{CC} and B = GND.
3/ Output voltages shall be either:
(a) H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 V and L < 1.5 V when using a high speed checker single comparator.
4/ The tests in subgroups 7 and 8 shall be performed in the sequence specified.
5/ Only a summary of attributes data is required.
6/ For device type 04, schematics incorporating a 4 kΩ base resistor in the clock input circuit, the minimum and maximum limits shall be -0.7 and -1.6 mA, respectively. For schematics incorporating a 6 kΩ base resistor in the clock input circuit, the minimum and maximum limits shall be -0.4 and -1.3 mA, respectively.

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Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Shift right	3 Input A	4 Input B	5 Input C	6 Input D	7 Shift left	8 GND	9 S_0	10 S_1	11 Clock	12 Q_D	13 Q_C	14 Q_B	15 Q_A	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
T_A = 25°C [1]	V_{OH}	3006	1	2.0 V		2.0 V	2.0 V	2.0 V	2.0 V		GND	2.0 V	2.0 V	A [1/]			-0.8 mA	-0.8 mA	4.5 V	Q_A	2.4		V
			2																	Q_B			
			3													-0.8 mA				Q_C			
			4												-0.8 mA					Q_D			
	V_{OL}	3007	5			0.8 V	0.8 V	0.8 V	0.8 V									16 mA		Q_A		0.4	
			6														16 mA			Q_B			
			7													16 mA				Q_C			
			8												16 mA					Q_D			
	V_{IC}		9	-12 mA																Clear		-1.5	
			10		-12 mA															Shift R			
			11			-12 mA														Input A			
			12				-12 mA													Input B			
			13					-12 mA												Input C			
			14						-12 mA											Input D			
			15							-12 mA										Shift L			
			16									-12 mA								S_0			
			17										-12 mA							S_1			
			18											-12 mA						Clock			
	I_{IL1}	3009	19	0.4 V															5.5 V	Clear	-0.4	-1.3	mA
			20		0.4 V								GND							Shift R			
			21			0.4 V						5.5 V	5.5 V							Input A			
			22				0.4 V													Input B			
			23					0.4 V												Input C			
			24						0.4 V											Input D			
			25							0.4 V		GND								Shift L			
	I_{IL2}		26									0.4 V								S_0	[7/]		
	I_{IL2}		27										0.4 V							S_1			
	I_{IL3}		28	5.5 V										0.4 V						Clock	-0.7	-1.6	mA

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.

Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Shift right	3 Input A	4 Input B	5 Input C	6 Input D	7 Shift left	8 GND	9 S_0	10 S_1	11 Clock	12 Q_D	13 Q_C	14 Q_B	15 Q_A	16 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
1 T_A = 25°C	I_{IH1}	3010	29	2.4 V							GND								5.5 V	Clear		40	µA
			30		2.4 V								5.5 V							Shift R			
			31			2.4 V						GND	GND							Input A			
			32				2.4 V													Input B			
			33					2.4 V												Input C			
			34						2.4 V											Input D			
			35							2.4 V		5.5 V								Shift L			
			36									2.4 V								S_0			
			37										2.4 V							S_1			
			38											2.4 V						Clock			
	I_{IH2}		39	5.5 V																Clear		100	
			40		5.5 V								5.5 V							Shift R			
			41			5.5 V						GND	GND							Input A			
			42				5.5 V													Input B			
			43					5.5 V												Input C			
			44						5.5 V											Input D			
			45							5.5 V		5.5 V								Shift L			
			46									5.5 V								S_0			
			47										5.5 V							S_1			
			48											5.5 V						Clock			

See footnotes at end of device type 05.

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Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Shift right	3 Input A	4 Input B	5 Input C	6 Input D	7 Shift left	8 GND	9 S_0	10 S_1	11 Clock	12 Q_D	13 Q_C	14 Q_B	15 Q_A	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	I_{OS}	3011	49 50 51 52	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V		GND	5.5 V	5.5 V	A	GND	GND	GND	GND	5.5 V	Q_A Q_B Q_C Q_D	-20	-57	mA
	I_{CC}	3005	53		5.5 V	GND	GND	GND	GND	5.5 V										V_{CC}		63	
2	Same tests, terminal conditions and limits as subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																						
7 T_A = 25°C 3/ 6/	Truth table test 5/	3014	54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98	B 2/ A B A

TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	Clear	Shift right	Input A	Input B	Input C	Input D	Shift left	GND	S_0	S_1	Clock	Q_D	Q_C	Q_B	Q_A	V_{CC}		Min	Max	Unit
7 T_A= 25°C 3/ 6/	Truth table test 5/	3014	99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123	A B	B	B	B	B	B	A A A B B B A A A B B B A A A A B	GND	B	A B	A B A A B A A B A A B A A B A B A B A B A B A B A B	L L H H H L L L H H H L L L H H H H L	L H H H H L L L L H H H H L L L L	L H H H H L L L L H H H H L L L L	L H H H H L L L L H H H H L L L L	5.0 V		4/		
8	Repeat subgroup 7 at T_A = 125°C and T_A = -55°C.																						
9 T_A = 25°C	f_{MAX}	(Fig 10)	124	B	IN	GND	GND	GND	GND	GND	GND	5.0 V	GND	IN				OUT	5.0 V	Q_A	11		MHz
	t_{PHL1}	3003 (Fig 10)	125 126 127 128	IN		5.0 V	5.0 V	5.0 V	5.0 V				5.0 V		OUT	OUT	OUT	OUT		Clear/Q_A Clear/Q_B Clear/Q_C Clear/Q_D	7	34	ns
	t_{PLH2}		129 130 131 132	B		IN 5.0 V 5.0 V 5.0 V	IN 5.0 V 5.0 V	IN 5.0 V	IN						OUT	OUT	OUT	OUT		Clock/Q_A Clock/Q_B Clock/Q_C Clock/Q_D		26	
	t_{PHL2}		133 134 135 136	5.0 V		IN 5.0 V 5.0 V 5.0 V	5.0 V IN 5.0 V	5.0 V 5.0 V IN	5.0 V 5.0 V 5.0 V IN						OUT	OUT	OUT	OUT		Clock/Q_A Clock/Q_B Clock/Q_C Clock/Q_D		32	

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 Shift right	3 Input A	4 Input B	5 Input C	6 Input D	7 Shift left	8 GND	9 S_0	10 S_1	11 Clock	12 Q_D	13 Q_C	14 Q_B	15 Q_A	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
10 T_A=125°C	f_{MAX}	(Fig 10)	137	B	IN	GND	GND	GND	GND	GND	GND	5.0 V	GND	IN				OUT	5.0 V	Q_A	9		MHz
↓	t_{PHL1}	3003 (Fig 10)	138	IN		5.0 V	5.0 V	5.0 V	5.0 V				5.0 V					OUT		Clear/Q_A	7	48	ns
	↓	↓	139	↓		↓	↓	↓	↓				↓			OUT	OUT	↓		Clear/Q_B	↓	↓	↓
			140																	Clear/Q_C			
			141												OUT					Clear/Q_D			
	t_{PLH2}		142	B		IN	↓											OUT		Clock/Q_A		36	
	↓		143	↓		5.0 V	IN	↓								OUT	OUT	↓		Clock/Q_B		↓	
			144			5.0 V	5.0 V	IN	↓											Clock/Q_C			
			145			5.0 V	5.0 V	5.0 V	IN						OUT					Clock/Q_D			
	t_{PHL2}		146	5.0 V		IN	5.0 V	5.0 V	5.0 V									OUT		Clock/Q_A		44	
	↓		147	↓		5.0 V	IN	5.0 V	5.0 V							OUT	OUT	↓		Clock/Q_B		↓	
			148			5.0 V	5.0 V	IN	5.0 V											Clock/Q_C			
			149			5.0 V	5.0 V	5.0 V	IN						OUT					Clock/Q_D			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																						

1/ A = normal clock pulse, except for subgroup 7 and 8 (see 3/).
2/ B = momentary GND, then V_{IN} (except for subgroups 7 and 8). For subgroups 1, 2 and 3, V_{IN} = V_{CC}; for subgroups 9, 10 and 11, V_{IN} = 3.0 V minimum (see figure 10).
3/ For subgroups 7 and 8, A = V_{CC} and B = GND.
4/ Output voltages shall be either:
(a) H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 V and L $<$ 1.5 V when using a high speed checker single comparator.
5/ The tests in subgroups 7 and 8 shall be performed in the sequence specified.
6/ Only a summary of attributes data is required.
7/ For device type 05, schematic circuits A and B, the minimum and maximum limits shall be -0.4 and -1.3 mA, respectively. For schematic C, the minimum and maximum limits shall be -0.7 and -1.6 mA, respectively.

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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
				Clear	J	$\overline{K}$	Input A	Input B	Input C	Input D	GND	Shift/ load	Clock	$\overline{Q}_D$	Q_D	Q_C	Q_B	Q_A	V_{CC}				
T_A = 25°C 1/	V_{OH}	3006	1	2.0 V			2.0 V	2.0 V	2.0 V	2.0 V	GND	0.8 V	A 1/					-0.8 mA	4.5 V	Q_A	2.4		V
			2														-0.8 mA			Q_B			
			3													-0.8 mA				Q_C			
			4												-0.8 mA					Q_D			
			5				0.8 V	0.8 V	0.8 V	0.8 V				-0.8 mA						$\overline{Q}_D$			
	V_{OL}	3007	6															16 mA		Q_A		0.4	
			7														16 mA			Q_B			
			8													16 mA				Q_C			
			9												16 mA					Q_D			
			10				2.0 V	2.0 V	2.0 V	2.0 V				-.8 mA						$\overline{Q}_D$			
	V_{IC}		11	-12 mA																Clear		-1.5	
			12		-12 mA															J			
			13			-12 mA														$\overline{K}$			
			14				-12 mA													A			
			15					-12 mA												B			
			16						-12 mA											C			
			17							-12 mA										D			
			18									-12 mA								Shift/ load			
			19										-12 mA							Clock			
	I_{IL1}	3009	20	0.4 V								5.5 V							5.5 V	Clear	-0.4	-1.3	mA
	I_{IL2}		21	GND	0.4 V							5.5 V								J			
			22	B 2/	5.5 V	0.4 V	5.5 V					5.5 V	A							$\overline{K}$			
			23				0.4 V					GND								A			
			24					0.4 V												B		7/	
			25						0.4 V											C			
			26							0.4 V										D			
			27									0.4 V								Shift/ load			
	I_{IL3}		28	5.5 V									0.4 V							Clock	-0.7	-1.6	mA

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06.- Continued.

Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Max	Unit
				Clear	J	$\overline{K}$	Input A	Input B	Input C	Input D	GND	Shift/ load	Clock	$\overline{Q}_D$	Q_D	Q_C	Q_B	Q_A	V_{CC}				
1 T_A = 25°C	I_{IH1}	3010	29	2.4 V							GND								5.5 V	Clear		40	µA
			30	5.5 V	2.4 V		5.5 V					GND	A							J			
			31	GND		2.4 V						GND								$\overline{K}$			
			32				2.4 V					5.5 V								A			
			33					2.4 V												B			
			34						2.4 V											C			
			35							2.4 V										D			
			36									2.4 V								Shift/ load			
			37	GND									2.4 V							Clock			
	I_{IH2}		38	5.5 V																Clear		100	
			39	5.5 V	5.5 V		5.5 V					GND	A							J			
			40	GND		5.5 V						GND								$\overline{K}$			
			41				5.5 V					5.5 V								A			
			42					5.5 V												B			
			43						5.5 V											C			
			44							5.5 V										D			
			45																	Shift/ load			
			46	GND									5.5 V							Clock			

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 J	3 $\overline{K}$	4 Input A	5 Input B	6 Input C	7 Input D	8 GND	9 Shift/load	10 Clock	11 $\overline{Q}_D$	12 Q_D	13 Q_C	14 Q_B	15 Q_A	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_A = 25°C	I_{OS}	3011	47	5.5 V			5.5 V	5.5 V	5.5 V	5.5 V	GND	GND	A					GND	5.5 V	Q_A	-20	-57	mA
			48																	Q_B			
			49													GND	GND			Q_C			
			50												GND					Q_D			
			51	GND										GND						$\overline{Q}_D$			
	I_{CC}	3005	52	5.5 V	GND	GND	GND	GND	GND	GND		GND	A							V_{CC}		63	
2	Same tests, terminal conditions and limits as subgroup 1, except T_A = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as subgroup 1, except T_A = -55°C and V_{IC} tests are omitted.																						
7 T_A = 25°C 3/ 6/	Truth table test 5/	3014	53	B	B	B	B	B	B	B	GND	A	A	H	L	L	L	L	5.0 V	4/			
			54	A	A	A							A			L		L					
			55										B					L					
			56										A					H					
			57										B				H						
			58										A			H							
			59										B										
			60										A			H							
			61										B		L								
			62										A	L	H								
			63		B								A										
			64										B										
			65										A										
			66										B										
			67										A										
			68										B										
			69										A										
			70										B										
			71			B							A										
			72										A										
			73										B										
			74										A					L					
			75										B				L						
			76										A			L							
			77										B										
			78										A			L							
			79										B		L								
			80				A		A			B	A	H	L								
			81				A		A				A										
			82				A		A				B										
			83				A	A	A	A			A			H		H					
			84				B	A	B	A			A			H		H					
			85				B		B				B		L	L		L					
			86				B		B				A	L	H	L	H	L					
			87				A		A				A			L		L					
			88										B			L		L					
			89										A			H		H					
			90	B									B	H	L	L	L	L					
			91	B									A	H	L	L	L	L					

See footnotes at end of device type 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated are open, high level or low level)

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 Clear	2 J	3 $\overline{K}$	4 Input A	5 Input B	6 Input C	7 Input D	8 GND	9 Shift/load	10 Clock	11 $\overline{Q}_D$	12 Q_D	13 Q_C	14 Q_B	15 Q_A	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_A = 25°C	Truth table test 5/	3014	92	B	A	A	A	A	A	A	GND	B	A	H	L	L	L	L	5.0 V	4/			
			93	A								B											
			94	B								A											
			95	A																			
			96	A			B	B	B	B			B										
			97	A			B	B	B	B			A					H					
8	Repeat subgroup 7 at T_A = 125°C and T_A = -55°C.																						
9 T_A = 25°C	f_{MAX}	(Fig 11)	98	B	5.0 V	GND					GND	5.0 V	IN					OUT	5.0 V	Q_A	15		MHz
	t_{PHL1}	3003 (Fig 11)	99	IN			5.0 V	5.0 V	5.0 V	5.0 V		GND				OUT	OUT	OUT		Clear/Q_A	7	30	ns
			100																	Clear/Q_B			
			101																	Clear/Q_C			
			102												OUT	OUT				Clear/Q_D			
	t_{PLH2}		103	B			IN	IN	IN	IN		IN					OUT	OUT		Clock/Q_A		24	
			104																	Clock/Q_B			
			105																	Clock/Q_C			
			106												OUT	OUT				Clock/Q_D			
	t_{PHL2}		107				IN	IN	IN	IN							OUT	OUT		Clock/Q_A		29	
			108																	Clock/Q_B			
			109																	Clock/Q_C			
			110												OUT	OUT				Clock/Q_D			
10 T_A = 125°C	f_{MAX}	(Fig 11)	111		5.0 V	GND						5.0 V						OUT		Q_A	12		MHz
	t_{PHL1}	3003 (Fig 11)	112	IN			5.0 V	5.0 V	5.0 V	5.0 V		GND				OUT	OUT	OUT		Clear/Q_A	7	34	ns
			113																	Clear/Q_B			
			114																	Clear/Q_C			
			115												OUT	OUT				Clear/Q_D			
	t_{PLH2}		116	B			IN	IN	IN	IN		IN					OUT	OUT		Clock/Q_A		28	
			117																	Clock/Q_B			
			118																	Clock/Q_C			
			119												OUT	OUT				Clock/Q_D			
	t_{PHL2}		120				IN	IN	IN	IN							OUT	OUT		Clock/Q_A		34	
			121																	Clock/Q_B			
			122																	Clock/Q_C			
			123												OUT	OUT				Clock/Q_D			
11	Same tests, terminal conditions and limits as subgroup 10, except T_A = -55°C.																						

1/ A = normal clock pulse, except for subgroups 7 and 8 (see 3/).
2/ B = momentary GND, then V_{IN}, except for subgroups 7 and 8. For subgroups 1, 2 and 3, $V_{IN} = V_{CC}$; for subgroups 9, 10 and 11, V_{IN} = 3.0 V minimum (see figure 11).
3/ For subgroups 7 and 8, A = V_{CC} and B = GND.
4/ Output voltages shall be either:
(a) H = 2.4 V minimum and L = 0.4 V maximum when using a high speed checker double comparator, or
(b) H $\geq$ 1.5 V and L $<$ 1.5 V when using a high speed checker single comparator.
5/ The tests in subgroups 7 and 8 shall be performed in the sequence specified.
6/ Only a summary of attributes data is required.
7/ For device type 06, schematic circuits A and B, the minimum and maximum limits shall be -0.4 and -1.3 mA, respectively. For schematic circuit C, the minimum and maximum limits shall be -0.7 and -1.6 mA, respectively.

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3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 5 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).
   1. Test condition D or E, using the circuit shown on figure 5, or equivalent.
   2. T_A = 125°C minimum.
b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.
c. Percent defective allowable (PDA) - The PDA for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for group A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for group A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.
b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

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4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.
b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B and C devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.
c. Operating life test (method 1005 of MIL-STD-883) conditions, or equivalent:
   1. Test condition D or E, using the circuit shown on figure 8, or equivalent.
   2. T_A = 125°C minimum.
   3. Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 MIL-STD-883; end-point electrical parameters shall be specified in table II herein.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in e appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. rrents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of e operating life or burn-in test condition, they shall be cooled to room temperature prior to removal the bias. Alternately, the bias may be removed during cooling if the case temperature is reduced room temperature within 30 minutes after removal of the test condition.

4.6 Packaging inspection. The sampling and inspection of the preservation-packaging, packing, and ntainer marking shall be in accordance with the requirements of MIL-M-38510, except that the rough ndling test shall not apply.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for use for overnment microcircuit applications (original equipment) and logistic support.

6.3 Ordering data. Procurement documents should specify the following:

a. Complete part number (see 1.2).
b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.
c. Requirements for certificate of compliance, if applicable.
d. Requirements for notification of change of product or process to procuring activity in addition to notification to the qualifying activity, if applicable.
e. Requirements for packing and packaging.
f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.
g. Requirements for product assurance options.
h. Requirements for carriers, special lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.
i. Requirement for "JAN" marking.

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6.4 Abbreviations, symbols,and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - Electrical ground (common terminal)

V_{IN} - - - - - - - - - - - - - - Voltage level at an input terminal

V_{IC} - - - - - - - - - - - - - - Input clamp voltage

I_{IN} - - - - - - - - - - - - - - Current flowing into an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits procured for Government logistic support will be procured to device class B (see 1.2.2) and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification shall functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	5495, 7495
02	5496, 7496
03	54164, 74164
04	54165, 74165
05	54194, 74194
06	54195, 74195

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 85, 99
NASA - NA
DLA - ES

User activities:
Army - SM, WC
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-38510/9D

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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